Progress in Aerospace Sciences 143 (2023) 100956

Contents lists available at ScienceDirect

Progress in Aerospace Sciences

journal homepage: www.elsevier.com/locate/paerosci

Review on the aerodynamic issues of the exhaust system for scramjet and turbine based combined cycle engine

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ARTICLE INFO

Keywords:
Scramjet
Turbine based combined cycle
Exhaust system
Design
Variable geometry
Mode transition

ABSTRACT

A review of research progress in the design of the exhaust system for the scramjet and turbine based combined cycle (TBCC) engine is presented. Firstly, the technical challenges encountered in designing the exhaust system for a hypersonic propulsion system are highlighted and discussed, and the performance parameter definition as well as the theoretical thrust prediction for the exhaust system is introduced. The review of scramjet nozzle focuses on three aspects: 1) the design method of the single expansion ramp nozzle (SERN) for the integration of the airframe with the propulsion system, in which the design method developments of the two-dimensional (2D) SERN, SERN with lateral expansion and three-dimensional (3D) SERN with shape transition are all summarized; 2) the unique flow phenomena of the scramjet nozzle, including the nonuniform inflow and chemical nonequilibrium flow in SERN; 3) the coupling and interaction of the internal flow with the external freestream. Besides, the design and flow researches of the TBCC exhaust system is also reviewed for three parts: 1) variable geometry design for wide flight range, in which both a 2D and 3D exhaust system are described; 2) the over-expanded flow separation mechanism and its control at low flight Mach number; 3) mode transition from low-speed flowpath (LSF) to high-speed flowpath (HSF) for over-under exhaust system. Through the above summary and analysis, the current status, bottlenecks, and development trend of the exhaust system for an airbreathing hypersonic propulsion system can be further clarified.

1. Introduction

1.1. Development history of the air-breathing hypersonic propulsion system

For sustained hypersonic (Ma>5) atmospheric flight, the key is the propulsion system with high performance [1–3], and the development of the hypersonic technology starts from the research of the supersonic combustion and scramjet [4]. Compared with the common rocket, scramjet has lower operating temperature and pressure, which usually relates to improved reliability and reusability, and it can also capture the oxygen in the atmosphere with superior safety and higher specific impulse (Fig. 1) [5–7]. Furthermore, the scramjet relies on the forward speed of the vehicle to compress the air in the inlet instead of the mechanical compressor of a gas turbine, which makes it to be simple in structure, light in weight and large in payload [1]. Therefore, as the optimum power choice of the hypersonic flight, the scramjet has become a research hotspot [8–10].

In the 1950s, researchers have carried out the conceptual and basic studies on the supersonic combustion [11], in particular Ferri [12] experimentally proved that the stable combustion without strong shock waves could be realized in the supersonic flowfield of Mach 3.0, verifying the feasibility of the supersonic combustion, and setting off the upsurge of the research on the scramjet. Thereby, major aerospace powers have formulated the corresponding research projects (Fig. 2). As the leader of hypersonic technology, the United States has been committed to the scramjet research, and has successively launched the Hypersonic Research Engine (HRE) program [13], Supersonic Combustion Ramjet Missile (SCRAM) [14], National Aero-Space Plane (NASP) [15,16], Hypersonic Technology (HyTech) [17,18], Hyper-X [19,20], Hypersonic Flight Missile (HyFly) [21], Force Application and Launch from Continental United States (FALCON) [22,23], X-51A [24], and several hypersonic international cooperation projects with Australia, including HyShot [25,26], Hypersonic Collaborative Australia/United States Experiment (HyCAUSE) [27,28], and Hypersonic

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https://doi.org/10.1016/j.paerosci.2023.100956
Received 14 February 2023; Received in revised form 21 September 2023; Accepted 6 October 2023
Available online 22 October 2023
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Nomenclature

A	Area
a	Speed of sound
C_{fx}	Thrust coefficient, the symbol with 1 represents the thrust coefficient based on exit surplus momentum, and the symbol with 2 represents the thrust coefficient based on net thrust
F	Thrust
f	A weighted function
G	Asymmetric factor
I	Surplus momentum
Ma	Mach number
$\dot{m}$	Mass flow rate
p	Pressure
R	Gas constant
R_x	x-direction component of net thrust
u	x-direction component of velocity
v	y-direction component of velocity
w	Width
α	Mach angle
δ	2D/Axisymmetric/Quasi-2D flow switch
δ_L	Initial expansion angle of cowl
δ_U	Initial expansion angle of ramp
φ	under-expanded factor
V	Velocity
ρ	Density
φ	Inclination angle
γ	Specific heat ratio
ξ	Proportion of expansion flow along divergent section to exit surplus momentum
θ	Flow angle
χ	Flow parameter
$\overline{\chi}$	f-weighted parameter

Subscripts

a	Ambient
b	Back
e	Exit
in	Entrance
no	Without secondary injection
s	Isentropic
sec	Secondary injection
sep	Separation
t	Total
$with$	With secondary injection
x	Component along x-direction

Acronyms

AFRE	Advanced Full Range Engine
AR	Aera Ratio
CCE	Combined Cycle Engine
CFD	Computational Fluid Dynamics
DMRJ	Dual-mode Ramjet
ERJ	Ejector ramjet
FALCON	Force Application and Launch from Continental United States
FaCET	Falcon Combined-cycle Engine Technology
FSS	Free Shock Separation
HIFiRE	Hypersonic International Flight and Experimentation
HRE	Hypersonic Research Engine
HSF	High-speed Flowpath
HyCAUSE	Hypersonic Collaborative Australia/United States Experiment
LSF	Low-speed Flowpath
MOC	Method of Characteristics
NASP	National Aero-Space Plane
NPR	Nozzle Pressure Ratio
RBCC	Rocket Based Combined Cycle
RPNCC	Rocket Plug Nozzle Combined Cycle
RSS	Restricted Shock Seapration
RTA	Revolutionary Turbine Accelerator
SAC	Sustained Aerodynamic Choke
SCRAM	Supersonic Combustion Ramjet Missile
SERN	Single Expansion Ramp Nozzle
TA	Throat Area
TBCC	Turbine Based Combined Cycle
TIC	Truncated Ideal Contour
TOC	Thrust Optimized Contour
TOP	Thrust Optimized Parabolic
TRRE	Turbo-aided Rocket-augmented Ramjet Combined Cycle Engine
2D	Two-dimensional
3D	Three-dimensional

International Flight and Experimentation (HIFiRE) [29–31]. As the successor of the Soviet Union's scientific and military power, Russia has always been in an important position of the hypersonic technology field, and has also formulated a series of research projects, such as Kholod program [32,33], IGLA demonstrator flight [34,35], and Raduga-D2 vehicle [9]. In addition, some other countries, such as France, Germany and Japan, have also carried out the corresponding hypersonic technology research programs: ESOPE [3], PREPHA [36], JAPHAR [37], PROMETHEE [38], Sanger [39] and Hope-X [40,41].

Since the supersonic combustion concept was put forward, researchers have conducted many researches on theory, numerical simulation and ground wind tunnel test, and obtained a large number of results. As a result, the scramjet study started from the basic research, key component design and ground test to the stage of the flight demonstration (Table 1). Especially X-51A powered by scramjet with hydrocarbon fuel successfully achieved the autonomous flight test, and the flight Mach number and time reached 5 and 140s, respectively, marking the first hypersonic flight of the practical scramjet. After decades of the unremitting efforts, researchers have accumulated an incredible amount of experience in the fields of aerodynamic design, combustion, manufacture, thermal protection and integrated design, which lays a good foundation for the further engineering application.

Owing to the fact that the scramjet can only work at a certain flight Mach number, it cannot achieve the purpose of the horizontal takeoff and landing of the hypersonic vehicle. At present, the operating Mach number ranges of the conventional turbine, ramjet and scramjet are 0–3, 2–5 and greater than 5, respectively. Therefore, to satisfy the mission demand within a wide range, it is necessary to combine various engines with different operation ranges to obtain a combined cycle engine (CCE), such as turbine based combined cycle (TBCC) (Fig. 3). In an over-under configuration, the low-speed turbine flowpath (LSF) is located parallel to and above the high-speed ramjet/scramjet flowpath (HSF). During take-off stage, the turbine provides thrust to accelerate the vehicle through the low supersonic flight range, and the continuous acceleration through supersonic regime, the propulsion system will transition from turbine to ramjet/scramjet.

Considering the advantages of a TBCC engine in the field of the hypersonic propulsion system, researchers have given high expectations for its military and civil values. The first TBCC engine put into use is the J58 powering the SR-71 aircraft. It is composed of a turbine and an

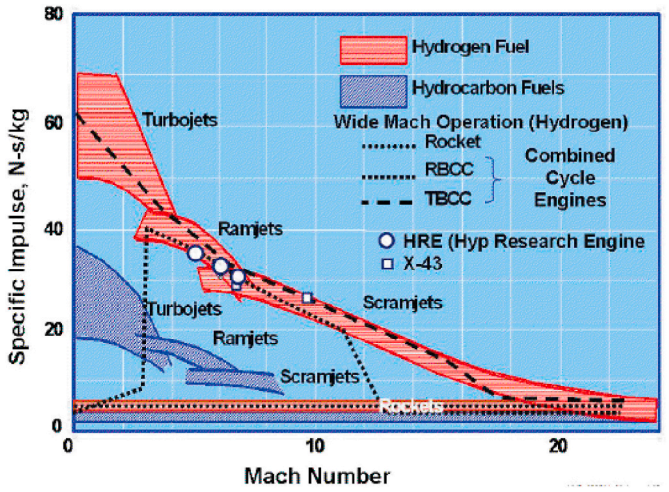


Fig. 1. Propulsion efficiency and operating regimes for a variety of flight systems [6].

afterburner/ramjet combustor, which can make full use of the optimum performance of the turbine and ramjet in different working ranges and accelerate the aircraft to Mach 3+ [43,44]. At Mach 3.0, the turbine produces only 17% of the total thrust, and the ramjet is the main power system. In the following decades, with the continuous progress of the hypersonic technology, the study on TBCC is more and more in-depth. Major countries have also formulated the corresponding research plans, such as NASP [15,45], Revolutionary Turbine Accelerator (RTA) [46–48], Falcon Combined-cycle Engine Technology (FaCET) [22,23,49,50], Quicksat [51,52], TriJet [53,54], SR-72 [55], Advanced Full Range Engine (AFRE), Hyscram in the United States, LAPCAT [56–58], Sabre [59,60] in Europe and Turbo-aided Rocket-augmented Ramjet Combined Cycle Engine (TRRE) [61,62] in China. Fig. 4 summarizes the detailed information of the TBCC research plans, and the over/under configuration is the key research direction, including NASP, FaCET, Quicksat, SR-72, TRRE, Sabre and Hyscram. It is worth mentioning that the wrap-around configuration of Sabre and Hyscram is a specific over/under concept with dual-flowpath. Furthermore, to settle the problem of the insufficient thrust during low flight Mach number and mode transition from the turbine to the dual-mode scramjet (DMRJ), the rocket or ejector ramjet (ERJ) is employed to bridge the thrust gap between the available turbine and DMRJ operation, such as Quicksat, TriJet and TRRE.

Table 1
Flight demonstration tests of scramjet.

Date	Country	Project	Flight Mach number	Operation time	Fuel
November 1991	Russia	Kholod	5.6	/	Hydrogen
July 2002	United States/ Australian	HyShot	7.6	short duration	Hydrogen
March 2004	United States	X-43A	6.8	10s	Hydrogen
November 2004	United States	X-43A	9.86	10s	Hydrogen
May 2010	United States	X-51A	5.0	140s	Hydrocarbon

In summary, as far as a hypersonic vehicle powered by an air-breathing propulsion system is concerned, the key is the breakthrough of scramjet, and the CCE based on the scramjet is crucial to the development of the propulsion system for a wide operation range. Therefore, major countries have formulated detailed research plans for scramjet and CCE, aiming to settle a series of problems in designing the hypersonic propulsion system.

1.2. Technical challenges encountered in designing the exhaust system

For a scramjet, it consists of forebody inlet, combustor and afterbody nozzle (Fig. 5), and the nozzle is a single flowpath. However, for an over-under TBCC exhaust system, the integrated exhaust system is adopted for both low-speed turbine and high-speed cycle engine (Fig. 3). As an indispensable component of the propulsion system, it not only determines the engine operation state, but also dominates the expansion of the exhaust gas with high-temperature and high-pressure from the combustor to provide the thrust for the vehicle. Furthermore, considering the design requirement of the high integration of the airframe with its propulsion system, the afterbody usually acts as the expansion ramp of the exhaust system, which results in its asymmetric structure and forms the single expansion ramp nozzle (SERN). Therefore, the exhaust system also produces the lift and pitching moment to control the balance and stability of the vehicle (Fig. 6) [63].

In the airbreathing propulsion system, the thrust is the difference

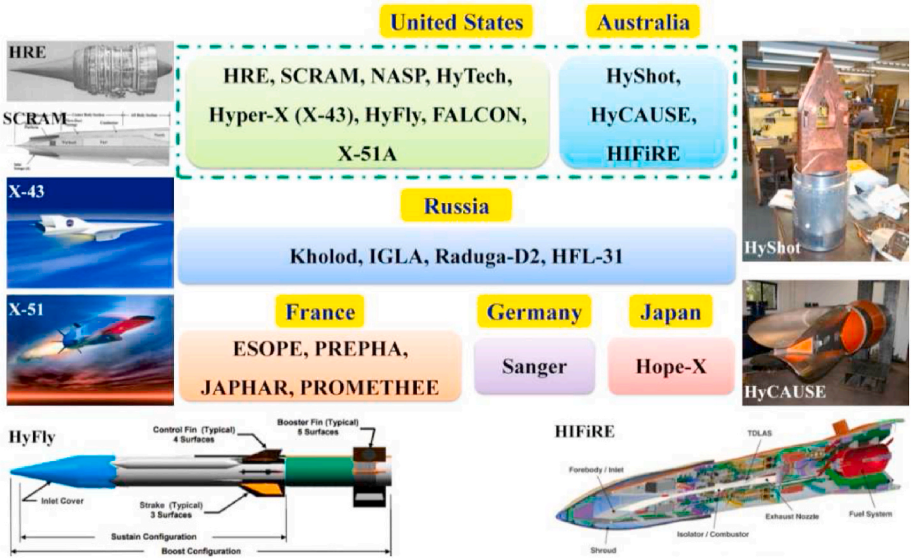


Fig. 2. Summary of scramjet research projects in major aerospace power.

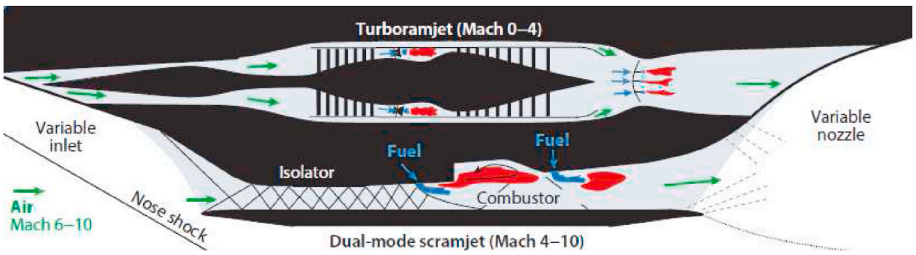


Fig. 3. Sketch of TBCC concept [42].

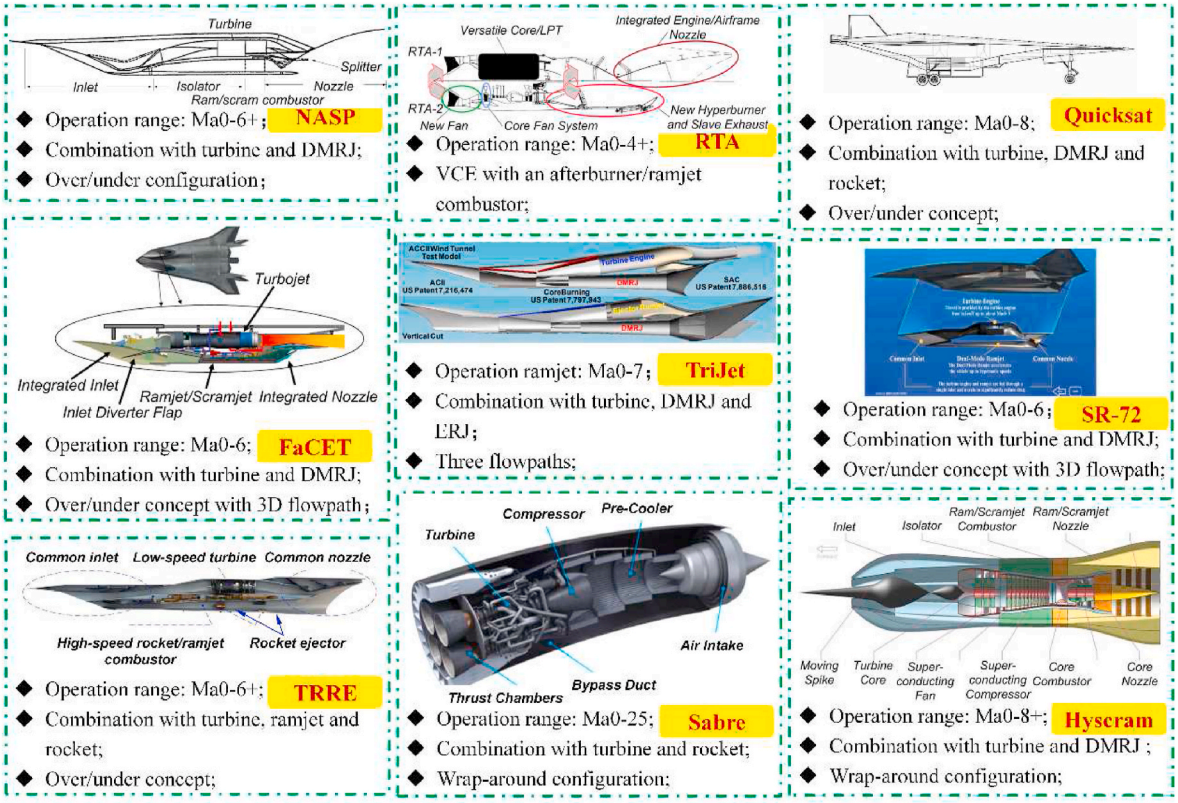


Fig. 4. Summary of typical TBCC research projects.

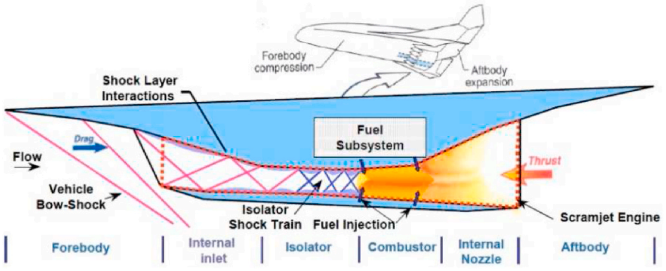


Fig. 5. Representative scramjet engine [63].

between the surplus momentums of the airflow at the exhaust system exit and that of the captured inflow at the intake entrance, and the magnitudes of the two momentums are identical. Therefore, the exhaust system is the main component to produce the net thrust. In particular, the exhaust system can provide 70% of the net thrust of the overall propulsion system at the flight Mach number beyond 6. Besides, a 1% decrease in the thrust coefficient of the exhaust system will result in a reduction of 4% in the installation net thrust of the propulsion system [64,65]. In general, the design technique level of the exhaust system directly determines the performance of the whole propulsion system.

The ability to operate at a wide range is one of the main characteristics of a hypersonic vehicle powered by the TBCC propulsion system (Fig. 7), and the flight Mach number has been extended from the subsonic, transonic, supersonic, and the hypersonic. Therefore, the propulsion system experiences the working process with various engines: the separate operation of the turbine at low flight Mach number, the mode transition from turbine to ramjet, the relay process from ramjet to scramjet, and the separate operation of the scramjet at hypersonic cruise state. As one of the key components of the propulsion system, the exhaust system also undergoes complex and changeable operation states, including:

(1) In the takeoff and transonic stage, the nozzle pressure ratio (NPR, i.e., the ratio of total pressure at nozzle entry to static ambient pressure) is low, and the highly overexpanded flow occurs in the exhaust system, resulting in the complex unsteady separation phenomenon and rapid decrease in the aerodynamic performance. Thereby, it is very important to improve the performance of the exhaust system at low flight Mach number.

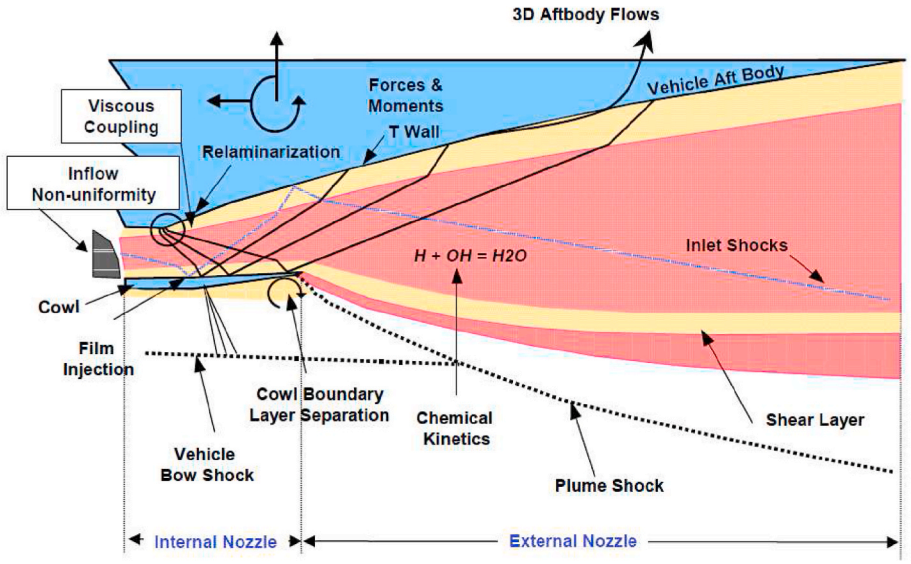


Fig. 6. Summary of important scramjet nozzle physics [63].

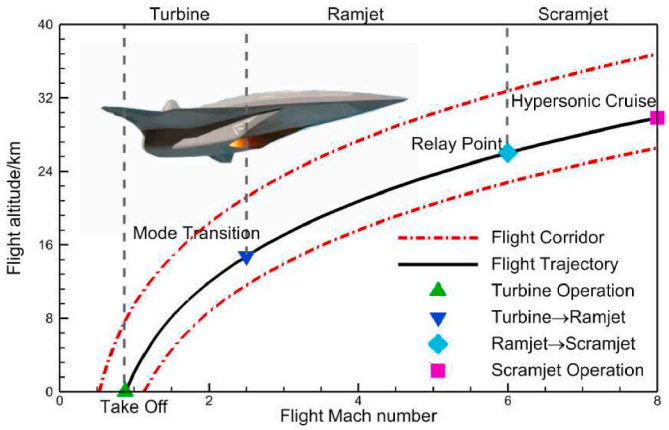


Fig. 7. Flight envelope of typical hypersonic vehicle.

(2) During the mode transition, the exhaust system is at the parallel operation of LSF and HSF. The flowfields and performances of the LSF and HSF are affected by each other, and the thrust, lift and pitching moment of the exhaust system are varied greatly. Consequently, it is difficult to satisfy the design requirement of smooth performance variation in the entire mode transition process.

(3) At the relay point, the flow in the exhaust system is changed from the subsonic at the entrance to the supersonic in the full flowfield, and the complicated inflow with shock and expansion waves is emerging at the entrance, which causes the obviously nonuniform characteristics of the inflow (Fig. 6).

(4) At the cruise condition, the exhaust system is working at the design point, the underexpanded flow and chemical nonequilibrium effect result in the complexity of the performance estimation [1] (Fig. 6).

In a word, the design of the TBCC exhaust system should not only consider the performance at the cruise condition, but also settle the problems encountered in the entire flight envelope to improve the performance at the off-design operations. Moreover, it also needs to consider the compatibility with the combustor exit, geometric constraints, weight saving and variable geometry, which causes the study to become one of the key technologies to be resolved in TBCC propulsion system research. Although major aerospace powers have formulated detailed researches plans for scramjet and TBCC, most of the research focus on supersonic combustion, inlet design and flow mechanism, and the study of the exhaust system is relatively lagging. Therefore, researches on the design and relevant issues of the exhaust system for scramjet and TBCC are of great significance.

Due to the wide operation of the TBCC exhaust system, the NPR is varied with the flight Mach number. Considering the optimum efficiency of the propulsion system in the overall flight envelope, the cruise condition generally acts as the design point of the exhaust system. At this moment, the propulsion system is at the separate operation of scramjet, while in other states, the exhaust system is operating at the off-design conditions. Consequently, the scramjet nozzle is the basis and core for designing the TBCC exhaust system. The following will be divided into two parts to review the research state of the exhaust systems: (1) the state of the research on the scramjet nozzle; (2) the state of the research on the TBCC exhaust system.

2. Performance parameters definition for the exhaust system

2.1. Thrust coefficient based on exit surplus momentum

Fig. 8 (a) and (b) show the nozzle configurations in different applications, and the nozzle with physical throat is applied in the ramjet or turbine, while the nozzle without physical throat is applied in the scramjet. Given that the exhaust system acts as the main component for producing the thrust, the axial thrust coefficient is always adopted to quantify the performance. As stated above, in the airbreathing propulsion system, the engine thrust is the difference between the surplus momentums at the nozzle exit and inlet capture plane. Hence, in terms of the overall engine performance, the exhaust thrust can be defined as the x-direction component of the integral sum of the outflow momentum and gauge pressure force acting on the exit, namely,

$$F_x = \int_C^D \rho V_x \left(\vec{V} \bullet \vec{n}\right) dA + \int_C^D (p_e - p_a) dA sin\varphi \tag{1}$$

where φ and p_e represent the inclination angle and absolute pressure of the exit plane, respectively, and p_a is the ambient pressure.

The ideal thrust F_s that is employed for the solution of the thrust coefficient is calculated as

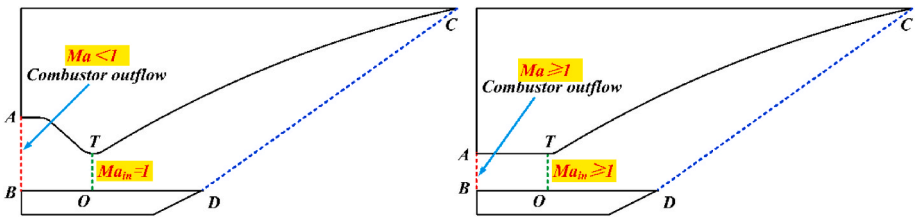


Fig. 8. Nozzle configurations in different application scenarios: (a) Nozzle with physical throat for ramjet; (b) Nozzle without physical throat for scramjet.

$$F_s = \dot{m}\sqrt{\frac{2\gamma}{\gamma-1}RT_t\left[1-\left(\frac{p_a}{p_t}\right)^{\frac{\gamma-1}{\gamma}}\right]} \tag{2}$$

where $\dot{m}$ is the mass flow rate through the nozzle. Besides, T_t and p_t are the temperature and pressure at stagnation state, respectively. Hence, the thrust coefficient is computed as

$$C_{fx,1} = \frac{F_x}{F_s} \tag{3}$$

This thrust coefficient can reflect the overall engine performance, and therefore, it is usually employed to quantify the exhaust system performance within the flight range.

2.2. Thrust coefficient based on net thrust

On the other hand, according to the momentum theorem, the outflow surplus momentum contains the surplus momentum at the divergence entry and the effect of the expansion flow along the divergent section, that is

$$F_x = I_{in} + R_x \tag{4}$$

$$I_{in} = \int_O^T \rho V_x\left(\vec{V}\bullet\vec{n}\right)dA + \int_o^t (p_{in}-p_a)dA\sin\varphi \tag{5}$$

where R_x is the x-direction component of the net thrust F, and the net thrust F is the integral force of the relative pressure and viscous shear stress over the internal wall of the divergent section. In addition, I_{in} is the surplus momentum at the divergent section entry.

In Eq. (4), the former depends on the contraction section (the nozzle with physical throat in Fig. 8 (a)) or the supersonic outflow of the combustor (the nozzle without physical throat in Fig. 8 (b)), while the latter relies on the expansion flow along the divergent section. In most thrust nozzle design methods, the divergent section design is the only objective, while the convergent section is relatively simple, especially the design of the nozzle without physical throat (Fig. 8 (b)). Moreover, the sonic or supersonic inflow at the divergent section entry causes the momentum magnitude at the divergent section entry to be equal to that at the exit, and thus, the net thrust is just a small amount.

Fig. 9 displays the effect of the expansion flow along the divergent section on the exit surplus momentum, and the ordinate is ξ defined as

$$\xi = \frac{R_{x,s}}{F_s}\times 100\% \tag{6}$$

where the variables of $R_{x,s}$ and F_s are the ideal net thrust and isentropic thrust, respectively, and the calculation of $R_{x,s}$ will be introduced in next section. The sonic inflow corresponds to the nozzle with physical throat (Fig. 8 (a)), while the supersonic inflow corresponds to the operation of scramjet or rocket (Fig. 8 (b)). As the inflow Mach number increases, the inflow surplus momentum is increased, and the proportion of the net thrust is decreased. However, with an increase of the NPR, the proportion of the net thrust is increased. It is worth mentioning that the net thrust proportion is less than 30%, although the NPR is up to 500.

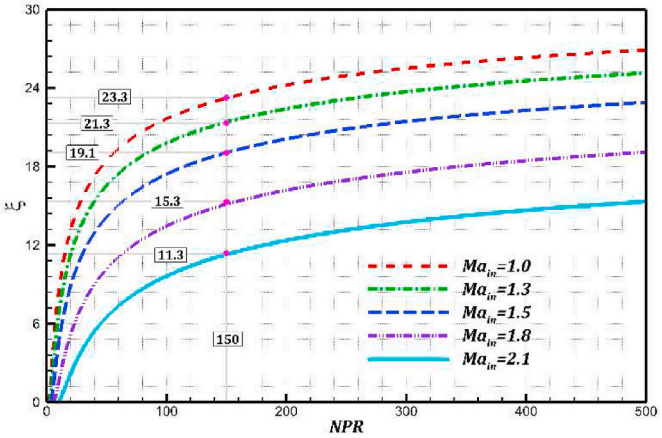


Fig. 9. Effect of the expansion flow along the divergence on the nozzle exit momentum.

Therefore, in terms of nozzle only, the thrust coefficient based on the exit surplus momentum cannot reflect the design level of the nozzle divergent section well, and the net thrust coefficient that only considers the expansion flow along the divergent section can reflect the nozzle design level more clearly.

According to Eq. (4) and Eq. (5), the axial thrust coefficient based on the net thrust can be defined as

$$C_{fx,2} = \frac{R_x}{F_s - I_{in}} \tag{7}$$

Note that the variation range of $C_{fx,1}$ is 0–1, while the value of $C_{fx,2}$ is less than 1, and it can be negative at the highly over-expanded state.

Furthermore, when the propulsion system is at the parallel operation, the axial thrust coefficient based on the exit surplus momentum can be defined as

$$C_{fx,1} = \frac{F_{x,LSF}+F_{x,HSF}}{F_{s,LSF}+F_{s,HSF}} \tag{8}$$

where $F_{x,LSF}$ and $F_{s,LSF}$ are the exit surplus momentum and ideal thrust of the LSF nozzle, respectively, and the variables $F_{x,HSF}$ and $F_{s,HSF}$ represent the exit surplus momentum and ideal thrust of the HSF nozzle, respectively.

In the same way, the axial thrust coefficient based on the net thrust is calculated as

$$C_{fx,2} = \frac{R_{x,LSF}+R_{x,HSF}}{F_{s,LSF}-I_{in,LSF}+F_{s,HSF}-I_{in,HSF}} \tag{9}$$

where $R_{x,LSF}$ and $R_{x,HSF}$ are the x-direction net thrusts of the LSF and HSF nozzles, respectively, and $I_{in,LSF}$ and $I_{in,HSF}$ are the entry surplus momentums at the divergence entrances of the LSF and HSF nozzles, respectively.

The exhaust system also generates lift and pitching moment for the

vehicle, and lift L is the y-direction component of the net thrust F from integrating the relative pressure and viscid shear stress on the internal wall. The reference point for calculating the pitching moment is the center of the HSF nozzle entry.

2.3. Theoretical prediction of the scramjet nozzle thrust

The combustion chamber of the scramjet is approximately equal in cross section, and the flow through the combustor remains supersonic. Then, the high-temperature and high-pressure gas expands through the nozzle with a divergent channel. Therefore, the nozzle is the main component to generate the net thrust in the scramjet, and the overall engine thrust can be predicted by the nozzle thrust. As described in Section 2.2, the nozzle net thrust R_x is the integral force of the relative pressure and viscous shear stress on the internal wall, and according to Eq. (4), it can be calculated as the difference between the surplus momentums at the nozzle exit and entry, namely,

$$R_x = [\dot{m}V_e + (p_e - p_a)A_e] - [\dot{m}V_{in} + (p_{in} - p_a)A_{in}] \tag{10}$$

Besides,

$$F_s - I_{in} = \dot{m}\sqrt{\frac{2\gamma}{\gamma-1}RT_t\left[1-\left(\frac{p_a}{p_t}\right)^{\frac{\gamma-1}{\gamma}}\right]} - [\dot{m}V_{in} + (p_{in} - p_a)A_{in}] \tag{11}$$

Then, Eqs. (10) and (11) can be calculated as

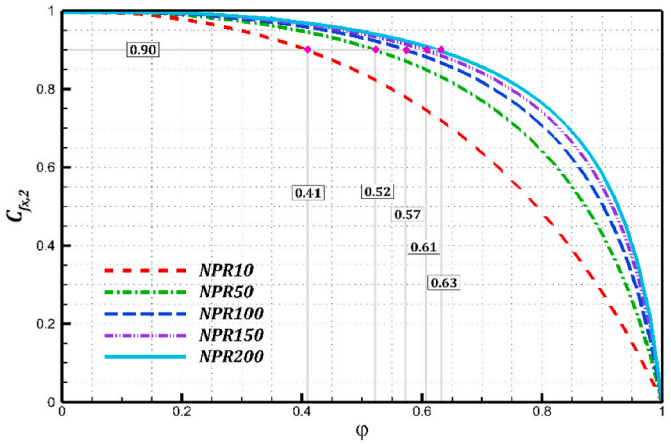


Fig. 10. Effect of the under-expanded factor on the nozzle net thrust coefficient.

coefficient can maintain 0.90 at the under-expanded factor of 0.61. This conclusion can provide a basic for the geometric constraints of the afterbody nozzle.

However, prior to designing a scramjet nozzle, in addition to the above parameters, another important parameter, namely, nozzle length, should be also taken into account for integrated design. Consequently,

$$\frac{R_x}{p_t A_{in}} = \gamma\left\{[(1-\varphi)AR_s + 1]Ma_e^2\left(1+\frac{\gamma-1}{2}Ma_e^2\right)^{-\frac{\gamma}{\gamma-1}} - Ma_{in}^2\left(1+\frac{\gamma-1}{2}Ma_{in}^2\right)^{-\frac{\gamma}{\gamma-1}}\right\} + \left[\left(1+\frac{\gamma-1}{2}Ma_e^2\right)^{-\frac{\gamma}{\gamma-1}} - \frac{1}{NPR}\right][(1-\varphi)AR_s + 1]$$
$$- \left[\left(1+\frac{\gamma-1}{2}Ma_{in}^2\right)^{-\frac{\gamma}{\gamma-1}} - \frac{1}{NPR}\right] \tag{12}$$

$$\frac{F_s - I_{in}}{p_t A_{in}} = \gamma Ma_{in}\left(1+\frac{\gamma-1}{2}Ma_{in}^2\right)^{-\frac{\gamma+1}{2(\gamma-1)}}\sqrt{\frac{2}{\gamma-1}\left[1-\left(\frac{1}{NPR}\right)^{\frac{\gamma-1}{\gamma}}\right]} - \gamma Ma_{in}^2\left(1+\frac{\gamma-1}{2}Ma_{in}^2\right)^{-\frac{\gamma}{\gamma-1}} - \left(1+\frac{\gamma-1}{2}Ma_{in}^2\right)^{-\frac{\gamma}{\gamma-1}} + \frac{1}{NPR} \tag{13}$$

where NPR and AR_s are nozzle pressure ratio ($NPR = p_t/p_a$) and ideal area ratio ($AR_s = A_{e,s}/A_{in}$), respectively. Due to the high integration of the afterbody with the exhaust system, the strong geometric constraints are usually imposed on the nozzle design, which leads to the condition that the nozzle always operates at the underexpanded condition at the design point. In order to reflect the effect of the under-expansion, the under-expanded factor φ is introduced into Eq. (12), and it is defined as

$$\varphi = 1 - \frac{AR - 1}{AR_s - 1} \tag{14}$$

The value range of φ is 0–1, and the nozzle operates at the ideal-expanded condition with $\varphi = 0$, while the nozzle operates at the under-expanded condition with $\varphi > 0$. According to Eqs. (12) and (13), it is easy to know that the nozzle net thrust coefficient is the function of φ, Ma_{in} and NPR, namely,

$$C_{fx,2} = f(\varphi, Ma_{in}, NPR) \tag{15}$$

Therefore, the thrust coefficient can be predicted with the known parameters of φ, Ma_{in} and NPR roughly.

Fig. 10 shows the effect of the underexpanded factor on the nozzle net thrust coefficient. With the increase of the underexpanded factor, the under-expanded loss is serious, and the net thrust coefficient is decreased gradually. The tendency is flatly at first and then steeply. For example, when the nozzle operates at the NPR of 150, the net thrust

the above prediction method ignores the effect of nozzle length, and the efficient expansion flow is assumed to be involved in the nozzle. Besides, the viscosity loss and unique flow with the entry non-uniform parameters, chemical nonequilibrium also cannot be considered in Eq. (15). Many factors lead to the situation that the net thrust coefficient in actual state is less than the prediction. Considering the above factors, how to realize the efficient expansion flow in nozzle is the key to the design.

Furthermore, the exit surplus momentum I_e and ideal thrust F_s can be calculated as

$$\frac{I_e}{p_t A_{in}} = \gamma Ma_e^2\left(1+\frac{\gamma-1}{2}Ma_e^2\right)^{-\frac{\gamma}{\gamma-1}}[1 + (1-\varphi)(AR_s - 1)]$$
$$+ \left[\left(1+\frac{\gamma-1}{2}Ma_e^2\right)^{-\frac{\gamma}{\gamma-1}} - \frac{1}{NPR}\right][1 + (1-\varphi)(AR_s - 1)] \tag{16}$$

$$\frac{F_s}{p_t A_{in}} = \gamma Ma_{in}\left(1+\frac{\gamma-1}{2}Ma_{in}^2\right)^{-\frac{\gamma+1}{2(\gamma-1)}}\sqrt{\frac{2}{\gamma-1}\left[1-\left(\frac{1}{NPR}\right)^{\frac{\gamma-1}{\gamma}}\right]} \tag{17}$$

Therefore, the thrust coefficient based on exit surplus momentum is also the function of φ, Ma_{in} and NPR, namely,

$$C_{fx,1} = f(\varphi, Ma_{in}, NPR) \tag{18}$$

According to Eqs. (15) and (18), the relationship between $C_{fx,1}$ and $C_{fx,2}$ can be obtained, as shown in Fig. 11. It is obvious that the relation

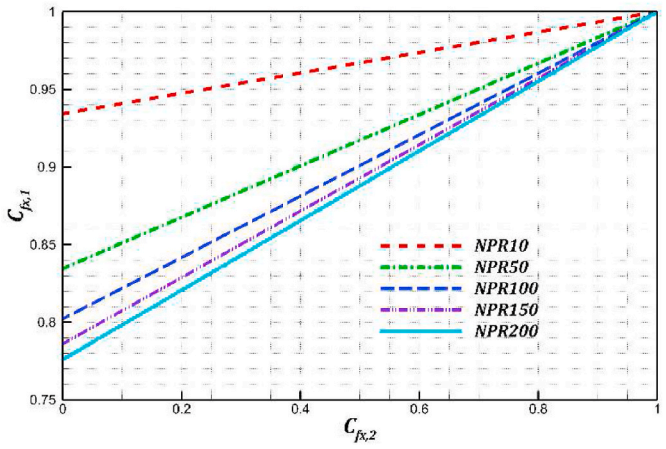


Fig. 11. Relationship between $C_{fx,1}$ and $C_{fx,2}$.

of $C_{fx,1}$ and $C_{fx,2}$ is approximately linear, namely,

$$C_{fx,1} = C_{fx,2,0} + kC_{fx,2} \tag{19}$$

where $C_{fx,2,0}$ represents the contribution of the surplus momentum at the divergent section entry to that at the nozzle exit, and it depends on the outflow of the combustor or the contraction section. The slope k is related to *NPR*, and the larger *NPR* corresponds to the larger slope. Hence, the surplus momentum at the nozzle exit can be obtained by the known net thrust coefficient $C_{fx,2}$ from Eq. (19).

3. State of the design and flow research on the scramjet nozzle

As one of the indispensable components of scramjet, the SERN performance determines the operational efficiency of the propulsion system directly. Since the propulsion system is highly integrated with the airframe, the SERN design should consider strong geometric constraints, compatibility with combustor exit and weight saving, which is in conflict with the high expansion ratio requirement resulting from the high operational NPR. Therefore, the ideal expansion cannot be realized in the nozzle, and it is difficult to organize the efficient expansion flow and achieve the goal of optimal performance, greatly increasing the difficulty of the SERN design. Besides, the unique flow phenomena of the scramjet nozzle, such as the nonuniform inflow at the entry, chemical nonequilibrium flow and coupling of the internal flow with the external freestream, are also the problem to be taken into account in the study of the SERN.

Table 2
Summary of SERN in major scramjet research projects.

Projects	Country	Flight Mach number	Configuration
SCRAM [14]	United States	Mach 5–7.3	3D SERN with shape transition
NASP [15]	United States	Mach 6-25	2D SERN
HyTech [17]	United States	Mach 4-8	2D SERN
JAPHAR [37]	France	Mach 4-8	SERN with lateral expansion
HyShot [25]	United States/ Australia	Mach 7.6	2D SERN
HyCAUSE [27]	United States/ Australia	Mach 10	3D SERN with shape transition
HIFiRE [29]	United States/ Australia	Mach 8	2D SERN
FALCON [22]	United States	Mach 6	3D SERN with shape transition
X-43A [19]	United States	Mach 6.8 and 9.86	2D SERN
X-51A [24]	United States	Mach 5	2D SERN

3.1. Research progress on SERN design

Considering the different application scenarios and requirements, the various SERN configurations of the two-dimensional (2D) planar SERN, SERN with lateral expansion and three-dimensional (3D) SERN with shape transition have been developed rapidly. Table 2 summarizes the SERN configurations in major scramjet research projects. The 2D SERN with simple structure is convenient to design and analyze, and thus, employed in many projects, including NASP [15], HyTech [17], HyShot [25], HIFiRE [29], X-43A [19] and X-51A [24]. Besides, the 2D configuration facilitates variable geometry for a wide operation range. However, the single expansion direction will lead to the low expansion efficiency. Therefore, the expansion along the sidewall is introduced in the 2D configuration to form the SERN with lateral expansion, just like the nozzle configuration in JAPHAR [37]. The 3D SERN with shape transition is conducive to be compatible with the combustor exit, especially the round or elliptical combustor, such as the configurations in projects of SCRAM [14], HyCAUSE [27] and FALCON [22]. Overall, the development of the SERNs with various configurations aims to settle the SERN design under complex constraints.

3.1.1. The 2D SERN design method

The entrance of the 2D SERN is rectangular or nearly rectangular, and the clear expansion of the exhaust gas with high-temperature and high-pressure is conducted along the ramp and cowl. The expansion flow can be calculated by the method of characteristics (MOC) directly, and it is convenient for analysis and design. Moreover, as stated above, it is easy to realize the variable geometry design for a wide range operation. Therefore, it has become one of the hot topics for the scramjet nozzle.

Due to the simple structure of the 2D SERN, the specific length and exit height always act as the geometric constraints for the integration design. At present, the design methods of SERN under geometric constraints are mainly reflected in four categories, including (1) geometric fusion method based on ideal nozzle, (2) optimization design based on computational fluid dynamics (CFD), (3) inverse design and (4) direct design method on the basic of maximum thrust theory. In the first method, the compressing or truncating perfect nozzle is applied to satisfy the geometric constraints, which usually cannot realize the optimal performance under geometric constraints. Optimization over the flight envelope can be applied in obtaining a SERN with high performance, but it is restrained by large number of calculations due to the use of full CFD simulation. Based on the given optimal parameter distribution, such as the flow parameter distribution along the ramp or at the nozzle exit, the third method utilizes the MOC to achieve the inverse design of the SERN. However, the specific parameter distribution should be obtained through optimization. The fourth method combines with the maximum thrust theory to design SERNs under geometric constraints directly, and it realizes the compatibility design of geometric constraints and performance optimization.

3.1.1.1. Geometric fusion method based on ideal nozzle. In the traditional design methods, the uniform and parallel outflow with the desired Mach number maximizes the nozzle thrust by applying the MOC [66], namely, the ideal nozzle (Fig. 12 (a)). However, the nozzle length is significantly beyond the acceptable range for the propulsion system, which is detrimental to the integration design and increases the weight. To decrease the ideal nozzle length, some approaches are developed, such as applying a minimum length nozzle [67,68], or increasing the expansion along the cowl by introducing an asymmetric factor [69,70]. The minimum length nozzle design is mainly reflected in increasing the expansion rate of the initial expansion section, which can reduce the length by 50%, compared with the traditional nozzle. The design method of SERN with asymmetric factor increases the expansion degree of the lower cowl, and a part of the gas expands through the cowl, thus reducing the nozzle length (Fig. 12 (b)). The asymmetric factor is defined as the ratio

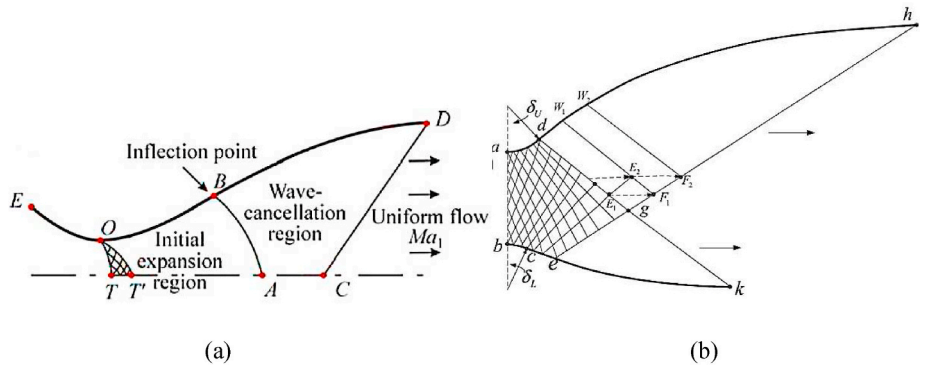


Fig. 12. Ideal nozzle design [69,71]: (a) Traditional method; (b) Asymmetric factor in designing SERN.

of the initial expansion angle of the cowl to that of the ramp, namely, $G = \delta_L/\delta_U$, and the nozzle length will gradually decrease as the increase of the asymmetric factor. Even though these approaches can achieve the desired results to a certain extent, the ideal nozzle length is still unacceptable in the highly integrated hypersonic vehicle, resulting from the high operational NPR.

For further study, to satisfy the geometric constraints of the afterbody nozzle, the methods of compressing and truncating perfect nozzles are proposed. In other words, the geometric constraints are satisfied by sacrificing performance.

(1) Compression method

The linearly compressed perfect nozzle was investigated initially, and Hoffman [72] indicated that an optimally compressed perfect nozzle was a good propulsive nozzle. Due to the high pressure acting on the initial expansion section and large expansion angle at the throat downstream, the axial component of the internal force integral on this expansion section is large. Therefore, the initial expansion section is the main part on producing the axial thrust. Based on the above analysis, the nonlinear compression method was introduced [73], in which the initial expansion section was not compressed to ensure the thrust performance, while the compression of the trailing edge part with less contribution on thrust was increased to achieve the length constraint. Compared with the result of the linear compression method, the thrust performance of the nonlinearly compressed nozzle was greatly improved. In addition, the reduction of the thrust performance was very small in a wide working range, compared with the ideal nozzle.

(2) Truncation method

The design process of this method can be briefly generalized as below: The ideal SERN contour at the design point is determined by MOC initially, and the asymmetric factor G is employed to control the internal expansion. Then the final configuration of the SERN is generated by truncating the ramp and cowl of the ideal nozzle in an identical rate to meet the geometric constraints (Fig. 13). Previous study denoted that the performance of the SERN truncated by 70% was decreased by less than 1% at Mach 6.5 [74]. However, it deteriorated rapidly with decrease of NPR. Furthermore, considering the relationship between thrust and friction drag on the internal wall, a novel truncation method based on the balance of thrust and friction was proposed [75], and the thrust coefficient was increased by 1.5% with the decrease of 25% and 34% in the length and weight, respectively. Due to the relative simplicity of the truncation method, it is widely adopted in designing the SERN.

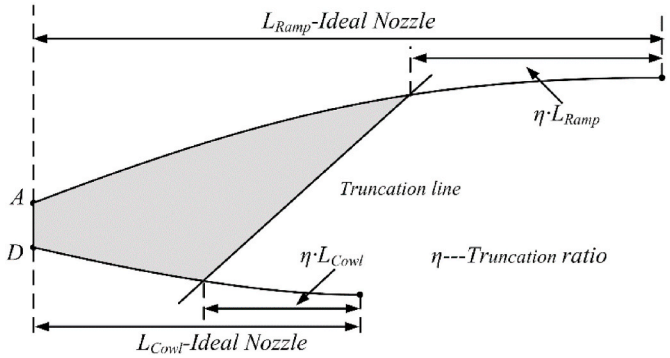


Fig. 13. Sketch of truncation method for SERN [69].

3.1.1.2. Optimization design method. With the rapid development of the computer performance, the SERN obtained through a large number of calculation and optimization has become a choice for the designers. Damira et al. [76] and Li et al. [77] studied the effects of the geometric parameters such as divergent angle, cowl length and nozzle length on the SERN performance, which pointed out the direction for the optimization design. Gao et al. [78] optimized a SERN contour based on the B-spline curve and sequential quadratic programming, while Refs. [79–81] conducted the multi-objective optimization design based on the cubic curve, and the correctness of the design results were proved by numerical simulation. Moreover, aiming to improve the problem of the large pitching moment difference between cold and hot states, the SERN cowl was optimized by combining the multi-objective optimization algorithm with CFD. Fig. 14 displays the Mach number contour and experimental schlieren of the optimized SERN.

In terms of the optimization algorithm, Huang et al. [83] employed the data mining technology to analyze the influence of the geometric parameters on the SERN thrust and lift, and Song et al. [84] and Li et al. [85] applied the discrete adjoint method and chaotic particle swarm to optimize the SERN, respectively.

From the above studies, researchers employed different curves and optimization algorithms to optimize the SERN from single-to multi-objective, which can solve the SERN design problem under geometric constraints. However, it is undeniable that the optimization design requires a large amount of computation resources and time. Therefore, it is not applicable to the vehicle design that needs multiple iterations.

3.1.1.3. Inverse design method. Considering that the pressure distribution along the ramp determines the nozzle performance, Tang et al. [86,87] imposed the ramp pressure distribution in advance, combined with the MOC to design the SERN contour inversely, and finally obtained the optimal configuration through optimization. Besides, the inverse design method based on outflow parameters was also developed [88,89], and the exit height and aerodynamic parameters should be specified in advance. Fig. 15 shows the sketch of the SERN inverse design method based on exit parameters. However, it is worth mentioning that the former cannot control the SERN exit height, while the latter cannot

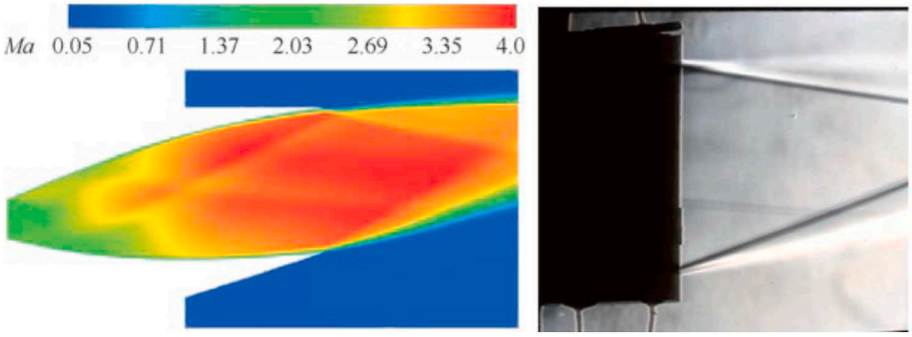


Fig. 14. Mach number contour and experimental schlieren of SERN [82].

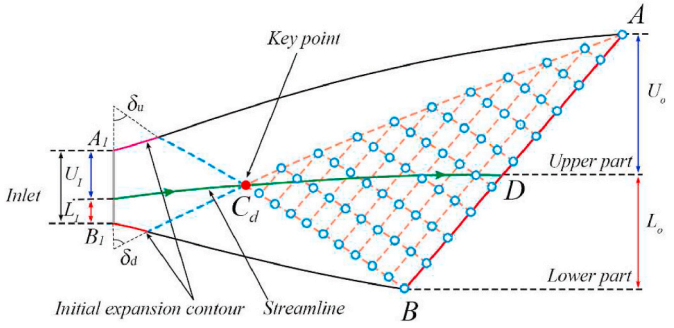


Fig. 15. Sketch of SERN inverse design method based on exit parameters [88].

constrain the length. Consequently, this method does not realize the SERN design under fully geometric constraints, and the optimal parameter distribution with meeting the physical law should be optimized.

3.1.1.4. Direct design method based on maximum thrust theory. On the basic of summarizing the previous design methods, Lv et al. [90] proposed a novel design method in line with the optimal thrust theory to determine a SERN with full geometric constraints directly. In his initial design idea, the expansion flow along the ramp introduced the optimal thrust theory, and the lower cowl was a straight contour without expansion (Fig. 16 (a)). In the field of design principle, the maximum thrust theory of the axial nozzle developed by Rao [91] was acting as a reference for obtaining the optimal thrust theory of a 2D planar nozzle. Fig. 16 (a) shows the general feature of a 2D SERN with maximum thrust. The control surface *IC* is the last left-running characteristic line of the flowfield, and the angle φ is associated with the flow angle θ and Mach angle α through the below relation

$$\varphi = \theta + \alpha \tag{20}$$

Besides, two additional conditions are generated along the control surface *IC*, they are

$$\frac{Vcos(\theta - \alpha)}{cos\ \alpha} = C_1 \tag{21}$$

$$\rho V^2\ sin^2\ \theta\ tan\ \alpha = C_2 \tag{22}$$

where C_1 and C_2 are constants.

In the design process, the SERN flowfield was split into three regions, including the regions of Mach, kernel and transition. The Mach and kernel regions could be calculated by the unit processes of the interior, wall, and symmetric points directly [92]. Then the key was combining with the maximum thrust theory to choose a suitable point *I* in the kernel region to satisfy the geometric constraints, and the detailed process is presented in Ref. [90]. Furthermore, the design method of the SERN with expansion along the cowl was also developed (Fig. 16 (b)), and the calculation of the Mach region was relatively complex. Finally, the parametric study was performed to optimize the SERN configuration (Fig. 17).

Compared with the common nozzle, the SERN designed by the proposed method increased the axial thrust coefficient, lift and pitching moment by 5.5%, 1098.2% and 20.3%, respectively, at design point. Besides, the proposed SERN produced positive lift with significant increases in axial thrust coefficient and in pitching moment at off-design points. All of the results verified the superiority of the proposed method.

For further study, the wind tunnel experiment was also conducted to examine the generality of the new method, in which the flowfield structures and performance parameters at both overexpanded and underexpanded operations were obtained (Figs. 18 and 19). The overexpanded flow resulted in the existence of a shock wave at the trailing edge of the ramp (Fig. 18 (b)), and the underexpanded flow loss might be more serious, compared with the overexpanded results (Fig. 19).

As a conclusion, the method proposed by Lv et al. [90] realizes the direct design of SERN with geometric constraints, and the SERN performance is greatly increased, by comparison with the common nozzle result. In relation to the methods of geometric fusion, optimization method and inverse design, the direct method well resolves the

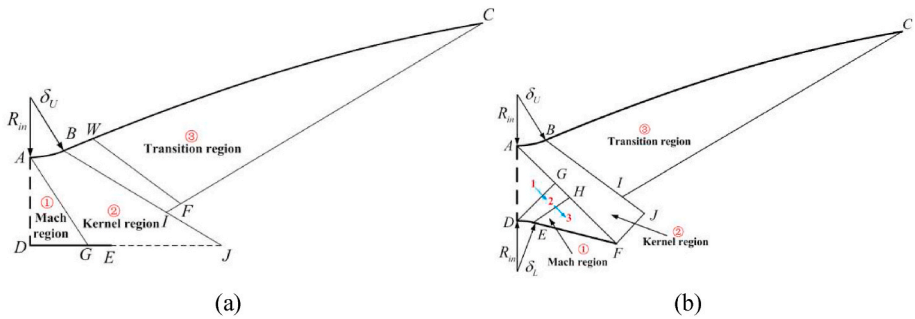


Fig. 16. Direct design method for SERN under geometric constraints: (a) SERN without expansion on the cowl; (b) SERN with expansion on the cowl.

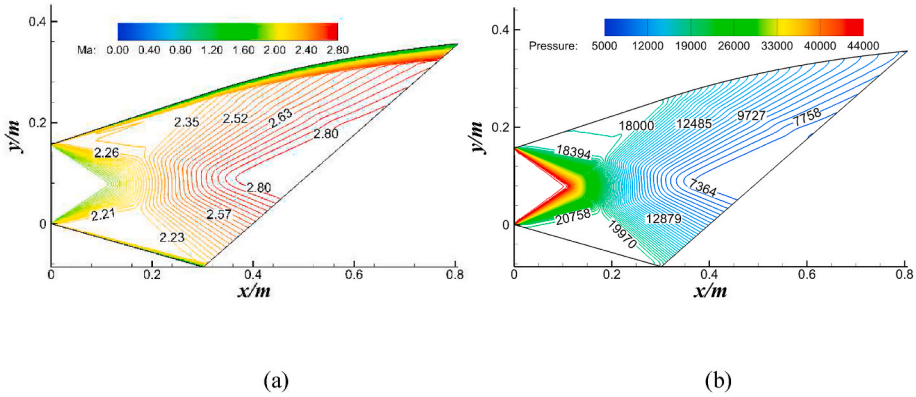


Fig. 17. Mach number and pressure contours of SERN designed by Lv et al. [90]: (a) Mach number; (b) Pressure.

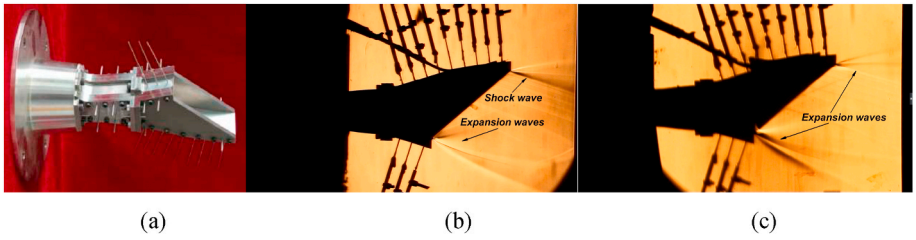


Fig. 18. SERN Flowfield structures at various operations [93]: (a) Experimental model; (b) *Ma*4.0 (NPR38.98, overexpanded operation); (c) *Ma*6.0 (NPR103.34, underexpanded operation).

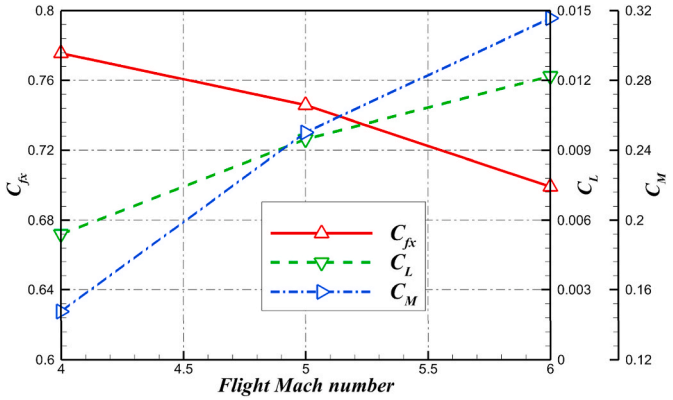


Fig. 19. SERN Performance at various operations [93].

compatibility design of the performance optimization theory and geometric constraints. However, in the Lv nozzle, the optimal thrust theory is only led into the expansion flow along the ramp, and thereby, the method has a shortcoming that the cowl with a straight wall is not an optimized contour. The further study should extend the initial method and introduce the optimal thrust theory into both expansion flows along the ramp and cowl under geometric constraints.

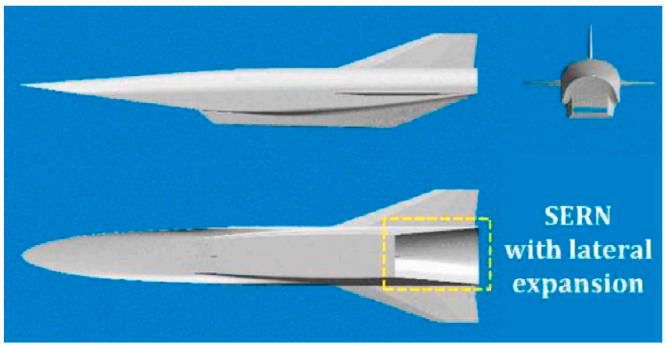


Fig. 20. SERN configuration in JAPHAR program [94].

3.1.2. SERN design method with lateral expansion

For the hypersonic propulsion system, the afterbody nozzle is composed of the ramp, cowl and sidewalls. In order to realize the integrated design with the afterbody, make full use of the afterbody space and increase the expansion efficiency in a limited space, the sidewalls can also have a certain angle to introduce the lateral expansion, such as the configuration of the exhaust system in JAPHAR program of French [94] (Fig. 20). Therefore, it can improve the thrust performance and weight saving of the afterbody nozzle.

Fig. 21 displays the SERN configuration with lateral expansion. The ramp and cowl contours are identical to those of a 2D SERN, and the sidewall is set as the function of the coordinate x, namely, $w(x)$. The flow with lateral expansion is assumed to be quasi-2D [95], and its control equations, including the conservations of mass, momentum, and energy, can be derived through the analysis of the differential control volume (Fig. 22).

The conservation of mass was calculated as

$$\rho\frac{\partial u}{\partial x}+\frac{\partial \rho}{\partial x}u+\rho\frac{\partial v}{\partial y}+\frac{\partial \rho}{\partial y}v+\delta\rho u=0 \tag{23}$$

where

$$\delta=\frac{d(lnw)}{dx}$$

The momentum equations along x and y directions were

$$\rho u\frac{\partial u}{\partial x}+\rho v\frac{\partial u}{\partial y}+\frac{\partial p}{\partial x}=0 \tag{24}$$

$$\rho u\frac{\partial v}{\partial x}+\rho v\frac{\partial v}{\partial y}+\frac{\partial p}{\partial y}=0 \tag{25}$$

The energy equation was written as

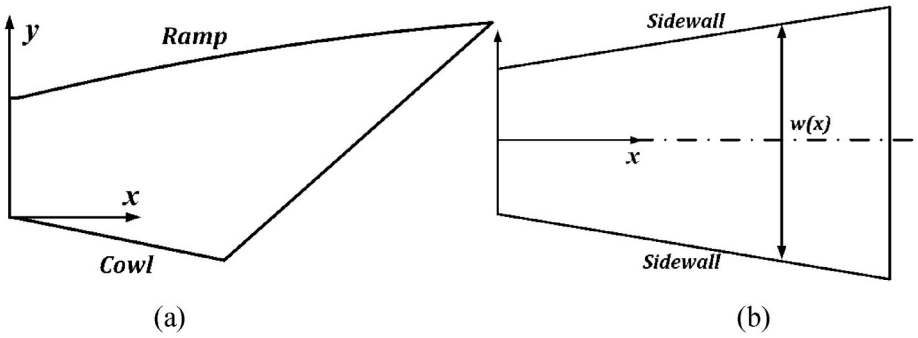


Fig. 21. Configuration of SERN with lateral expansion: (a) Ramp and Cowl; (b) Sidewalls.

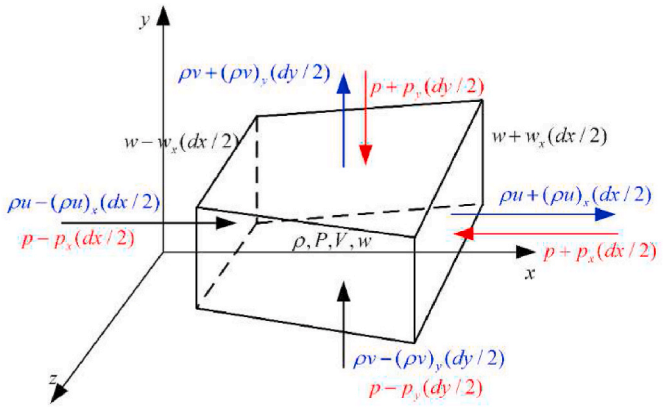


Fig. 22. Differential control volume for supersonic flow with lateral expansion [95].

$$\frac{\partial p}{\partial x}u+\frac{\partial p}{\partial y}v-a^2\left(\frac{\partial \rho}{\partial x}u+\frac{\partial \rho}{\partial y}v\right)=0 \tag{26}$$

where a is the speed of sound.

Then, the characteristic and compatibility equations can be obtained from the above flow control equations [95]. The equations for the left and right-characteristics (C_+ *and* C_-) are as follows:

$$\frac{dy}{dx}=tan(\theta\pm\alpha) \tag{27}$$

where θ and α are the angles of the flow and Mach, respectively. The compatibility equations along the two Mach lines are calculated below:

$$\frac{\sqrt{Ma^2-1}}{\rho V^2}dp_\pm\pm d\theta_\pm+\delta\left[\frac{cos\,\theta dx_\pm}{Macos(\theta\pm\alpha)}\right]=0 \tag{28}$$

The equation for the streamline C_0 is shown below:

$$\frac{dy}{dx}=tan\,\theta \tag{29}$$

and the compatibility equations along it are displayed as follows:

$$\rho VdV+dp=0 \tag{30}$$

$$dp-a^2d\rho=0 \tag{31}$$

Based on the above characteristic and compatibility equations for quasi-2D supersonic flow, the design method of the ideal nozzle with lateral expansion was developed, and it was compared with the optimization method. The nozzle configurations obtained by the two methods basically coincided, and the thrust and lift deviations were less than 0.60% and 8.07%, respectively. The identical conclusion was also reflected in the SERN pressure contours (Fig. 23). Moreover, the design method could reduce the time consuming from 30.5 h to 2.0 h. The results verified the reliability and superiority of the quasi-2D MOC.

Considering the lateral expansion contour should be specified in advance, it is necessary to study the effect, such as the influences of the straight and curved sidewalls (Fig. 24). The SERN performance was optimal with the straight sidewall of 6°, and the larger in the initial expansion angle of the convex curve was used to improve the thrust [96].

Even though the above method has realized the design of the ideal nozzle with lateral expansion, the nozzle is too long to integrate with the afterbody. Consequently, Lv et al. [97] proposed a novel method for SERN considering lateral expansion and geometric constraints. This method consulted the design method of 2D SERN under geometric constraints, and the maximum thrust theory for SERN with lateral expansion is derived initially. Just like the optimal thrust theory of the 2D SERN, the control surface *IC* (Fig. 16 (a)) is the last left-running characteristic line of the flowfield, and along it, the angle φ is the sum of the flow angle θ and Mach angle α, namely,

$$\varphi=\theta+\alpha \tag{32}$$

At the same time, two additional conditions obtained along the control surface *IC* are written as

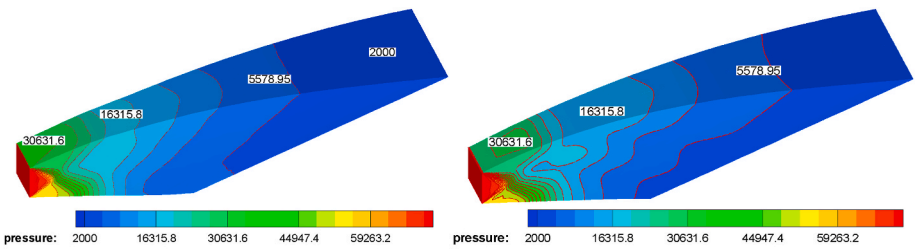


Fig. 23. Pressure contours of SERNs with lateral expansion [95]: (a) MOC; (b) Optimization.

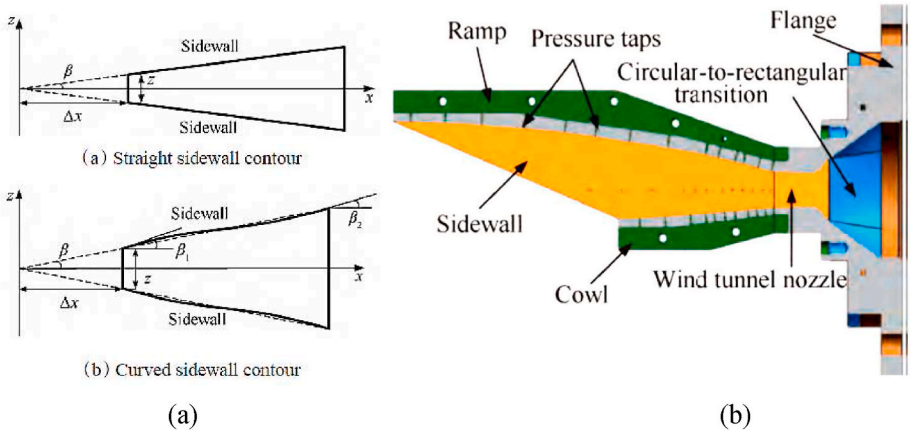


Fig. 24. Research on the effect of the lateral expansion contour [96]: (a) Lateral expansion contour; (b) Experimental model.

$$\frac{Vcos(\theta-\alpha)}{cos\ \alpha}=C_1 \tag{33}$$

$$w\rho V^2\ sin^2\ \theta\ tan\ \alpha=C_2 \tag{34}$$

where C_1 and C_2 are constants. Note that the effect of the lateral expansion contour $w(x)$ is led into the maximum thrust theory by Eq. (34).

The design process of the SERN considering lateral expansion and geometric constraints is similar to that of the 2D SERN under geometric constraints, and the key is also combining with the maximum thrust theory to choose a suitable point I in the kernel region to satisfy the geometric constraints. With the completion of the parametric studies of the lateral expansion contour, cowl length and initial expansion angle, the final configuration of the SERN is determined (Fig. 25). The pressure is decreased smoothly with the expansion, without compression or shock waves involved in the flowfield. Compared with the common nozzle, the axial thrust coefficient, lift and pitching moment of the proposed SERN are increased by 6,86%, 203.15% and 642.86%, respectively, at the design point. Besides, the lateral expansion can contribute 22.46% of the entire axial thrust, but has no contribution on producing the lift and pitching moment (Fig. 26).

In the design of the SERN with lateral expansion, previous studies realized the efficient expansion of the flow from the flow theory, but it is undeniable that the 3D flow was assumed to be quasi-2D. Fortunately, from the study of Niu et al. [95], the small difference between the results of the quasi-2D MOC and optimization indicated that the assumption was reasonable, and the novel method proposed by Lv et al. [97] realized the integrated design of the SERN with lateral expansion. For further study, this method should also be extended, in which the maximum theory will be introduced into the expansion flows along both ramp and cowl to optimize the ramp and cowl.

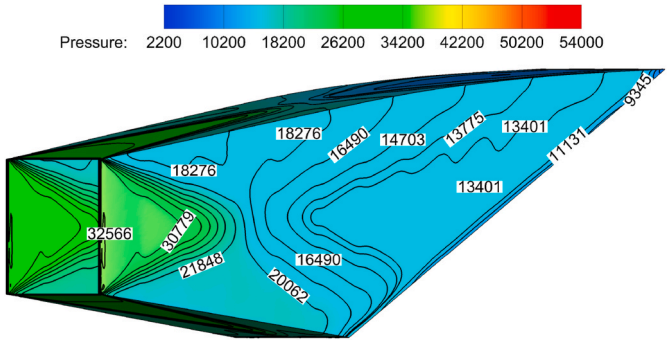


Fig. 25. Pressure contour of SERN with lateral expansion at Mach 6.0 [97].

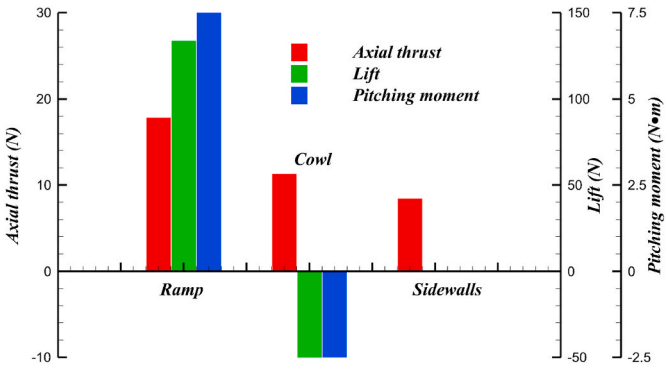


Fig. 26. Contributions of the expansion flows along the ramp, cowl and sidewalls on axial thrust, lift and pitching moment in SERN with lateral expansion [97].

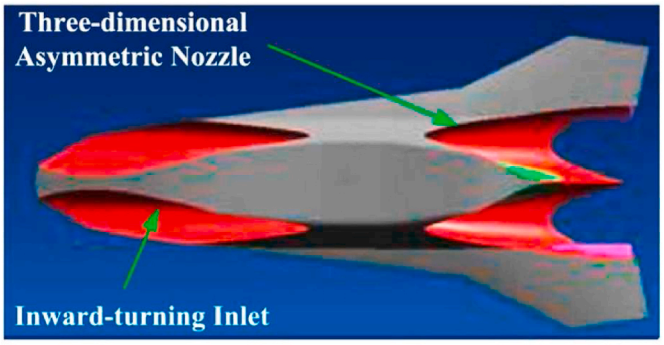


Fig. 27. Propulsion system with inward-turning inlet and 3D SERN [98].

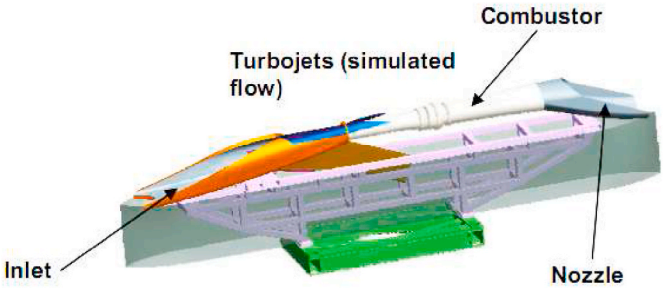


Fig. 28. Scramjet configuration in the FaCET program [22].

3.1.3. 3D SERN design method with shape transition

In recent years, the propulsion system combining with an inward-turning inlet, circular combustor and 3D SERN (Fig. 27) has become a research hotspot for hypersonic vehicle, such as the scramjet configuration in the FaCET program [22] (Fig. 28). For a scramjet with 3D flowpath, it is easier to realize the integration with the airframe, and the flowpath has low friction drag and high structural strength, which can effectively reduce the weight. Compared with the 2D SERN, the 3D SERN can better integrate with the round or elliptical combustor, and has higher expansion efficiency resulting from the lateral expansion. Besides, for the hypersonic vehicle powered by the modular scramjet, the section shapes of the inlet capture and nozzle exit should be rectangular so that the modules can be installed side-by-side. Consequently, the 3D SERN that can transition from the round entry to a rectangular exit is very appealing, and the complex configuration design of the 3D SERN with shape transition is the key problem to be solved.

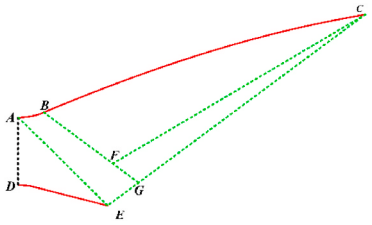
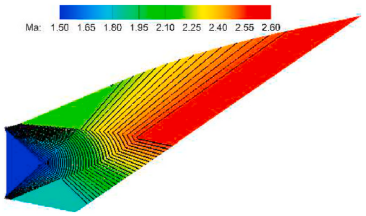
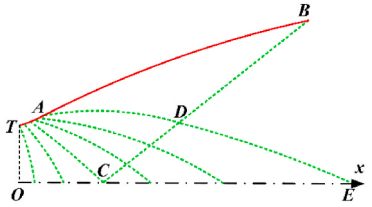
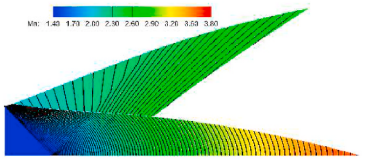
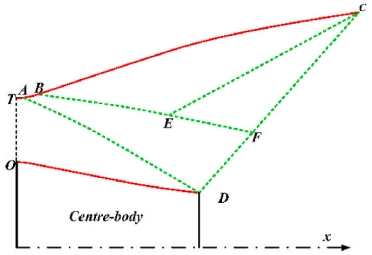
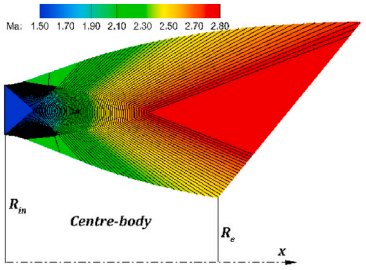
The early researchers focused on the mathematical methods, such as the calculus of variations, which was employed to optimize a 3D maximum thrust nozzle under specific geometric constraints initially [99]. Following this method, the application of the calculus of variations was developed to design a non-axisymmetric maximum thrust nozzle for specified inflow parameters and ambient pressure [100]. Besides, the reference plane method [101] was applied in the calculation of the highly complicated jet flowfield of the nozzle. However, both the calculus of variation and reference plane method are too complex to be applied in practice. With the enhancing of the computing capability, the hybrid optimization method was combined with CFD solvers to optimize a 3D elliptic nozzle [102,103] and a quasi-axisymmetric 3D nozzle operating at Mach 8.0 [104,105]. Although the optimization method can well solve the engineering design problem, it requires a long computational time with full 3D simulation. Therefore, at present, the 3D SERN design mainly adopts the streamline tracing method.

The streamline tracing technique [106] was widely applied in designing hypersonic vehicles and their propulsion system, such as the waverider, inward-turning inlet, and 3D SERN. A 3D SERN with specific entrance or exit shape can be obtained by tracing streamlines in a known or easily calculated flowfield (e.g., the planar, axisymmetric, or annular flowfield), and the resulting nozzle has the same inviscid flowfield characteristic and performance as the parent flowfield [107]. Thereby, the chosen parent flowfield directly determines the 3D SERN performance. Table 3 summarizes the common parent flowfields, including the 2D planar, axisymmetric, and annular flowfields. Compared with the 2D planar flowfield, the axisymmetric and annular flowfields can incorporate lateral expansion into the 3D SERN, thereby resulting in higher expansion efficiency.

In the 2D planar flowfield design, Lv et al. [90] proposed the compatibility design method of the geometric constraints with the optimal performance, and realized the direct design of the SERN under geometric constraints. Taking a page from Lv's method, the annular flowfield under geometric constraints can be optimized, and the lateral expansion in the streamline-traced SERN can be controlled by adjusting the parameters of R_{in} and R_e. Besides, the axisymmetric flowfield with maximum thrust has been developed by Rao [91].

Initially, based on the axisymmetric maximum thrust flowfield, Lu et al. [108] employed the streamline tracing method to design a 3D SERN with specific entry, and the designed nozzle increased the thrust and lift by 2.7% and 69.5%, respectively, compared with the straight wall configuration. However, the method of Lu et al. [108] has a

Table 3
Summary of parent flowfield.

Configurations	Design sketch	Designed by MOC
2D planar flowfield		
Axisymmetric flowfield		
Annular flowfield		

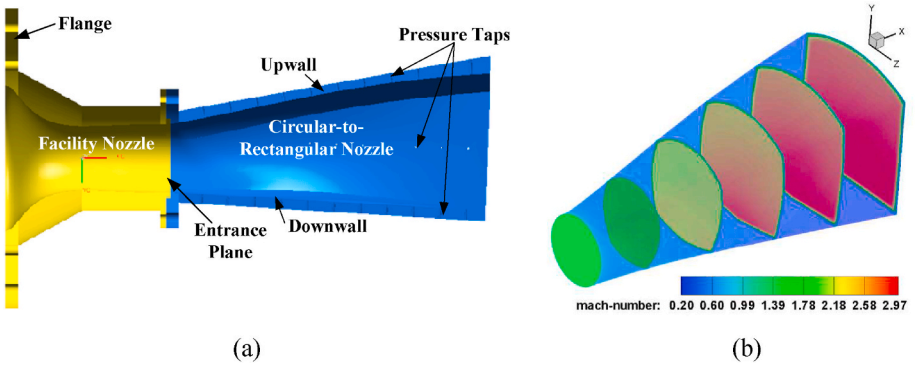


Fig. 29. 3D SERN with circular to rectangular shape transition [109]: (a) Experimental model; (b) Mach number contour.

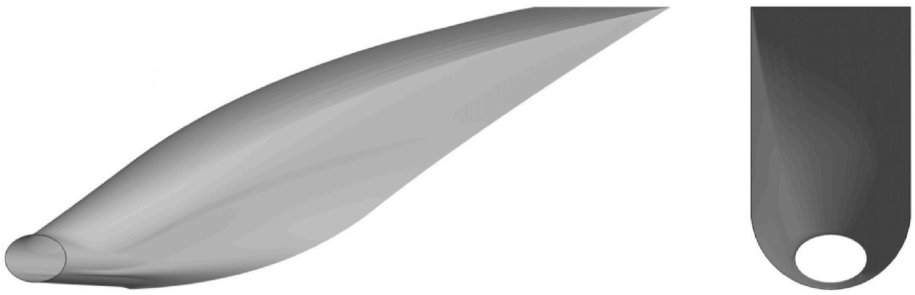


Fig. 30. 3D SERN with elliptical to rectangular shape transition [110].

shortcoming that the exit shape of the 3D SERN was uncontrollable. Consequently, Mo et al. [109] blended two streamline-traced nozzles along both flow and reverse directions in an annular flowfield to optimize a 3D SERN with circular to rectangular shape transition (Fig. 29), and the thrust and lift experienced an increase of 1.0% and 92.5%, respectively, relative to the conventional thrust optimized nozzle. In addition, the centerbody radius of the annular flowfield had a large effect on the general shape of the streamline-traced nozzle, and could be adopted to tailor the nozzle to a variety of vehicle shapes (Fig. 30) [110]. Due to the negative contribution of the lower cowl, the 3D streamline-traced SERN always produced negative lift and small nose-down pitching moment [109]. Consequently, it can be trimmed along the last characteristic line of the parent flowfield to improve the performance and weight [111], and the lift and pitching moment of the trimmed nozzle (Fig. 31) are increased by 427.00% and 10.80%, respectively, while the weight could be reduced by as much as 37.51%.

In the design method of Mo et al. [109], the method of tracing streamlines along the positive and reverse flow directions can well realize the design of 3D SERN with shape transition. However, owing to the fact that the final nozzle is formed by blending two streamline-traced nozzles, the feature of inheritance will be compromised to a certain extent. Although the conclusion of Kunze et al. [110] indicates that the blended nozzle can capture the expansion characteristic of the parent flowfield to a high degree, large differences become apparent in the regions where the integration with the combustor dominates the nozzle shape. Therefore, the design method of the 3D SERN with shape transition, which can organize efficient expansion flow in the 3D SERN, should continue to be developed, such as the complex 3D MOC, and the key is to reduce the complexity of the 3D MOC design with reasonable assumptions and simplifications.

3.2. Research progress on nonuniform inflow and chemical nonequilibrium flow

3.2.1. Nonuniform inflow effect in SERN

In the actual operation of a scramjet, the complex shock waves produced by the forebody, inlet and isolator, the mixing of the fuel and supersonic flow in the combustor, and the unstable combustion, all of those lead to the nonuniform flow at the combustor exit (Fig. 5). In addition, the scramjet nozzle has no contraction section and physical throat to fair the nonuniform inflow, resulting in the existence of the shock and expansion waves at the entry, which certainly will have an impact on the nozzle flowfield structure and performance.

The inflow aerodynamic parameters of the scramjet nozzle include Mach number, static pressure, temperature and flow angle. For a specific nonuniform flow profile, the equivalent uniform flow can be acquired by averaging the flow properties of the nonuniform flow in such a way that the values represent the nonuniform flow and are self-consistent with

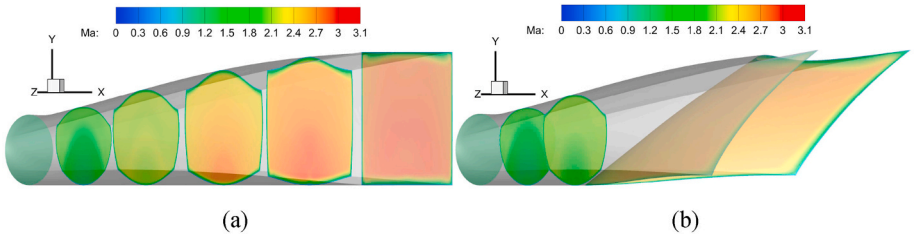


Fig. 31. 3D SERN with ellipse to rectangular shape transition [111]: (a) Without trimming; (b) With trimming.

the governing equations of the uniform flow, and then the result of the equivalent flow can be compared to that obtained from the nonuniform flow.

To average the flow properties of a nonuniform flow, a weighted function, f, is defined as follows

$$\overline{\chi} = \frac{1}{f}\int \chi df \tag{35}$$

where χ is the flow parameter, and $\overline{\chi}$ is the w-weighted parameter. Typically, the static temperature, Mach number and flow angle are based on the mass-weighted average, while static pressure is by the area-weighted average.

Initially, Snelling [112] calculated and analyzed the scramjet nozzle performance under the uniform and nonuniform inflow operations, and the properties of the nonuniform inflow included both Mach number and static temperature. Due to the higher compression on the ramp front and lower pressure acting on the aft resulting from the compression and expansion waves in the nonuniform flowfield, the thrust and pitching moment were increased and decreased, respectively. In other words, compared with the uniform case, the nonuniform inflow can alter the pressure acting on the ramp, and thereby, influence the performance. Considering the temperature nonuniformity had little effect on the pressure field [113], the study of Snelling actually reflected the influence of Mach number nonuniformity on the pressure distribution. However, the temperature nonuniformity can affect the temperature field as well as the shear stress on the ramp. Therefore, the variations in both pressure and shear stress on the ramp caused by the nonuniform inflow are the major factor to affect the SERN performance.

Moreover, the one-dimensional analysis was applied in characterizing the effect of the nonuniform flow [114], in which two streams with different initial Mach number were established without considering the mixing and viscid losses. The nonuniform loss was measured by the difference between the cases of one average stream and two adjacent streams, and the nearly 1% thrust reduction caused by the Mach number nonuniformity was denoted in the analysis, which was quite different from the conclusion of Snelling [112]. Therefore, Quan et al. [115] conducted the detailed investigations to clarify the mechanism of the Mach number nonuniformity. The expansion waves near the entrance were almost straight in the uniform case (Fig. 32 (a)), while they were curved in the nonuniform flowfield (Fig. 32 (b)), resulting in the significant difference of the pressure distribution along the ramp near the entry (Fig. 32 (c)). The thrust and pitching moment could be decreased by 2.92%–5.02% and 4.2%–6.7%, respectively, while the negative lift could be increased up to 17.2%, and all of the conclusions greatly depended on the NPR and nonuniform degree.

In the study of Quan et al. [115], the uniform flow angle coupled with the nonuniform Mach number acts as the inflow condition. However, due to the complex shock and expansion waves in the combustor, the flow angle at its exit is nonuniform actually. The nonuniform flow angle is accompanied by the shock and expansion waves in the entrance flowfield (Fig. 33 (b) and (c)) [116]. Compared with the uniform case, the Mach line is more curved under the shock wave entrance (SWE) condition, and flatter under the expansion wave entrance (EWE) operation. Therefore, the pressure distribution along the ramp appears rising in the SWE case, whereas decreases like stairs in the EWE case. The decrease trends in thrust and pitching moment are also confirmed in the study, and the trend in the negative lift depends on the nonuniformity form. For example, the negative lift can decrease by 2.8%–7.0% in the SWE case, and increase by 4.4%–8.2% under EWE operation. The conclusions stated above indicate that the effect of the nonuniform inflow cannot ignore the shock and expansion waves in the inflow. In addition to the nonuniformities of Mach number, temperature and flow angle, the pressure nonuniformity is equally deserving of attention. The pressure distribution of the inflow directly affects the pressure-field of the nozzle, and then influences the performance. However, the related impact mechanism and characteristic have not yet been disclosed in public literature and are worthy of further research.

The nonuniform inflow leads to the unfavorable expansion flow involved in the nozzle, and changes the internal pressure-field and shear stress along the ramp, and thereby, causes the degraded performance. The design method of the SERN considering the nonuniform inflow may be a way to eliminate the adverse effects, so that the efficient expansion flow can be organized to adapt the nonuniform inflow. In the study of Mo et al. [69], a rotational MOC program which takes into account the nonuniform inflow effects was developed for designing SERN. The nonuniform inflow serving as the design condition is imposed on the entrance, and the computational domain is divided into three regions of Mach, kernel and transition (Fig. 34). Compared with the SERN with uniform inflow, the SERN with nonuniform inflow has a slightly shorter ramp and smaller exit area (Fig. 35). For a specified nonuniformity of Mach number (Fig. 36 (a)), a SERN is designed and analyzed by both numerical and experimental methods (Fig. 36 (b)), and the nozzle can increase the thrust, lift and pitching moment by 1.75%, 6.51% and 6.35%, respectively, which indicates the superiority of the design method.

According to the published literature, there are still few studies on the nonuniform inflow effect, but most of the conclusions are consistent that the Mach number nonuniformity will decrease the thrust and pitching moment, and the trend of the lift depends on the shock and expansion waves coupled with the inflow. Due to the significant performance degradation under the nonuniform inflow operation, the SERN design should take into account the nonuniform inflow condition, which is of great significance to the scramjet nozzle design in practical engineering application.

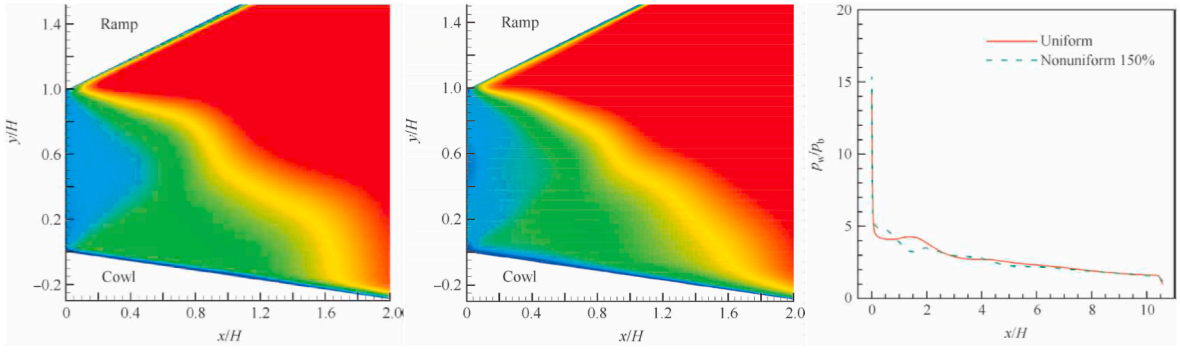


Fig. 32. Flowfield Mach number contour and pressure distribution on ramp: (a) Uniform case; (b) Nonuniform case; (c) Comparison of pressure distribution.

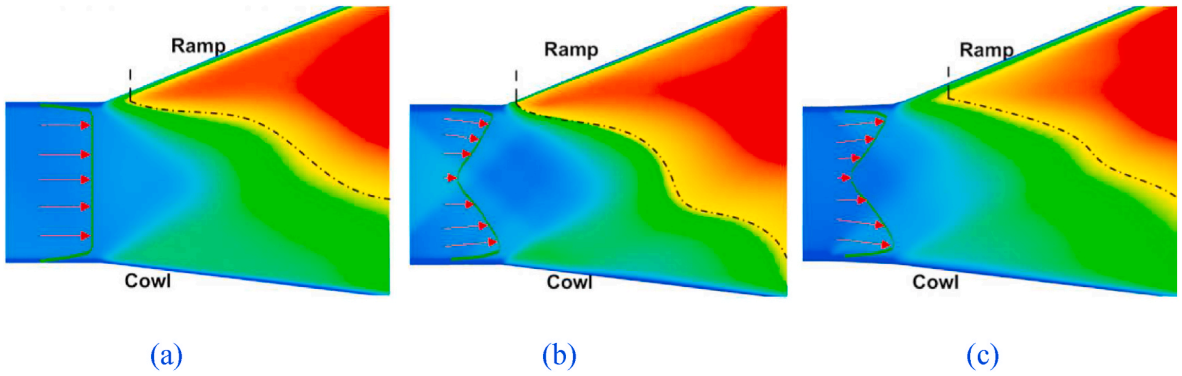


Fig. 33. Mach number contours of SERN flowfield under different operations [116]: (a) Uniform; (b) Nonuniform Mach number coupling with shock wave; (c) Nonuniform Mach number coupling with expansion wave.

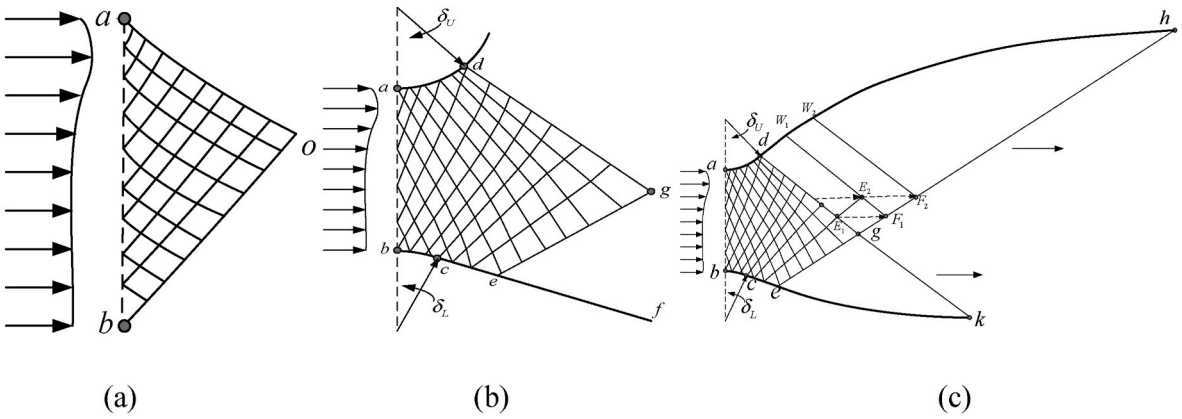


Fig. 34. Design process of SERN considering nonuniform inflow: (a) Mach region; (b) Kernel region; (c) Transition region [69].

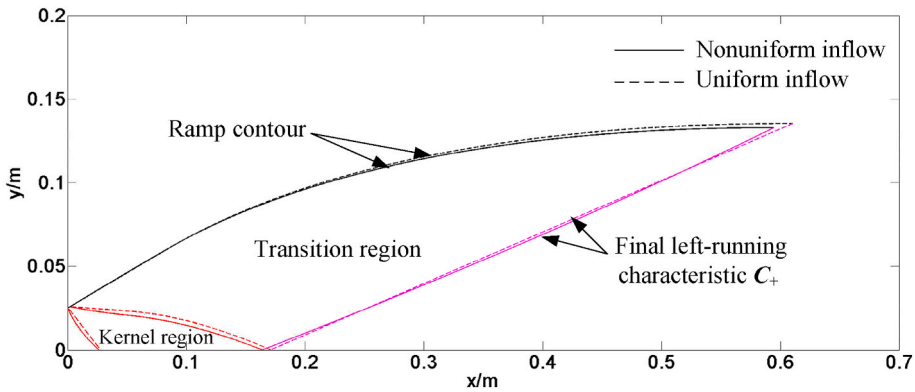


Fig. 35. Comparison of nozzle contour based on uniform and nonuniform inflows [69].

3.2.2. Chemical nonequilibrium flow effect in SERN

Due to the rapid expansion in the rocket and ramjet nozzles, the frozen flow is always assumed in the performance estimation. However, for the hypersonic propulsion system, in particular for a scramjet operating above Mach 10, the residence time of supersonic flow in the combustor is generally in the order of milliseconds [117], which causes the fuel not to be fully burned [118,119] and may continue to burn in the nozzle. Moreover, with the increase of the flight Mach number, the radical recombination is accompanied by the gradual expansion process in the long nozzle resulting from the increased NPR and inflow Mach number. Owing to the continuous combustion of the unburned fuel and radical recombination, it is necessary to consider the chemical nonequilibrium effect in SERN performance estimation and its design.

In the study of Rizkalla et al. [120], a finite-rate chemistry, MOC program was developed to analyze the chemical nonequilibrium flow in a scramjet nozzle, and the result indicated that a frozen-chemistry assumption would result in significantly lower performance estimates. Based on the CHEMKIN code, Sangiovanni et al. [121] developed a computer model for describing quasi-one-dimensional flow of a mixture with area change, and simultaneous gas phase chemical reactions with the fuel of hydrogen. Furthermore, a parabolized Navier-Stokes code coupling with a modified MacCormack's method was applied in calculating the finite-rate chemistry kinetics in a nozzle flow [122]. Both results also showed that beneficial chemical process was maintained throughout the overall nozzle length, and finite-rate chemistry should not be neglected in nozzle simulations. Through the

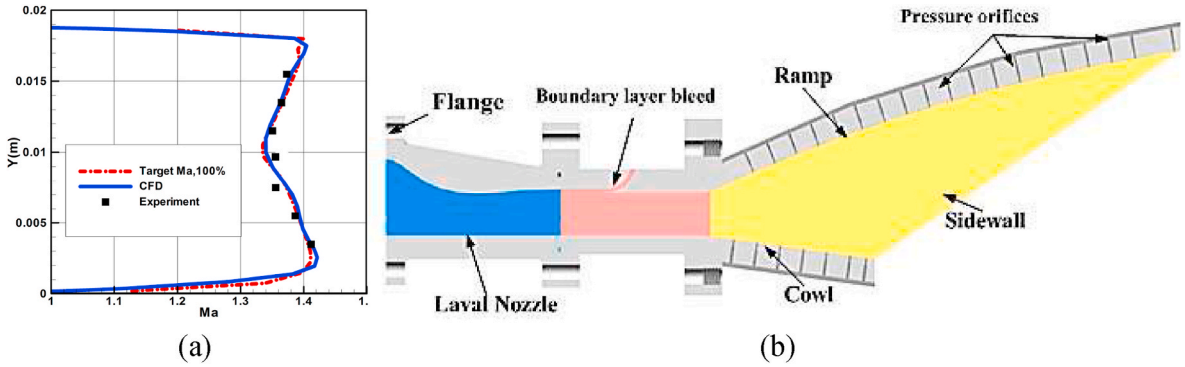


Fig. 36. Experimental study of a SERN considering nonuniform inflow: (a) Mach number profile; (b) Experimental model [69].

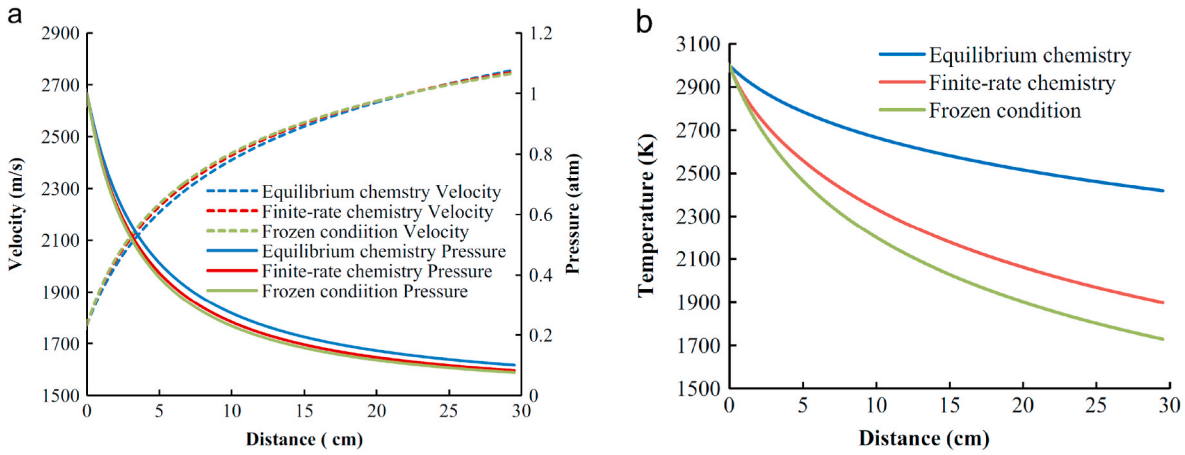


Fig. 37. Flow variables variation with axial distance in HyShot scramjet nozzle [123].

quasi-one-dimensional analysis for the three chemical flows of frozen, nonequilibrium and equilibrium in HyShot scramjet nozzle, Huang et al. [123] showed that the result of the chemical nonequilibrium flow was between those of the frozen and equilibrium flows, and was closer to that of the frozen flow. The conclusion can be clearly obtained from the aerodynamic parameter history along the flow in Fig. 37. The similar conclusion was also confirmed by Sangiovanni et al. [121] and Gronland et al. [124], namely, the difference in gross thrust between the frozen and equilibrium flows may be as high as 5%, and the change in the nozzle flows between the finite-chemistry and equilibrium was 3%.

As stated above, due to the lower combustor efficiency and high combustion temperature of the scramjet, both the continuous

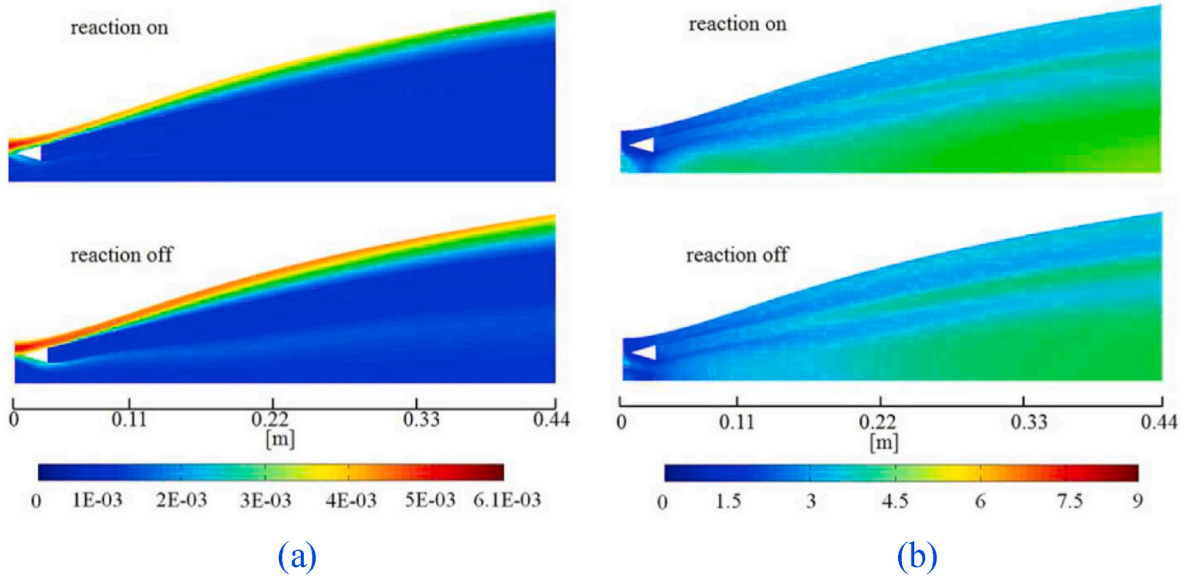


Fig. 38. Flowfields of optimal configuration comparing cases of reaction on and reaction off [127] : (a) Hydrogen mass fraction; (b) Mach number.

combustion reaction and radical recombination exist in the nozzle. The continuous combustion in the nozzle plays a role in the afterburning of the combustor [125], and the effect of reaction heat release on thrust depends on the degree of combustion completeness [123]. Stalker et al. [126] established a combination model of combustor and nozzle to study the continuous combustion in the nozzle, and calculation showed that the thrust was within 6% of the finite-rate chemistry result. The kinetic afterburning led to the presence of compression waves in the nozzle, which increased the pressure distribution along the ramp and thus improved the thrust performance. Therefore, the kinetic afterburning was partly responsible for thrust production and offered a means of extending the operating Mach number range of a fixed geometry scramjet. In the study of Candon et al. [127], the supersonic afterburning with strut fuel injection (Fig. 38) was introduced into a scramjet nozzle to increase the thrust. Due to a combination of contributions from aerodynamic and combustion effects, the thrust was found to be augmented significantly. Besides, to eliminate the problem of inlet thin layer in the calculation analysis of a single nozzle, Chen et al. [128] also established a strut combustor-SERN model to study the effect of the continuous combustion in SERN. The chemical reaction existed in the overall nozzle, especially it was particular significant near the entrance, and the nonequilibrium effect was more obvious with the higher equivalent ratio of combustor. For example, the increments of the thrust and lift could be up to 9.41% and 16.39%, respectively, with the equivalent ratio of 0.8.

Except for the effect of the continuous combustion of the unburned fuel in SERN, attention should also be paid to the radical recombination accompanied by the expansion flow. Although the radical recombination can also release tremendous amount of heat, it has relatively small effect on thrust compared with the case of the continuous combustion, because the heat is mainly employed to increase the gas temperature. Zhang et al. [129] denoted that the radical recombination became more significant with the increase of the flight Mach number, due to the higher temperature in the combustor, and it could improve be nearly 2% the nozzle performance at Mach 8.0. Just like the flow characteristic of the continuous combustion, the radical recombination is also mainly appearing in the front of the nozzle, resulting from the rapid expansion in this region. In the radical recombination process, the energy release depends on the different chemical reaction process and species group [130]. In the hydrogen-fueled scramjet nozzle flows, the thermomolecular reactions for hydrogen radical recombination exert the greatest influence on the chemical contribution to nozzle performance [121]. However, in the hydrocarbon-fueled scramjet nozzle flows, the reactions with C and H make the major contribution to the performance (Fig. 39 (a)) [130]. Furthermore, the energy release from the radical recombination is also affected by the inflow conditions and nozzle scales. The increase of the inflow static temperature can improve the chemical reaction processes and increase the potential dissociation energy that can be released ((Fig. 39 (b)). The increased static pressure of inflow can accelerate the chemical reactions but reduce the fraction of radical components, and thus, the two opposite effects lead to the slight effect of the finite-chemistry. The radical recombination can be improved with the increased nozzle scale.

In a word, due to the continuous combustion of the unburned fuel and radical recombination in the scramjet nozzle, especially the kinetic afterburning, the frozen flow assumption usually underestimates the scramjet nozzle performance, and the finite-rate chemistry cannot be ignored in the nozzle performance estimation. However, for the complex inflow condition with various species, it is also necessary to explore ways to realize the efficient expansion flow with finite-rate reaction in a scramjet nozzle. Therefore, the further study should develop the SERN design method considering chemical nonequilibrium flow. Rizkalla et al. [120] utilized the MOC coupled with finite-rate chemistry to analyze the chemical nonequilibrium flow in a scramjet nozzle, and as a reference, the MOC can also be adopted to design a SERN with finite-rate chemistry. The characteristic equations for the 2D flow with finite-rate chemistry are identical to those of the frozen flow, including Eqs. (28) and (30), and the chemical nonequilibrium is mainly reflected in the compatibility equations [92]:

$$\rho V dV + dp = 0 \tag{36}$$

$$dp - a_f^2 d\rho = \frac{\psi}{u} dx \tag{37}$$

$$\rho u d C_i = \sigma_i dx (i = 1, \ldots, n) \tag{38}$$

$$\frac{\sqrt{Ma^2 - 1}}{\rho V^2} dp_{\pm} \pm d\theta_{\pm} + \left[\frac{\delta \sin\theta}{yMa} - \frac{\psi}{\rho V^2 a_f} \right] \frac{dx_{\pm}}{\cos(\theta \pm \alpha)} = 0 \tag{39}$$

where $\delta = 0$ for planar flow and $\delta = 1$ for axisymmetric flow, and ψ is the nonhomogeneous term. Besides, C_i, σ_i and a_f are mass fraction, source function of species i and sound speed at frozen condition.

Along the streamlines, Eqs. (36)–(38) are applicable. Eq. (39) is applicable along both of the left and right-characteristics. The CHEMKIN

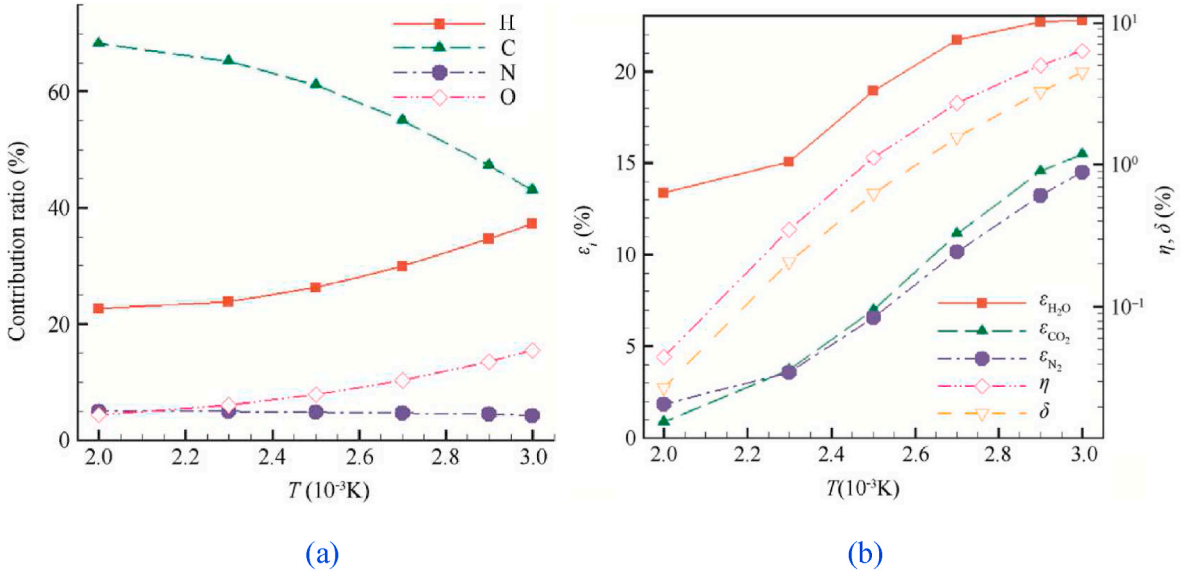


Fig. 39. Variations of the key parameters with the inlet static temperature: (a) Energy contribution ratios with; (b) other key parameters defined in Ref. [130].

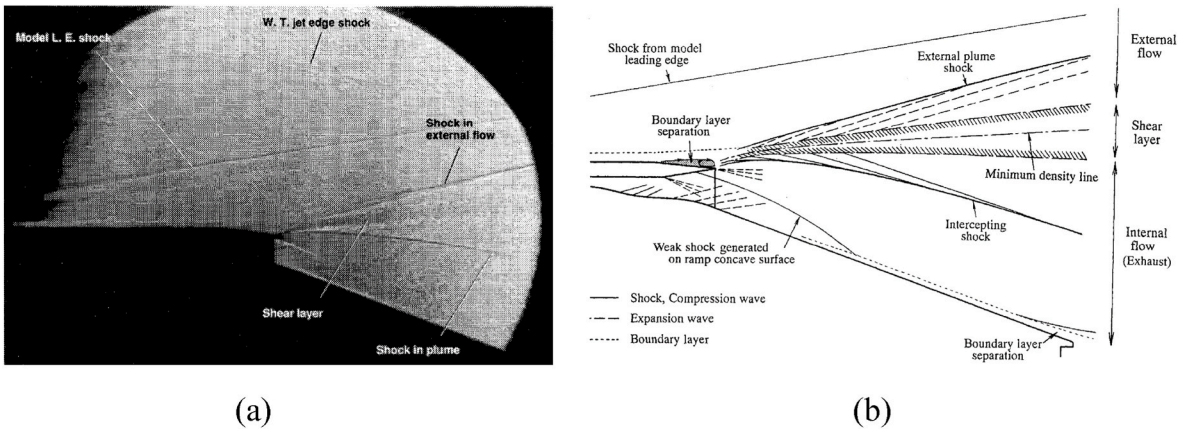


Fig. 40. Typical SERN flowfield characteristic with external flow [131,132].

code can be employed to solve the continuity equation of the species. Based on the MOC for the 2D flow with chemical nonequilibrium, it will be possible to achieve the flowfield analysis and design of the SERN with finite-rate chemistry.

3.3. Research progress on internal and external flow coupling in SERN

For the hypersonic vehicle, due to the integrated design requirement of the airframe with the propulsion system, the airframe afterbody usually acts as the SERN ramp, and the lower cowl is relatively short. Therefore, the complex flowfield interaction between the nozzle exhaust and the external freestream will exist near the cowl (Fig. 40). Since the scramjet net thrust is much less than the SERN thrust, it is necessary to study the coupling effect of the internal and external flow on nozzle performance. According to the public literature, owing to the limitation of the ground experimental equipment, most of the SERN experiments only focus on the internal flow characteristic, where there are few coupling experiments of the internal and external flows for the afterbody nozzle.

Initially, Spaid et al. [131,133] conducted the SERN experiment with external flow of Mach 7.3 in the 3.5-Ft Hypersonic Wind Tunnel of the NASA Ames Research Center. The experiment captured the detailed flow feature of the SERN flow with the hypersonic external freestream. Due to the underexpanded operation of the SERN, the nozzle exhaust pressure is greater than that of the freestream, the shear layer originating from the trailing edge of the cowl deflects toward the freestream, which results in an external plume shock as well as a separation bubble at the cowl outside. Besides, the nozzle flow with the rapid expansion at the trailing edge of the cowl encounters the shear layer, leading to an intercepting shock in the nozzle. As shown in Fig. 40, the external shock, shear layer and internal shock are all reflected in the experimental schlieren. The experiment also revealed that the highly 3D feature of the coupling flow (Fig. 41), and the shear layer thickness showed significant differences between vertical and spanwise traverses.

Then, Watanabe [132] investigated the interaction of the SERN flow with the freestream in a Mach 7.1 wind tunnel, and analyzed the effect of the external flow on the SERN flowfield and performance in detail. Owing to the absence of the impinging of the intercepting shock on the ramp at the underexpanded condition (Fig. 40), the external flow affected the model surface pressure only from the ramp sides, while the external flow directly affected the ramp pressure through a shock impingement upon the ramp at the overexpanded condition. The external flow effectively restrained the outward expansion of the exhaust flow, and delayed the boundary-layer separation on the ramp side regions and near the ramp aft end, and thus, it had significant effects on nozzle thrust, lift and pitching moment. The spanwise expansion was suppressed in the nozzle with a long-length side fence [132,134], reducing the effect of the external flow on the SERN internal flow, and the nozzle flowfield was fairly 2D. Therefore, a slight thrust increase was obtained with the case of long-length side fence.

For further study, both DLR (German Aerospace Center) and CARDC (China Aerodynamics Research and Development Center) also

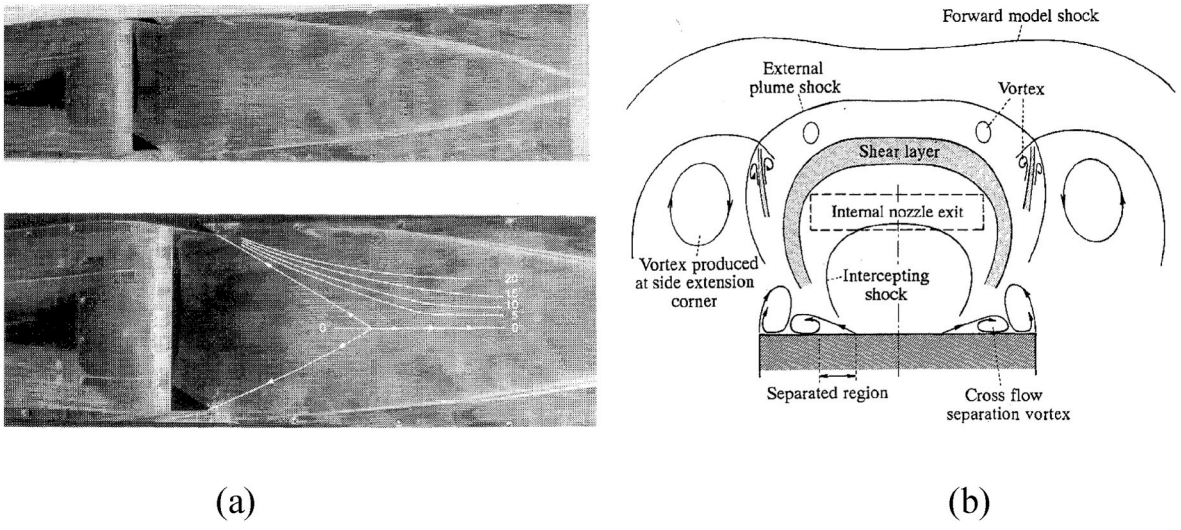


Fig. 41. 3D feature of the SRERN flow with external freestream: (a) Oil-flow photographs on ramp [131]; (b) Cross-sectional schematic of nozzle flowfield [132].

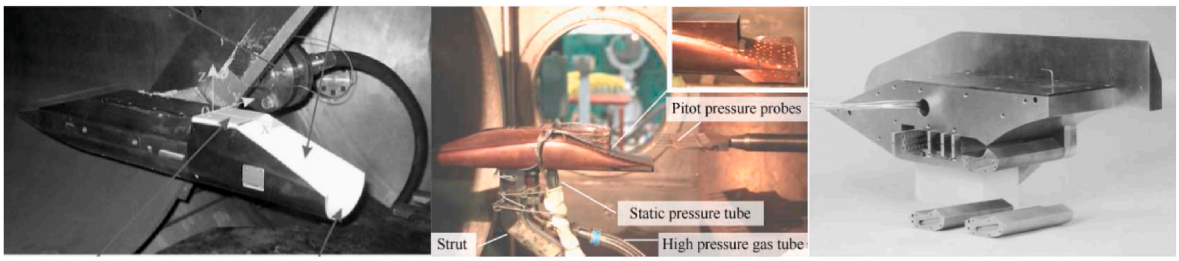


Fig. 42. SERN experimental with external freestream: (a) Mach 7.0 in DLR [135]; (b) Mach 5.0 in CARDC [138]; (c) Mach 6.0 in DLR [139].

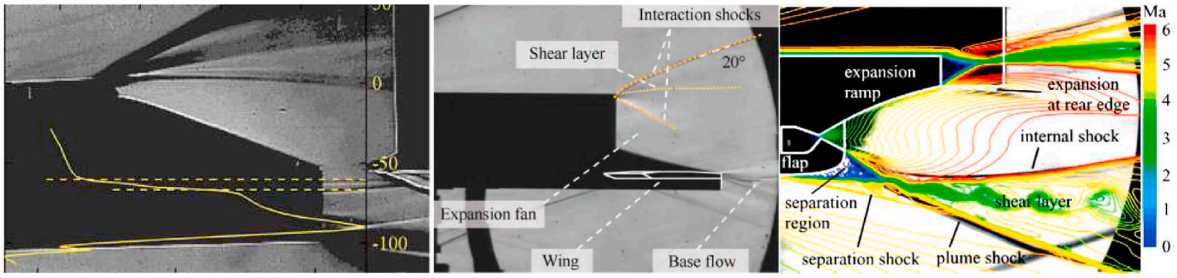


Fig. 43. SERN flowfield characteristics captured in experiment with external freestream: (a) Mach 7.0 in DLR [135]; (b) Mach 5.0 in CARDC [138]; (c) Mach 6.0 in DLR [139].

conducted the nozzle/afterbody experiment with external flow (Fig. 42 and Fig. 43). Hischen et al. [135–137] performed an experimental study on nozzle/afterbody flow in the hypersonic wind tunnel H2K of Mach 7.0 in DLR. Owing to the underexpanded operation of the nozzle, the internal flow characteristic greatly depended on the SERN geometry configuration, and was not affected by the external flow conditions, denoting that the nozzle thrust efficiency mainly depended on the geometric shape, and the efficient design method was the key to achieve the high performance. Both the NPR and freestream Reynolds number influenced the interaction of the internal jet with the external freestream, including the exhaust plume size, position of the shear layer, as well as external and internal shock, which indicated that the NPR and freestream Reynolds number affected the afterbody base drag. Moreover, the coupling of the internal jet with the freestream also had a significant impact on the pressure distribution as well as the flowfield structure of the vehicle tail and rudder, and thus it greatly affected the thrust, resistance, and lift characteristics [138]. The external flow also significantly affected the boundary layer separation on the outside of the SERN cowl, and the separation position was determined by the NPR and freestream Reynolds number [139,140], and the integrated afterbody/SERN performance could be improved by optimizing the external configuration of the cowl.

As a conclusion, the previous researches have confirmed that the interaction of the internal and external flows has a weak effect on the SERN internal flow at the underexpanded operation, and however, it greatly affects the propulsion and stability of the vehicle by influencing the pressure distribution of the tail and rudder. The coupling of the exhaust jet with the freestream depends on the NPR and freestream Reynolds number, while the SERN internal flow feature and its performance are mainly affected by the nozzle configuration. Therefore, just in terms of SERN design, the key is to organize the efficient expansion flow in the nozzle, and it can ignore the interaction of the internal and external flows. The thrust, resistance and lift characteristics of the integrated afterbody/nozzle should take into account the coupling effect, and can be improved by optimizing the configuration of afterbody.

4. State of the design and flow research on TBCC exhaust system

Due to the wide flight Mach number range from takeoff to hypersonic cruise, a TBCC engine experiences different engine operation modes. As one of its core components, the exhaust system is faced with a variety of complex working environments and tests. In the above sections, the status of the scramjet nozzle at the design point (i.e., the cruise state) was introduced. However, at the off-design points, the exhaust system should consider the variable geometry design for a wide flight range, overexpanded flow at low flight Mach number, and mode transition from LSF to HSF.

4.1. Variable geometry design for the TBCC exhaust system

As stated above, owing to the wide flight range of the propulsion system, the operating condition of the exhaust system is variable, and its NPR changes from 2 to 600 depending on the task. As a result, the variable geometry design of the exhaust system is crucial to adjust the mass flow rates and area ratios of both flowpaths. Some studies have confirmed that 54% of the fuel on board would be consumed to accelerate to the cruise Mach number of 7 [53], and the acceleration of propulsion system was essential to the success of a hypersonic vehicle. Thus, the exhaust system should operate efficiently in the acceleration regime. The SERN at the cruise state is determined initially in the design of the exhaust system, and then the variable geometry scheme designs of the LSF and HSF are conducted. The latter is directly related to the exhaust system performance within the whole flight envelope, and affects the vehicle acceleration. According to the SERN geometry configuration at the design point, the exhaust system is divided into 2D and 3D configurations. Owing to the simplicity of 2D regulation, most researches focus on the 2D exhaust system, while there are few studies on the 3D exhaust system.

4.1.1. 2D TBCC exhaust system

In the open literature, NASP was the first one to present the variable geometry scheme of a TBCC exhaust system [45], as shown in Fig. 44 (a).

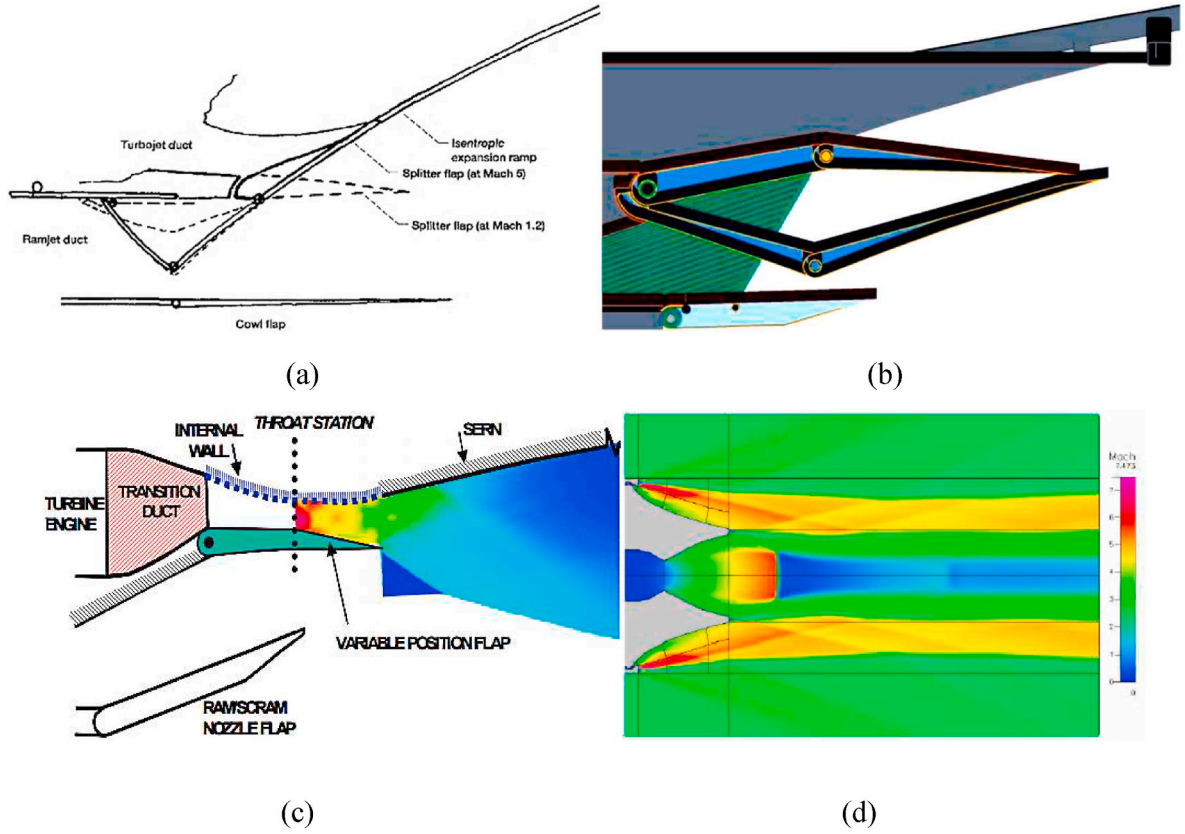


Fig. 44. Typical TBCC exhaust system configuration: (a) Scheme in NASP [45]; (b) Scheme of Gamble et al. [141]; (c) Scheme of Espinosa [142]; (d) RPNCC [143,144].

The ideal SERN contour designed by MOC was adopted to ensure the optimal performance at the cruise state of Mach 5. In the variable geometry features, the mass flow rate of the LSF was controlled by the rotation of the splitter, while that of the HSF was adjusted by the hinge structure, and the expansion ratio was adapted by the rotation of the cowl. Due to the uncontrollable area ratio in the LSF, the overexpanded flow in the exhaust system was inevitably at transonic flight condition, resulting in the low thrust performance. Therefore, the ideal configuration was that both the throat and exit areas of the two flowpaths could be adapted to the various flight conditions. As shown in Fig. 44 (b), the mass flow rate and area ratio were adjusted by the hinge structure, but the large and complex drive mechanisms with an associated weight penalty were inevitable [141].

Espinosa [142] indicated that the efficient operation of the propulsion system requires a highly integrated exhaust system. The desired throat areas of LSF and HSF could be established by the rotations of the splitter and cowl, respectively, as displayed in Fig. 44 (c), and the splitter position should be optimized in detail to minimize the flow interaction between the turbine nozzle exhaust and the ramjet nozzle jet, as well as minimize the internal volume and installation penalty.

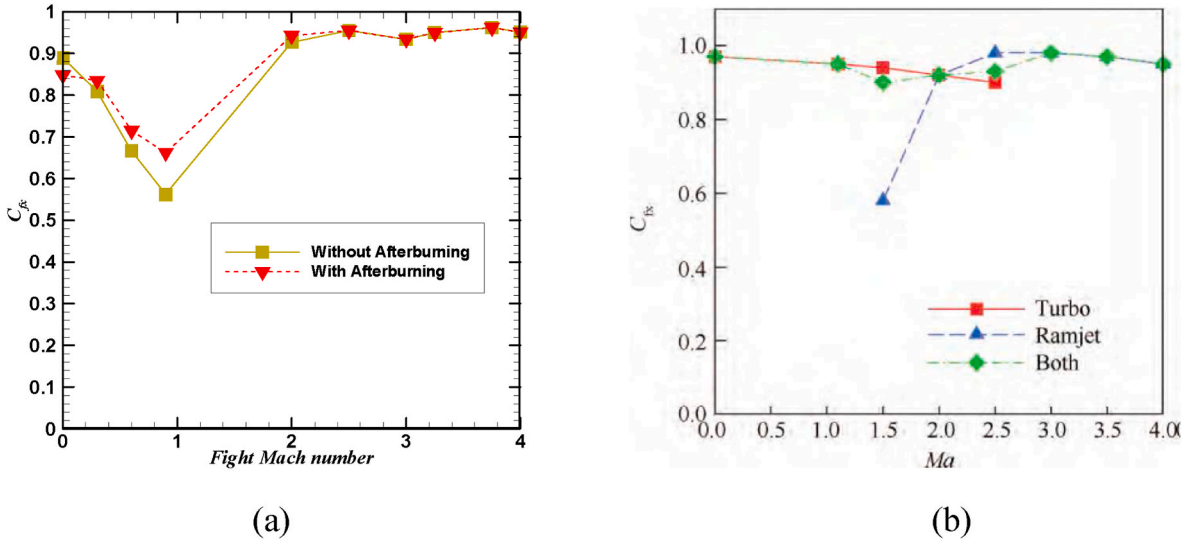


Fig. 45. Thrust coefficient of TBCC exhaust system operating at Mach 0–4: (a) Result of Mo et al. [145]; (b) Results of Niu et al. [146].

Except for the 2D planar exhaust system, the 2D axisymmetric configuration with dual-flowpath is still worth of attention. Wood et al. [143,144] proposed a new concept of Rocket Based Combined Cycle (RBCC), namely Rocket Plug Nozzle Combined Cycle (RPNCC) (Fig. 44 (d)). This configuration incorporated the performance enhancing effect of the altitudes compensating plug nozzle for HSF, while applying the exhaust plume from the LSF to provide the pressurization along the plug base, which could provide a reference for the axisymmetric exhaust system with dual-flowpath, such as Sabre and Hyscram. Unfortunately, the complex regulation scheme for the wide range was not presented in the study, and was required to be taken into account in the further study.

With the deepening of the research on the combined cycle engine, the design and performance analysis of the corresponding exhaust system have developed rapidly, with the studies of both the numerical simulation and wind tunnel experiment. Based on the regulation scheme proposed by Espinosa [142], Mo et al. [145] optimized an over-under TBCC exhaust system operating at Mach 0–4, and obtained the performance over the flight envelope (Fig. 45 (a)). The rotations of the splitter and cowl controlled the mass flow rates of the LSF and HSF, respectively, while the expansion areas of the two flowpaths could not be adapted to the various operations accurately (Fig. 44 (c)). Hence, the regulation scheme was obviously simple, relative to the configuration presented by Gamble et al. [141].

Besides, a new regulation scheme was also developed by Niu et al. [146] for an over-under exhaust system operating at Mach 0–4 (Fig. 46), and the results included the flowfield feature and performance over the flight envelope (Fig. 45 (b)). As displayed in Fig. 46 (a), the rotation of the common surface around the rear endpoint was employed to adjust the throat area of the LSF, and a part of the HSF SERN ramp acted as the common surface. Compared to the configuration presented by Espinosa [142], the complex mechanisms could be installed in the space between the airframe and common surface, which was beneficial to reduce the windward areas. Furthermore, the common surface was convenient for cooling resulting from only one side in the high temperature gas. However, it should bear large aerodynamic force as its large area, which was not conducive to its regulation.

Fig. 45 displays the performance comparison of the two configurations. An obvious thrust trap existed in the transonic phase resulting from the highly overexpanded flow in the result of Mo et al. [145], while the performance deterioration was not serious at the transonic speeds in the conclusion of Niu et al. [146]. The comparison indicated that the splitter rotation scheme was not conducive to the performance at the transonic stage, and the identical conclusion was also confirmed in the study of Wang [147]. In his study, the regulation scheme of Espinosa [142] was adopted to a TBCC exhaust system operating at Mach 0–7. The detailed flowfield features of the exhaust system at various operations were captured by the experiment (Fig. 47). Low performance with an obvious thrust trap at the transonic regime was also found (Fig. 48). In the separate operation phase of the HSF, the thrust coefficient was decreased with the increase of the flight Mach number, owing to the increase of the underexpanded loss. Even though the adjusting scheme and operational range of the exhaust systems studied by Mo et al. [145], Niu et al. [146] and Wang [147] were various, the flowfield characteristics were basically consistent. The flowfield structure was relatively simple at the separate operation of the LSF or HSF, while the flow was obviously complex with the interaction between the LSF jet and HSF exhaust at the parallel operation, and the internal shock, viscous shear layer and their interaction were captured in the flowfield.

To improve the overexpanded performance of the LSF for the regulation scheme of the splitter rotation, the splitter was shifted backward in the study of Lv et al. [148] (Fig. 49 (a)), and as stated by Espinosa [142], the performance could be improved by optimizing the splitter position. The expansion ratio of the LSF was decreased, thereby resulting in the increasing thrust coefficient at low flight Mach number. Besides, unlike the adjustment scheme of the cowl rotation, the cowl was rotating as well as sliding around a fixed point to meet the adjustment requirement of the HSF, and this scheme also had a certain ability to adjust the HSF area ratio. Therefore, the thrust coefficient could maintain above 0.96 at the separate separation of the HSF.

4.1.2. 3D TBCC exhaust system

In recent years, the propulsion system with 3D flowpath has become a hot topic for hypersonic vehicle, and the 3D exhaust system has been developed to satisfy the requirements of the integrated design and adapting to the combustor exit, such as the projects of FaCET [49,50], TriJet [53,54] and SR-72 [55]. In Section 3.1.3, the design method of the 3D SERN for scramjet is reviewed, and based on the 3D scramjet SERN, the 3D TBCC exhaust system with variable geometry is devised for wide flight range.

The TriJet concept combined with three engines of turbine, ejector ramjet (ERJ) and dual mode ramjet (DMRJ), and the EJR was supplemented to bridge the thrust gap between the available turbine and DMRJ. Due to three suites of engines in the concept, it was a challenge to design the common inlet and exhaust system integrated with three flowpaths. In terms of the inlet, the LSFs of the turbine and EJR were fused with the inward-turning inlet of DMRJ (Fig. 50 (a)), and similarly, it might be suitable to integrate both the turbine and ERJ nozzles with the streamline-traced SERN of DMRJ. However, in order to produce more thrust from the DMRJ before its nominal takeover speed, a sustained aerodynamic choke (SAC) was employed in the DMRJ nozzle (Fig. 51). The booster exhaust stream was injected into the DMRJ exhaust flow at the right location, and then an aerodynamic choke was established to prevent the flow between the DMRJ combustor exit and booster nozzle. The SAC converted the DMRJ into a pure ramjet with a large combustor flow area, thus generating more thrust. Considering that the turbine was shut down at the operation of SAC, the ERJ jet was fed to the nozzle flaps protruding into the DMRJ nozzle to establish the SAC condition. Therefore, the ERJ exhaust was divided into two flowpaths integrated with the DMRJ SERN, and the turbine nozzle was outside of the main nozzle (Fig. 50 (b)). However, in the studies of

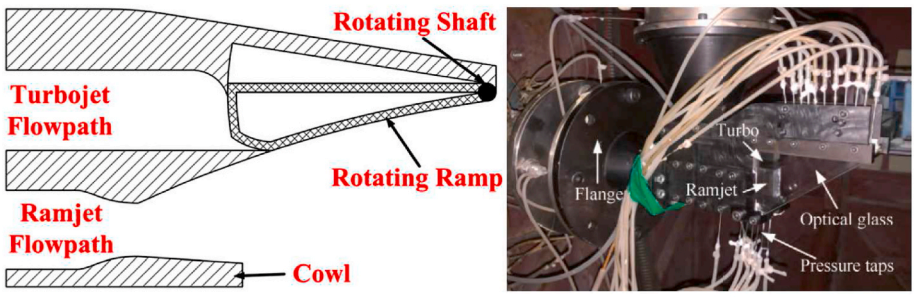


Fig. 46. TBCC exhaust system operating at Mach 0–4 developed by Niu et al. [146]: (a) Regulation scheme; (b) Experimental model.

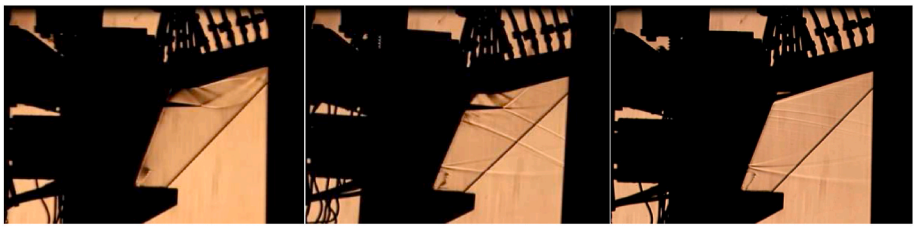


Fig. 47. Flowfield structures of TBCC exhaust system operating at Mach 0–7 developed by Wang [147]: (a) Flow structure at the separation operation of LSF; (b) Flow structure at the parallel operation; (c) Flow structure at the separation of HSF.

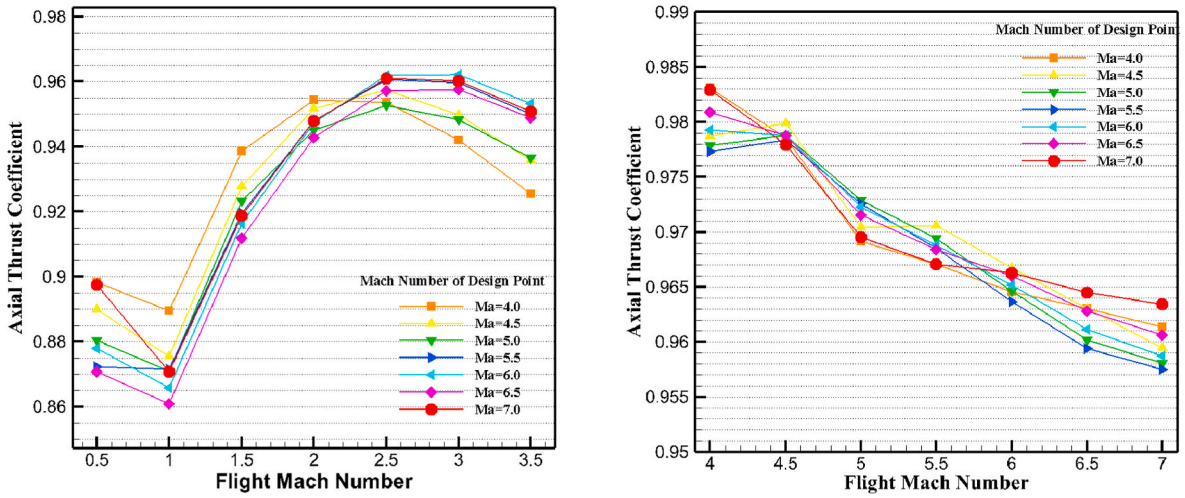


Fig. 48. Thrust coefficient of TBCC exhaust system operating at Mach 0–7 [147].

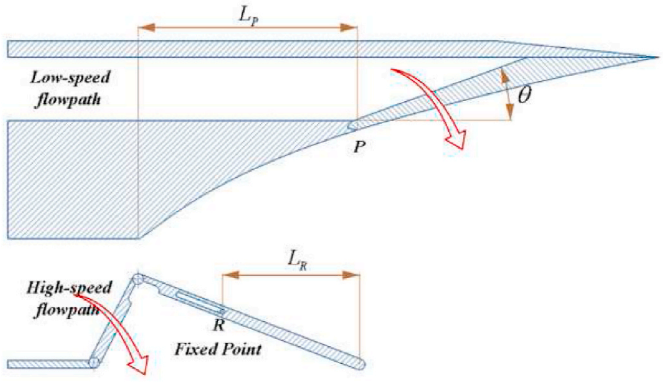


Fig. 49. Regulation scheme of exhaust system operating at Mach 0–6 developed by Lv et al. [148].

Siebenhaar et al. [54] and Bulman et al. [53], only the conceptual design was presented, and the detailed design and performance analysis of the 3D exhaust system were not described.

Based on a 3D SERN with circular to rectangular shape transition designed by the osculating method, Hua et al. [149] proposed a new scheme that the shared surface was sliding and rotating to adjust the LSF throat area and HSF expansion area, respectively (Fig. 52 (a)), and the 2D regulation was employed to maintain the reliable seal. Numerical and experimental results showed that this configuration of the 3D TBCC exhaust system was practicable for a wide flight range (Fig. 52 (b) and (c)). In addition, Xu et al. [150] also investigated a 3D TBCC exhaust system operating at Mach 0–4, in which a SERN with lateral expansion was employed to the HSF (Fig. 53 (a)), and the detailed design method of the SERN with lateral expansion has been summarized in Section 3.1.2. The regulation scheme of the LSF was identical to that applied in the study of Niu et al. [146], and the common surface with equal width was rotated around the rear endpoint to adjust the mass flow rate of the LSF. Although the thrust coefficient was decreased rapidly at Mach 1.5, it maintained above 0.91 over the flight envelope (Fig. 53 (b)). Experimental results also indicated that the flowfield feature was complex at the parallel operation, while it was relatively simple at the separate operation of the HSF (Fig. 53 (c) and (d)), which was similar to the flowfield characteristic of the 2D exhaust system.

Table 4 summarizes the TBCC exhaust system presented in the previous studies. The operating ranges of the exhaust system covers Mach number of 0–4, 0–6 and 0–7, and the exhaust system operating at Mach 0–4 is the most fully studied, including the 2D and 3D configurations. Owing to the fact that the 3D flowpath is beneficial for the highly integrated design of the propulsion system, the 3D SERN with variable geometry has developed rapidly in recent years and is very attractive for application in the hypersonic vehicle. The primary task of the regulation scheme is to ensure the requirement of the mass flow rate, and the rotations of the splitter ad cowl are widely used in LSF and HSF, respectively. Therefore, to reduce the complexity of the regulation mechanism, the expansion ratio without adjustment is reflected in most exhaust system configurations.

4.2. Overexpanded flow separation and control in SERN

The exhaust system NPR ranges from 2 to 600 resulting from the wide working range, and to ensure the optimal efficiency of the propulsion system, the design point of the exhaust system is generally set at the cruise state, which leads to the high design NPR as well as the large

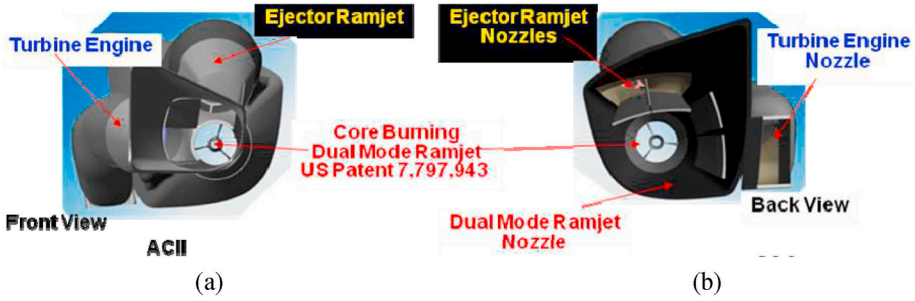


Fig. 50. Inlet and exhaust system configurations in TriJet concept [53]: (a) Inlet; (b) Exhaust system.

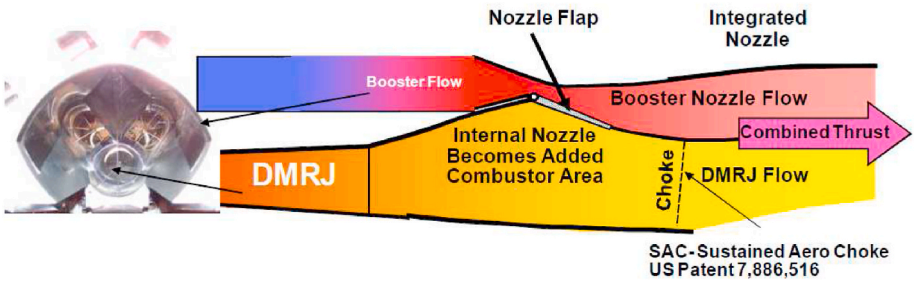


Fig. 51. Sustained aerodynamic choke (SAC) in TriJet exhaust system [53].

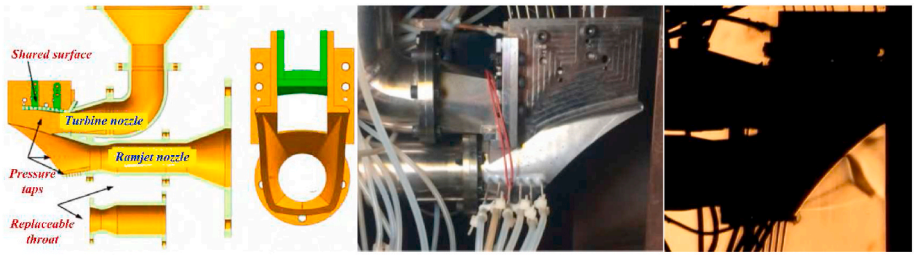


Fig. 52. 3D TBCC exhaust system operating at Mach 0–4 developed by Hua et al. [149]: (a) Experimental model; (b) Model installation; (c) Flowfield structure at the parallel operation.

expansion ratio. As a result, at low flight Mach number, the over-expanded flow and even separation will appear in the SERN, and tend to reduce performance and increase drag with low thrust and strong nose-up pitching moment, seriously affecting the acceleration and controllability of the vehicle. The flow control is critical to improve the over-expanded performance.

4.2.1. Overexpanded flow separation mechanism in SERN

The flow separation in the nozzle is usually accompanied by the interactions of shock with shock, shock with boundary layer, and shock with shear layer, resulting in the highly unstable and unsteady characteristics of the separated flow. Two separation patterns exist in the nozzle under highly overexpanded condition, namely free shock separation (FSS) and restricted shock separation (RSS). In FSS, the separated main flow does not reattach to the wall, and the wall pressure gradually tends to the ambient pressure downstream of the separation point. However, in RSS, the separated flow will reattach to the wall at a certain distance downstream of the separation point, and the main jet seems to be restricted to the wall. The previous researches mainly focus on the rocket nozzle, namely, the axisymmetric configuration, and the nozzle geometry configuration is one of the main factors affecting the separation pattern. Only FSS occurs in the truncated ideal contour (TIC) and conical contour, while both FSS and RSS exist in the thrust optimized contour (TOC) and thrust optimized parabolic (TOP), accompanied by the change of lateral force. It is considered that the pressure fluctuation and separation pattern transition are the main cause of the lateral load.

For the SERN, the asymmetric load under the separation state is the natural property, and the X-43A nozzle has confirmed that both FSS and RSS appeared in SERN [151]. Yu et al. [152–154] summarized the separation patterns in SERN in detail (Fig. 54 and Fig. 55), and RSS was divided into RSS on the ramp and RSS on the cowl, namely, RSS (Ramp) and RSS (Cowl). RSS (Ramp) is a common separation mode in SERN, and RSS (Cowl) appears in SERN with long cowl. FSS is a relatively specific separation pattern, and it only exists in the special configuration with a quite narrow NPR range.

Furthermore, the separation pattern transitions appear in the process of startup and shutdown [152,154]. During the startup process, the separation pattern transitions from RSS (Cowl), FSS to RSS (Ramp) (Fig. 56), and this process includes two separation pattern transitions of RSS (Cowl)-FSS and FSS-RSS (Ramp). The FSS is a transient intermediate state, from NPR of 3.2385–3.3095 (Fig. 56 (b)). The thrust, lift and moment are decreased by 1.09%, 114.57% and 11.21% in the first stage (Fig. 58 (a)), respectively, and they are decreased by 20.65%, 82.52% and 67.89 in the second stage (Fig. 58 (b)), respectively.

During the shutdown process, the separation pattern transitions from RSS (Ramp) to RSS (Cowl) in SERN (Fig. 57). FSS is only an

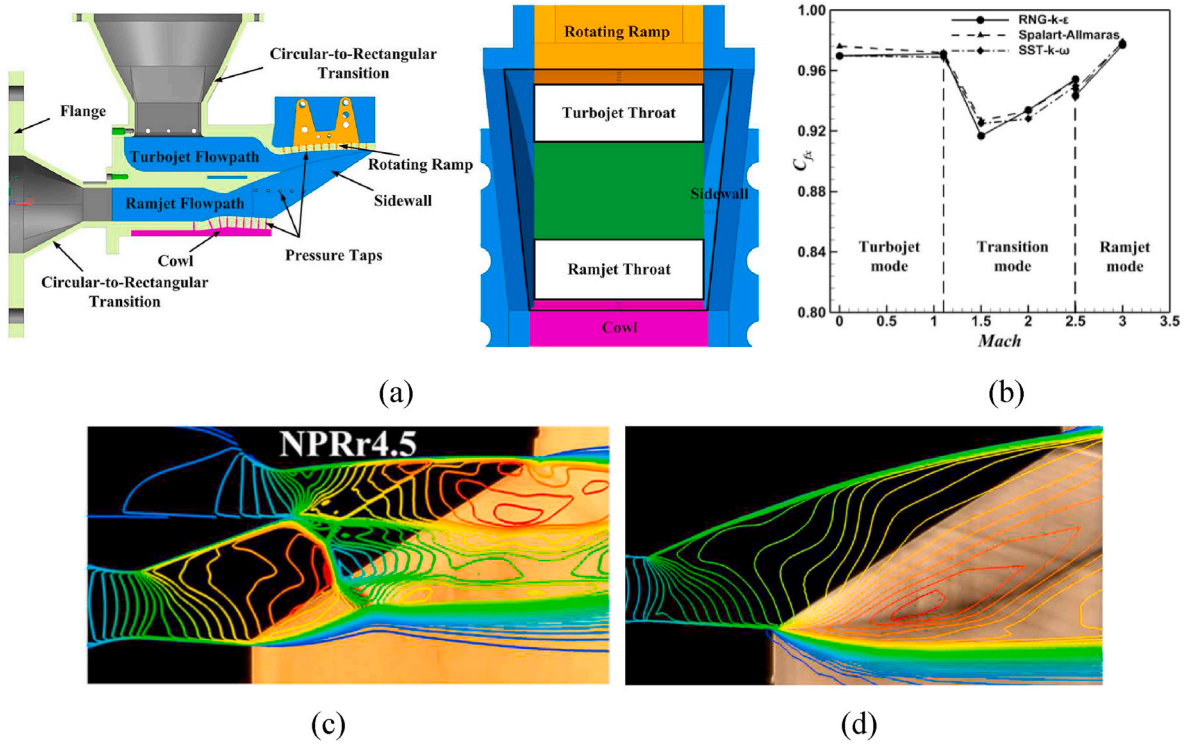


Fig. 53. 3D TBCC exhaust system operating at Mach 0–4 developed by Xu et al. [150]: (a) Experimental model; (b) Performance over the flight envelope; (c) Flow structure at the parallel operation; (d) Flow structure at the separation of HSF.

Table 4
Summary of TBCC exhaust system.

	Operating range	Configuration	Adjustment scheme			
			LSF		HSF	
			TA	AR	TA	AR
NASP [15,45]	Mach 0-6	2D	Splitter Rotation	/	Hinge structure	Cowl Rotation
Gamble et al. [141]	/	2D	Convergent flap Rotation	Divergent flap Rotation	Convergent flap Rotation	Divergent flap and cowl Rotations
Espinosa [142]	Mach 0–4+	2D	Splitter Rotation	/	Cowl Rotation	/
RPNCC [143,144]	/	Axisymmetric	/	/	/	/
Mo et al. [145]	Mach 0-4	2D	Splitter Rotation	/	Cowl Rotation	/
Niu et al. [146]	Mach 0-4	2D	Shared surface Rotation	/	/	/
Wang [147]	Mach 0-7	2D	Splitter Rotation	/	Cowl Rotation	/
Lv et al. [148]	Mach 0-6	2D	Splitter Rotation		Convergent flap Rotation	Cowl Rotation and Slide
TriJet [53,54]	Mach 0-7	3D	/	/	SAC	/
FaCET [22,23,49,50]	Mach 0–5+	3D	/	/	/	/
SR-72 [55]	Mach 0-6	3D	/	/	/	/
Hua et al. [149]	Mach 0-4	3D	Shared surface Slide	/	Shared surface Rotation	/
Xu et al. [150]	Mach 0-4	3D	Shared surface Rotation	/	/	/

TA and AR are the abbreviations of the throat area and area ratio, respectively.

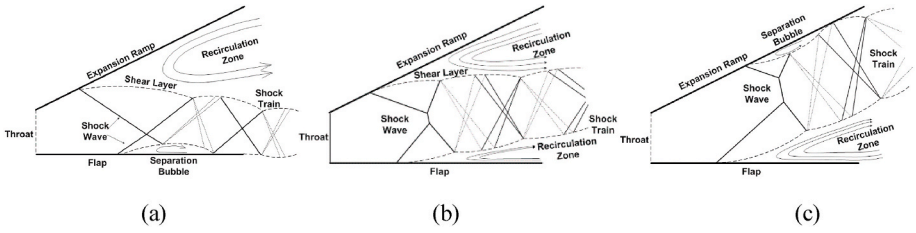


Fig. 54. Separation patterns in SERN: (a) RSS (Cowl); (b) FSS; (c) RSS (Ramp) [153].

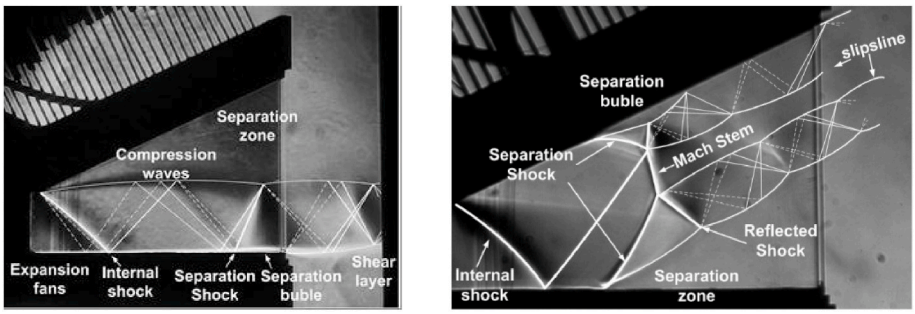


Fig. 55. Experimental schlieren of RSS mode: (a) RSS (Cowl); (b) RSS (Ramp) [154].

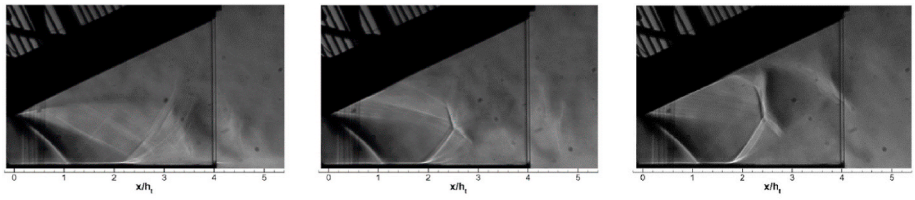


Fig. 56. Transition from RSS (Cowl), FSS to the RSS (Ramp) in startup process [154].

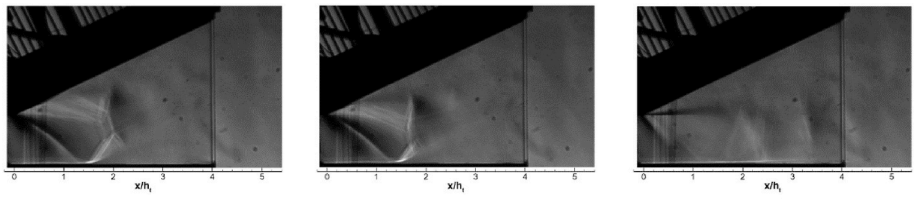


Fig. 57. Transition from RSS (Ramp) to RSS (Cowl) in shutdown process [154].

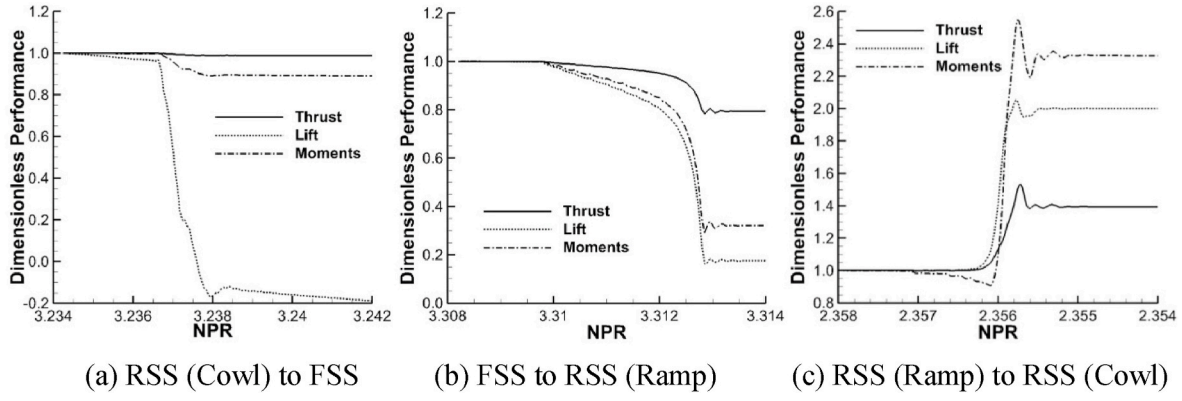


Fig. 58. Performance of SERN during separation pattern transition [152].

instantaneous process and is not stable. In this process, the duration of FSS is much shorter, and the thrust, lift and moment are increased by 39.31%, 100.08% and 132.62% in the overall process, respectively (Fig. 58 (c)).

In addition, the change tendency of SERN flowfield in the shutdown process is not the inverse process of the startup stage, and the critical NPR of the separation pattern transition in the shutdown process is lower than that of the startup stage (Fig. 59), denoting an obvious hysteresis effect during the separation pattern transition process.

The RSS (Ramp) in the SERN shows obvious unsteady characteristics [155]. The pressure in the separation intermittent zone has obvious oscillation, and the shock motion exhibits a low-frequency characteristic, with peak frequencies of 220Hz and 850Hz. He et al. [156] also observed the partially restricted shock separation (pRSS) in the SERN (Fig. 60), and the separated shear layer reattached the ramp intermittently. Therefore, the flow separation was neither RSS nor FSS. Besides, the separation pattern on the ramp was changed from RSS to FSS with the increase in NPR, and the pressure fluctuation near the separation had a relatively independent dominant frequency in RSS, while no obvious distinct dominant frequency existed in FSS and pRSS.

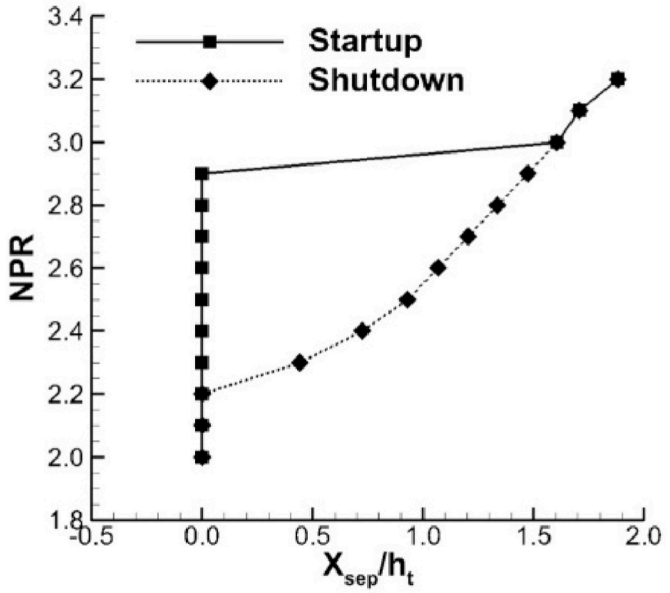


Fig. 59. Separation points on the ramp in the startup and shutdown processes [154].

Besides, the study results [157,158] also indicated that the separation prediction criteria proposed by Schmucker (Eq. (40)) were no longer applicable in RSS (Fig. 61 (a)). Ge et al. [158] developed the new separation prediction criteria for RSS in the SERN (Eq. (41)), and the prediction result agreed well with the experimental data with the root mean square error of 3.68% (Fig. 61 (b)). The complex separation characteristics in the SERN lead to the low calculation accuracy of general CFD software for separation point, separation zone and reattachment point. Therefore, Wang et al. [159] applied the method of optimizing the empirical parameters of SST (Shear Stress Transport) model to the SERN separation prediction, and the prediction errors of the position and pressure at the separation point were reduced to 1.99% and 4.38%, respectively.

$$\frac{p_{sep}}{p_b} = \left(1.88Ma_{sep} - 1\right)^{-0.64} \tag{40}$$

$$\frac{p_{sep}}{p_b} = -1.76\left(0.47Ma_{sep}^{0.45} - 1\right) \tag{41}$$

Researchers have conducted the detailed studies on the flow separation pattern, separation pattern transition and separation point prediction in the 2D SERN numerically and experimentally, which is beneficial to understand the flow separation mechanism in the 2D SERN. However, these knowledges may not be available in the 3D SERN, and the lateral expansion will increase the complexity of the flow separation, in particular the separation pattern transition and prediction, which are worthy for further study.

4.2.2. *Overexpanded flow control in the SERN*

The complex overexpanded flow in the SERN seriously threatens the flight performance and quality of the vehicle, and even induces the complicated nonlinear control problems. The sub-ambient pressure distribution acting on the ramp is the main cause of the low performance under the overexpanded condition. Therefore, previous researchers have put forward some approaches, including external burning, passive cavity concept and secondary injection, to increase the pressure distribution as well as improve the overexpanded performance.

(1) External burning

In the method of external burning, the fuel is injected into the external flow along the cowl, and then mixed and burned (Fig. 62 (a)). The external burning establishes the adverse pressure gradients on the ramp, and forces the boundary layer to separate, thereby pressuring the ramp [160,161]. However, it is inevitable that the additional drag resulting from the external burning and lower specific impulse causes more fuel consumption.

(2) Passive cavity

The passive cavity has the ability to control the flow separation by either alleviating or encouraging separation [162], and the former

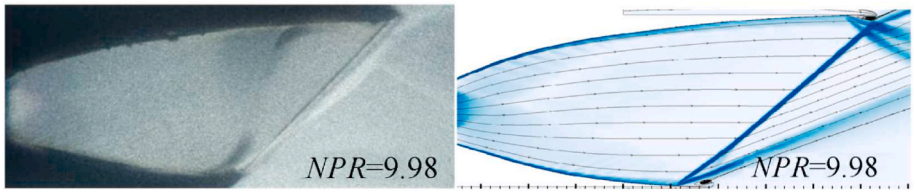


Fig. 60. Flowfield structure of pRSS in the SERN [156].

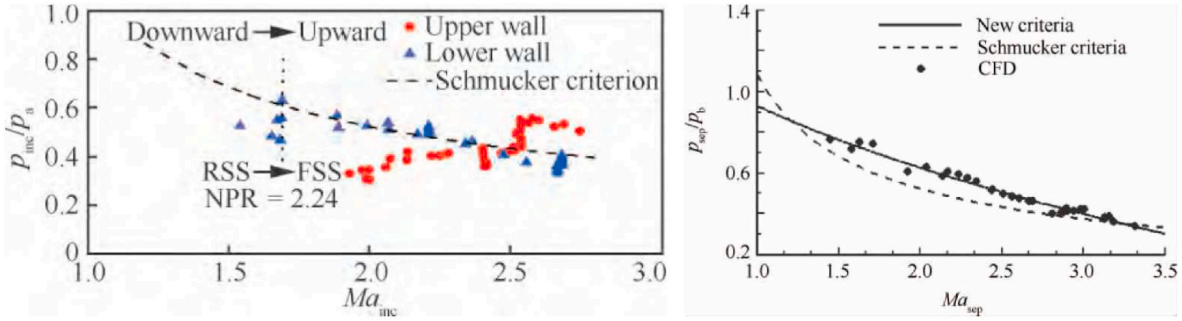


Fig. 61. Separation prediction criteria: (a) Result of He et al. [157]; (b) Result of Ge et al. [158].

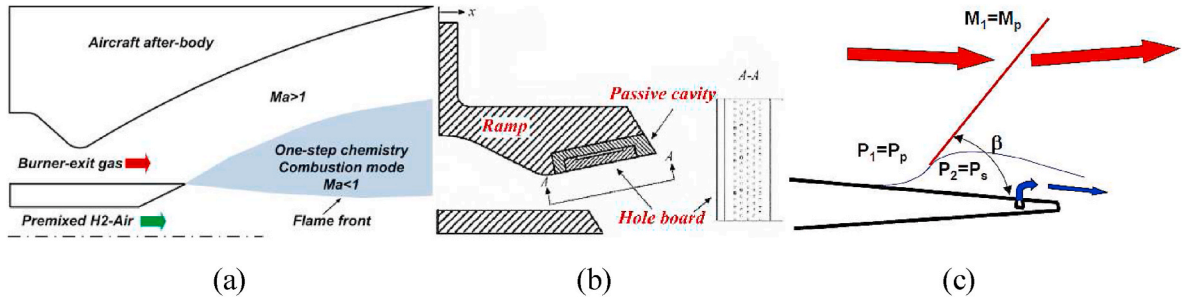


Fig. 62. Approaches to improve the SERN overexpanded performance: (a) External burning; (b) Passive cavity; (c) Secondary injection.

deteriorates the overexpanded thrust coefficient, while the latter can improve the off-design thrust coefficient. Zhou et al. [163] adopted the concept to the flow control of the overexpanded SERN (Fig. 62 (b)), and it could enhance flow separation and increase the ramp pressure at NPR of 3–13, which led to a thrust coefficient increase 1.75% under NPR of 8. Even so, the uncertainty of the passive cavity was also reflected in the SERN.

(3) Secondary injection

It is a good idea to introduce an incidence shock to promote the flow separation on the ramp (Fig. 62 (c)), and thereby, increase the pressure distribution. Based on this idea, Gamble and Haid [164] proposed the method of generating the incident shock by the interaction of the second injection with the main flow, namely, the supersonic transverse jet (Fig. 63). Significant improvement in SERN performance could be achieved by separating the overexpanded flow with secondary injection, and a net thrust based on the turbojet cycle had an increase of 0.3%. However, the SERN thrust coefficient might be decreased with the consideration of the effect of the secondary injection [165].

For further study, Lv et al. [166] revealed the flow mechanism of the secondary injection, and analyzed the influences of the key parameters in detail. The condition for improving the thrust coefficient via secondary injection was presented, namely,

$$R_{x,no} - R_{x,with} > C_{fx,no}F_{s,Sec} \tag{42}$$

where $C_{fx,no}$ and $F_{s,Sec}$ were the nozzle thrust coefficient without secondary injection and ideal thrust of secondary injection, respectively. Under the action of secondary injection, the x-direction component of the internal force acting on the nozzle internal wall was decreased, and under the condition that the reduction was greater than the value of the term $C_{fx,no}F_{s,Sec}$, the overexpanded performance could be improved. For example, the increments of the thrust coefficient, lift and pitching moment were 3.16%, 29.43% and 41.67%, respectively, at NPR of 10. The position, width and total pressure had considerable impacts on the performance (Fig. 64), and the low energy flow was suitable for introducing the secondary injection. Besides, the secondary injection was also employed to improve the performance of the TBCC exhaust system with dual-flowpath at low flight Mach number [167], and the thrust coefficient could be increased by 7.34%.

All of the three flow control methods stated above are to pressurize the ramp by introducing a shock wave into the overexpanded flow to promote flow separation, and thereby, improving the performance. The method of secondary injection is simple and effective, which may be becoming a smart choice. Unfortunately, no study exists on the sudden change in SERN performance resulting from the unsteady characteristics of the overexpanded separation, and the sudden performance change might be more detrimental to the flight safety of the vehicle. Therefore, methods to achieve the smooth performance need to be developed in the follow-up research.

4.3. Mode transition for the TBCC exhaust system

As one of the most critical problems for the TBCC propulsion system, the mode transition implements the switching from the turbine to the ramjet, in which the turbine is closed, while ramjet is started. Mode transition should be performed in a stable way, without sudden change of thrust, unstable change of moment and unstart of inlet. Due to the immaturity of combined cycle propulsion technology, AFOSR and NASA founded a new research center, and the center identified the mode transition from turbine to ramjet as critical to the successful operation of the propulsion system [168]. Exhaust system is one of the TBCC core components, and during the mode transition, the LSF and HSF are gradually shut down and opened up to control the operation conditions of the turbine and ramjet, respectively. Therefore, the research on its mode transition is a key link in the TBCC study.

Early study [169] focused on the steady method, including the numerical simulation and experiment, to study the flowfield feature and performance of the exhaust system at the typical points during the mode transition. Since the exhaust system operated at the parallel operation

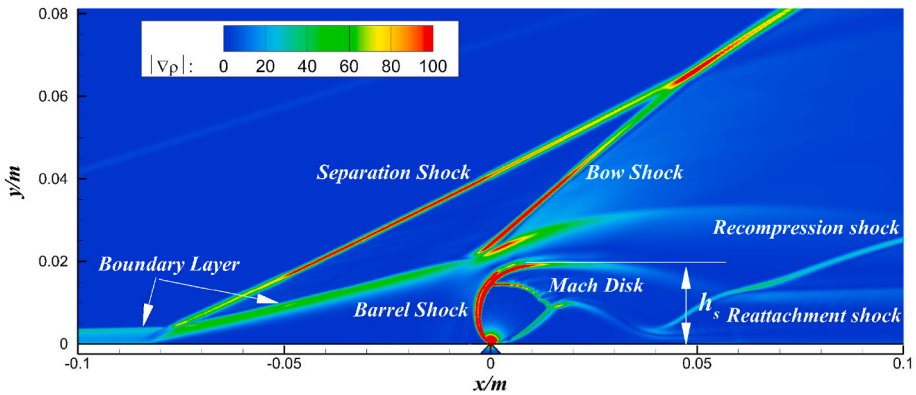


Fig. 63. Typically flowfield structure of supersonic transverse jet.

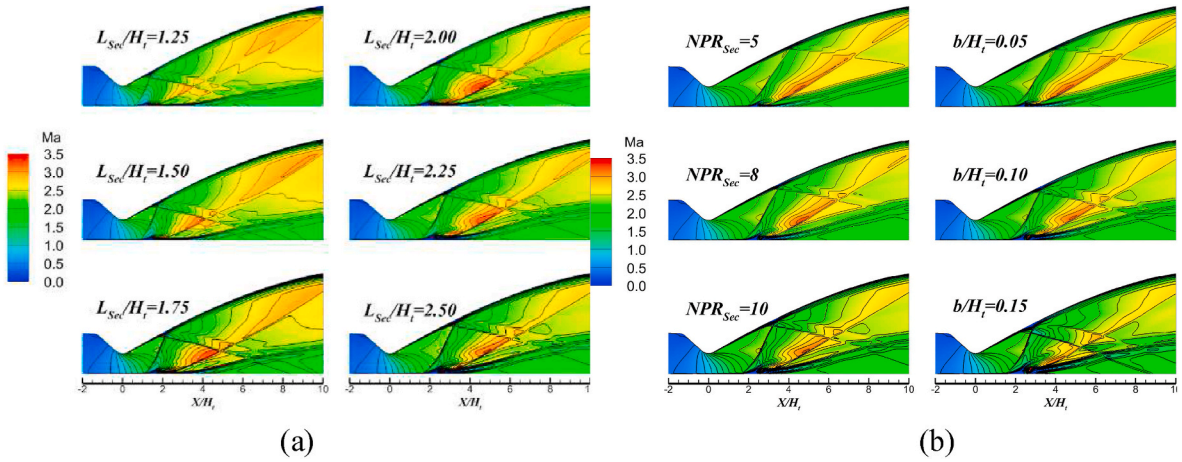


Fig. 64. Influence of key parameter of secondary injection: (a) Location; (b) Total pressure and width [166].

during the mode transition, the flowfield was complex with the interaction of the turbine exhaust with the ramjet jet. The thrust coefficient was changed from 0.9288 to 0.9675, increased by 4.17%. However, the mode transition is a dynamic process, the steady results may not reflect the unsteady properties of the flowfield and performance. Therefore, the unsteady numerical simulation combined with the dynamic mesh technology was employed to investigate the dynamic characteristics of the mode transition for the exhaust system [170]. In the study, the mode transition process was separated into three stages based on the operating condition of the turbine, including the afterburning stage (T_1), throttling stage (T_3), and transition stage between the afterburning state and throttling state (T_2). Fig. 65 summarizes the flowfield features of the exhaust system in the three stages. Due to the varieties in the entrance parameters and the different rotation patterns of the splitter and cowl in the three stages, the flow characteristics were complicated and various. At the initial state, the mass flow rate of the HSF was very small, but the steady flowfield of the exhaust system was quite complex. During the mode transition, the exhaust jets of the turbine and ramjet nozzles interacted with each other, and the resulting shock waves and shear layers moved along with the rotations of the splitter and cowl. The

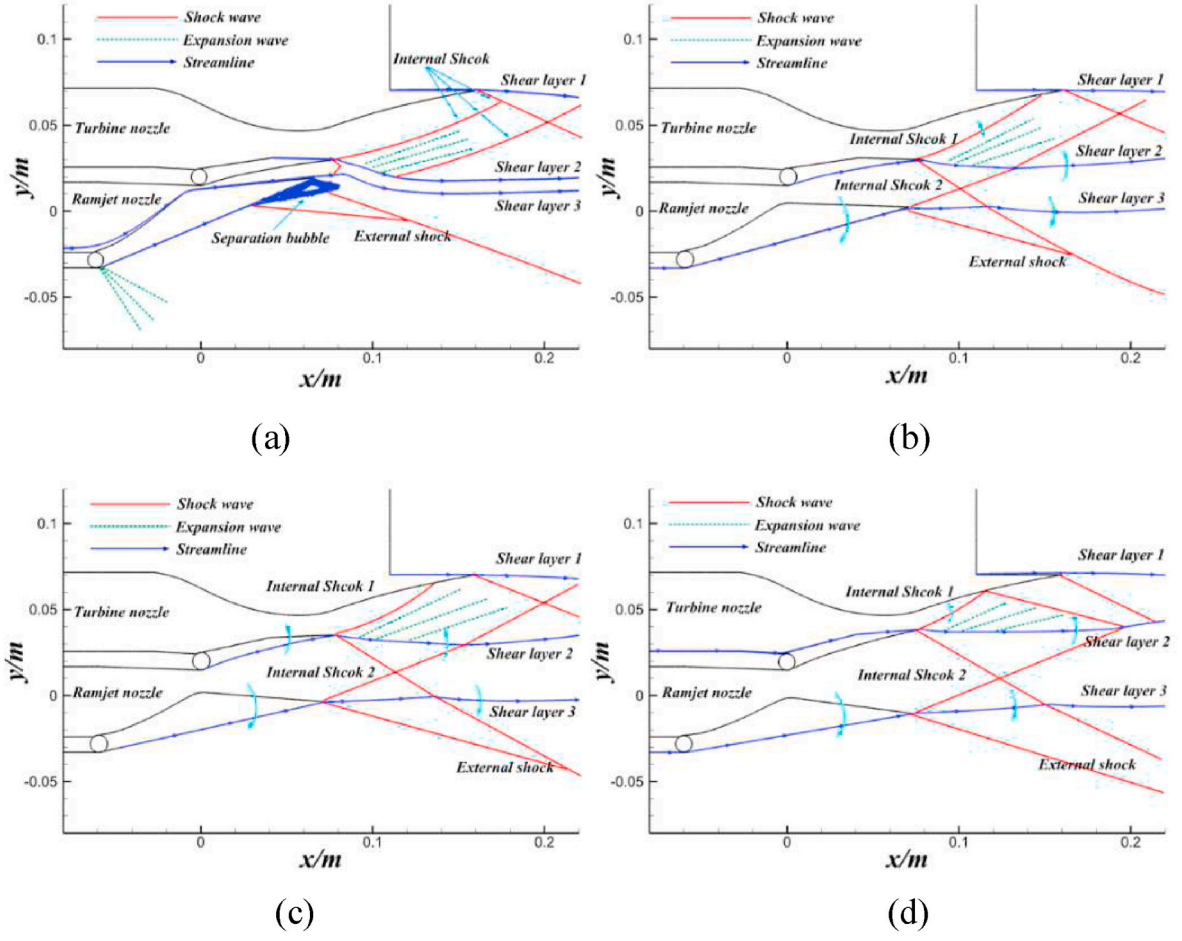


Fig. 65. Schematics of flowfields in different stages [170]: (a) initial state; (b) T_1 Stage; (c) T_2 Stage; (d) T_3 Stage.

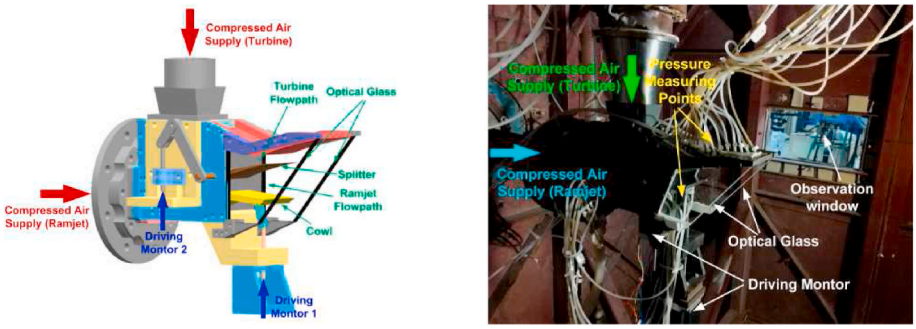


Fig. 66. Experimental model and its installation in wind tunnel [171].

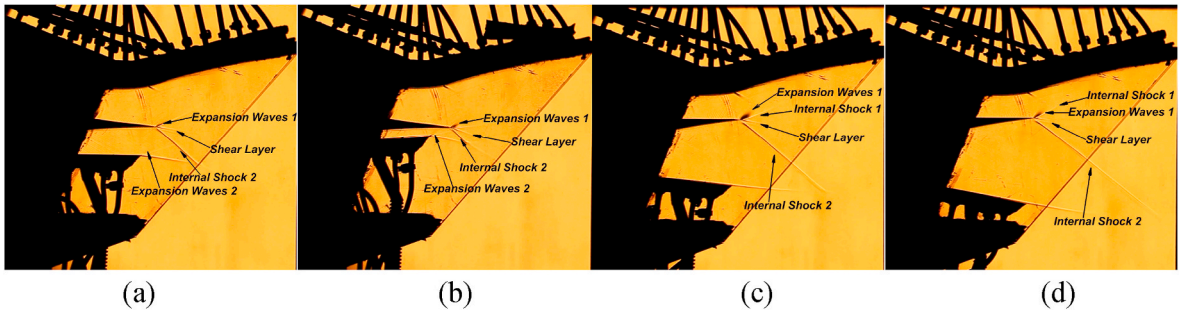


Fig. 67. Flowfield structure captured in experiment [171] : (a) Initial state; (b) Typical state in T_1 stage; (c) Typical state in T_2 stage; (d) Typical state in T_3 stage.

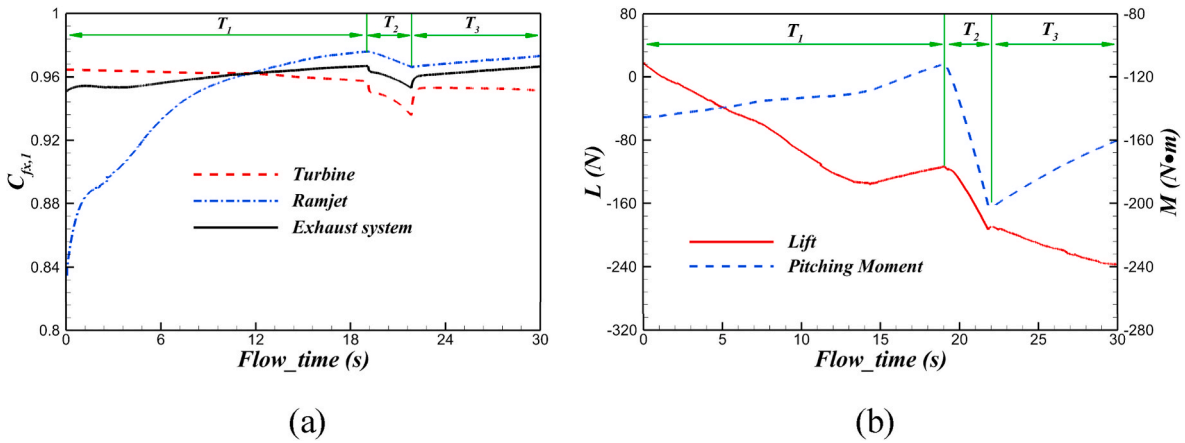


Fig. 68. Performance of the exhaust system in mode transition [170]: (a) Thrust coefficient; (b) Lift and pitching moment.

flowfield in the LSF was significantly influenced by the HSF plume. The dynamic flowfield characteristics were also captured in the experimental study (Fig. 66), especially the complex flow interaction of the jets between the LSF and HSF under the dynamic regulation (Fig. 67).

The complexity and variability of the flowfield in the mode transition were bound to have a direct impact on the performance (Fig. 68). The thrust coefficient changed abruptly at the beginning and end of the splitter rotation, indicating that the motion of the splitter had a negative influence on the thrust coefficient. Besides, both the lift and pitching moment reduced rapidly during the shutdown of the LSF, with the decrements of 67.15% and 80.92%, respectively, which had a negative effect on the stability of the vehicle. Therefore, improved approaches are very necessary to realize the smooth mode transition for the exhaust system. In the study of Wang et al. [172], the sudden change in lift was also reflected in the mode transition, and they denoted that the control law of the entrance parameters in the mode transition could be changed to ensure the smooth performance. However, this approach would affect the operational state of other component as well as the overall performance of the propulsion system. In our opinion, the efficient flow control might be the suitable method, which could use a reference from the flow control in the overexpanded SERN.

5. Development trend of the exhaust system for scramjet and TBCC

A larger amount of research on the aerodynamic issues of the exhaust system for scramjet and TBCC has been conducted in the previous investigations, including the design and the key issue, which can increase the physical understanding of the flow in the exhaust system. However, there are still many relevant aspects of the exhaust system that have not been addressed and are worth further exploration, including:

(1) The performance prediction of the nozzle only depends on the one-dimensional flow theory, and ignores the effects resulting

from geometric constraints, viscosity, chemical nonequilibrium and nonuniform inflow. For further study, the data driven technique may be employed to achieve a fast and accurate prediction that includes the multiple factors within the full envelope.

(2) Although some design methods have been proposed the 2D SERN, SERN with lateral expansion and 3D SERN with shape transition, it is undeniable that the design methods should continue to be developed in the future, such as the direct design for optimizing both ramp and cowl under geometric constraints in 2D SERN, and the 3D MOC for 3D SERN with shape transition.

(3) Previous studies have clarified the influence of the nonuniform inflow, however, the detailed action mechanism is not very clear, such as the variations of the pressure and shear stress distribution resulting from the nonuniform inflow.

(4) With the increase in the flight Mach number, the finite-rate chemistry should be taken into account in design and performance estimation for the SERN. It is necessary to reveal the mechanism of the chemical nonequilibrium flow coupling with shock and expansion waves in the SERN, and develop a design method that considers chemical nonequilibrium.

(6) Most of the experiments for the TBCC exhaust system only focus on the internal flow, and further study of the complex flow interaction between the multi-channel exhaust jets, as well as the coupling of the internal and external flows under the dynamic adjustment process. Besides, it is necessary to develop more configuration solutions for the TBCC exhaust system, such as the axisymmetric multi-channel configuration.

(7) Some key problems should be settled for the TBCC exhaust system at low flight Mach number and mode transition phase, such as the NPR prediction and flow control of the separation pattern transition, and the flow control of the exhaust system during mode transition.

6. Conclusions

The exhaust system is the core component of the hypersonic propulsion system, playing important roles of controlling the engine operating state and producing the thrust, lift and pitching moment for the vehicle. Due to the wide working range of the propulsion system, the design of the exhaust system should consider the performance at the cruise condition and many problems at the off-design conditions, namely the challenges for the scramjet and TBCC nozzles.

For the scramjet nozzle, the challenges are the integrated design requirement, the unique flow phenomena including the nonuniform inflow and chemical nonequilibrium flow, and the coupling of the internal and external flows.

(1) To satisfy the integrated design requirement, the design methods for the 2D planar SERN, SERN with lateral expansion, and 3D SERN with shape transition are developed. The direct design method for the 2D SERN resolves the coupling of the performance optimization theory and geometric constraints. Similarly, the combination of the quasi-2D MOC and optimal thrust theory achieves the efficient design of the SERN considering lateral expansion and geometric constraints. The streamline tracing technique is suitable for the complex contour design of the 3D SERN with shape transition, and the streamline-traced nozzle can be trimmed by the last-running characteristic of the parent flowfield to improve the performance and weight.

(2) The Mach number nonuniformity will be detrimental to the thrust and pitching moment, and the SERN design considering the nonuniform inflow can greatly improve the performance.

(3) The performance estimation of the scramjet nozzle cannot ignore the effect of the finite-rate chemistry, and the further study may focus on the SERN design method considering chemical nonequilibrium flow to realize the efficient expansion flow with finite-rate chemistry in a SERN.

(4) The SERN performance can be improved by optimizing the internal configuration without considering the coupling of the internal and external flows.

In terms of the TBCC, the exhaust system should take into account the variable geometry design for wide flight range, overexpanded flow separation, and mode transition from LSF to HSF.

(1) The variable geometry design, including the 2D and 3D exhaust system, has been developed for various requirements, and the adjustment scheme significantly affects the performance in the flight envelope.

(2) The various separation patterns and their transition exist in the highly overexpanded SERN, greatly decreasing the performance and increasing its instability, and the efficient flow control, such as the secondary injection, is necessary to overcome the problems.

(3) The exhaust system flowfield is complex and unsteady during the mode transition, resulting in the sudden change of the performance, and the effective approach, such as the flow control, should be introduced to ensure the stable mode transition.

Declaration of competing interest

The authors declared that they have no conflicts of interest to this work.

We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

Data availability

No data was used for the research described in the article.

Acknowledgments

We would like to acknowledge the support of 1912 Project on the contract numbers of 2019-JCJQ-DA-001-072, 2019-JCJQ-DA-001-073 and 2019-JCJQ-DA-001-141, as well as the support of NUAA (Nanjing University of Aeronautics and Astronautics) Basic Research on the contract numbers of NS2023008, and the authors are grateful to the editors and reviewers.

References

[1] M. Smart, Scramjet, Aeronaut. Sci. 10 (2017) 605–619.
[2] John J. Bertin, Russell M. Cummings, Fifty years of hypersonics: where we've been, where we're going, Prog. Aero. Sci. 39 (2003) 511–536.
[3] E.T. Curran, S.N.B. Murthy, Scramjet Propulsion, American Institute of Aeronautics and Astronautics, Virginia, 2001.
[4] A. Ferri, Review of scramjet propulsion technology, J. Aircraft 5 (1) (1968) 3–10.
[5] Andre W. Marshall, Ashwani K. Gupta, Mark J. Lewis, Critical Issues in TBCC Modeling, 2004. AIAA Paper 2004-3827.
[6] R. Charles, McClinton, high-speed/hypersonic aircraft propulsion technology development, RTO-EN-AVT 150 (2008).
[7] Weixuan Li, Xiong Chen, Dan Zhao, et al., Swirling effect on thermodynamic performance in a solid fueled ramjet with paraffin-polyethylene, Aero. Sci. Technol. 107 (2020), 106341.
[8] Ronald S. Fry, A century of ramjet propulsion technology evolution, J. Propul. Power 20 (1) (2004) 27–58.
[9] T. Edward, Curran, scramjet engines: the first forty years, J. Propul. Power 17 (6) (2001) 1138–1148.
[10] D. Szirozak, H. Smith, A review of design issues specific to hypersonic flight vehicles, Prog. Aero. Sci. 84 (2016) 1–28.
[11] R.A. Gross, W. Chinitz, A study of supersonic combustion, J. Aero. Sci. 27 (7) (1960) 517–524.
[12] A. Ferri, P.A. Libby, V. Zakkay, Theoretical and Experimental Investigations of Supersonic Combustion, Polytechnic Inst. of Brooklyn, New York, 1962. Chap. 2.
[13] E.H. Andrews, E.A. Mackley, Review of NASA's Hypersonic Research Engine Project, 1993. AIAA Paper 93-2323.

[14] F.S. Billig, SCRAM-A Supersonic Combustion Ramjet Missile, 1993. AIAA Paper 93-2329.
[15] A.H. Whitehead, NASP Aerodynamic, 1989. AIAA Paper 1989-5013.
[16] H. Wright, National Aero-Space Plane Technology Development Overview, 1989. AIAA Paper 1989-5003.
[17] H. Albert, Boudreau, Status of the U. S. Air Force HyTech Program, 2003. AIAA Paper 2003-6947.
[18] O.A. Powell, J.T. Edwards, R.B. Norris, K.E. Numbers, Development of hydrocarbon-fueled scramjet engines: the hypersonic technology (HyTech) program, J. Propul. Power 17 (6) (2001) 1170–1176.
[19] Charles R. McClinton, David R. Rausch, Hyper-X Program Status, 2001. AIAA Paper 2001-1910.
[20] R.C. Rogers, A.T. Shin, N.E. Hass, Scramjet Development Tests Supporting the Mach 10 Flight of the X-43, 2005. AIAA Paper 2005-3351.
[21] Lawrence O. Brase Jr., Darin P. Haudrich, Flutter and Divergence Assessment of the HyFly Missile, 2009. AIAA Paper 2009-2462.
[22] Steven Walker, Ming Tang, Sue Morris, Caesar Mamplata, Falcon HTV-3X - A Reusable Hypersonic Test Bed, 2008. AIAA Paper 2008-2544.
[23] Steven H. Walker, Fredrick Rodgers, Falcon Hypersonic Technology Overview, 2005. AIAA Paper 2005-3253.
[24] Joseph M. Hank, James S. Murphy, Richard C. Mutzman, The X-51A Scramjet Engine Flight Demonstration Program, 2008. AIAA Paper 2008-2540.
[25] Thomas Neuenhahn, Herbert Olivier, Development of the HyShot Stability Demonstrator, 2006. AIAA Paper 2006-2960.
[26] Michael K. Smart, Neal E. Hass, Flight data analysis of the HyShot 2 scramjet flight experiment, AIAA J. 44 (10) (2006) 2366–2375.
[27] Steven H. Walker, Frederick C. Rodgers, Aaron L. Esposita, Hypersonic Collaborative Australia/United States Experiment (HyCAUSE), 2005. AIAA Paper 2005-3254.
[28] Steven Walker, Frederick Rodgers, Alan Paull, HyCAUSE Flight Test Program, 2008. AIAA Paper 2008-2580.
[29] M.E. Roberts, M.K. Smart, M.A. Forst, HIFiRE 7: Design to Achieve Scientific Goals, 2012. AIAA Paper 2012-5841.
[30] Kevin R. Jackson, Mark R. Gruber, Salvatore Buccellato, Mach 6-8+ hydrocarbon-fueled scramjet flight experiment: the HIFiRE flight 2 project, J. Propul. Power 31 (1) (2015) 36–53.
[31] M.E. Roberts, M.K. Smart, M.A. Frost, HIFiRE 7: Design to Achieve Scientific Goals, 2012. AIAA Paper 2012-5841.
[32] Marc Bouchez, Alexandre S. Roudakov, Valery I. Kopchenov, Vyacheslav L. Semenov, French-Russian Anslysis of Kholod Dual-Mode Ramjet Flight Experiments, 2005. AIAA Paper 2005-3320.
[33] A.S. Roudakov, V.L. Semenov, M.V. Strokin, et al., The Prospects of Hypersonic Engines In-Flight Testing Technology Development, 2001. AIAA Paper 2001-1807.
[34] V.V. Kislykh, A.A. Kondratov, V.L. Semenov, The Program for the Complex Investigation of the Hypersonic Flight Laboratory (HFL) "IGLA" in the PGU of TSNIIMASH, 2001. AIAA Paper 2001-1875.
[35] A.S. Roudakov, V.L. Semenov, M.V. Strokin, et al., The Prospects of Hypersonic Engines In-Flight Testing Technology Development, 2001. AIAA Paper 2001-1807.
[36] M. Sancho, The French Hypersonic Research Program PREPHA: Progress Review, 1993. AIAA Paper 93-5160.
[37] Ph Novelli, W. Koschel, Progress of the JAPHAR Cooperation between ONERA and DLR on Hypersonic Airbreathing Propulsion, 2001. AIAA Paper 2001-1870.
[38] Laurent Serre, Francois Falempin, PROMETHEE: the French Military Hypersonic Propulsion Program-Status in 2002, 2003. AIAA Paper 2003-6950.
[39] S. Weingarther, Sanger-the Reference Concept of the German Hypersonics Technology Program, 1993. AIAA Paper 93-5161.
[40] Naoki Sato, Toshio Akimoto, Yoshikazu Miyazawa, Hope-X High Speed Flight Demonstration Program Phase I, 2001. AIAA Paper 2001-1806.
[41] Masaaki Yanagihara, Yoshikazu Miyazawa, Toshio Akimoto, et al., Hope-X High Speed Flight Demonstration Program Phase II, 2001. AIAA Paper 2001-1805.
[42] Javier Urzay, Supersonic combustion in air-breathing propulsion systems for hypersonic flight, Annu. Rev. Fluid Mech. 50 (2018) 592–627.
[43] Stephen Corda, Timothy R. Moes, Masashi Mizukami, et al., The SR-71 Test Bed Aircraft: A Facility for High-Speed Flight Research, 2000. NASA/TP 2000-209023.
[44] Stephen Corda, Bradford A. Neal, Timothy R. Moes, et al., Flight Testing the Linear Aerospike SR-71 Experiment (LASRE), 1998. NASA/TM 1998-206567.
[45] D.W. Lam, Use of the PARC Code to Estimate the Off-Design Transition Performance of an Over/Under Turboramjet Nozzle, 1995. AIAA Paper 95-2616.
[46] Paul A. Bartolotta, Nancy B. McNelis, Douglas G. Shafer, High Speed Turbines: Development of a Turbine Accelerator (RTA) for Space Access, 2003. AIAA Paper 2003-6943.
[47] Nancy McNelis, Bartolotta Paul, Revolutionary Turbine Accelerator (RTA) Demonstrator, 2005. AIAA Paper 2005-3250.
[48] Jinho Lee, Ralph Winslow, Robert J. Buehrle, The GE-NASA RTA Hypersonic Design and Development, 2005. NASA/TM 2005-213803.
[49] Caesar Mamplata, Ming Tang, Technical Approach to Turbine-Based Combined Cycle: FaCET, 2009. AIAA Paper 2009-5537.
[50] Steven Walker, Ming Tang, Caesar Mamplata, TBCC Propulsion for a Mach 6 Hypersonic Airplane, 2009. AIAA Paper 2009-7238.
[51] J.E. Bradford, A. Charania, J. Wallace, Quicksat: A Two-Stage to Orbit Reusable Launch Vehicle Utilizing Air-Breathing Propulsion for Responsive Space Access, 2004. AIAA Paper 2004-5950.
[52] D.R. Eklund, A.H. Boundreau, A Turbine-Based Combined Cycle Solution for Responsive Space Access, 2005. AIAA Paper 2005-4186.
[53] Melvin J. Bulman, Siebenhaar Adam, Combined Cycle Propulsion: Aerojet Innovations for Practical Hypersonic Vehicles, 2011. AIAA Paper 2011-2397.
[54] Siebenhaar Adam, J. Thomas, Bogar, Integration and Vehicle Performance Assessment of the Aerojet "TriJet" Combined-Cycle Engine, 2009. AIAA Paper 2009-7420.
[55] Weiwei Wang, Dan Li, Jun Zeng, Development of propulsion system for near-space HTHL high-speed vehicles abroad, Gas Turbine Exp.Res. 27 (1) (2014) 57–62 (in Chinese).
[56] J. Steelant, Achievement Obtained for Sustained Hypersonic Flight within the LAPCAT Project, 2008. AIAA Paper 2008-2578.
[57] J. Steelant, Sustained Hypersonic Flight in Europe: Technology Drivers for LAPCAT II, 2009. AIAA Paper 2009-7240.
[58] Francois Falempin, Marc Bouchez, V. Perrillat, LAPCAT 2-Axisymmetric Concept for a Mach 8 Cruiser Preliminary Design and Performance Assessment, 2009. AIAA Paper 2009-7437.
[59] Unmeel Mehta, Michael Aftosmis, Jeffrey Bowles, Shishir Pandya, Skylon Aerodynamics and SABRE Plumes, 2015. AIAA Paper 2015-3605.
[60] Barry M. Hellman. Two Stage to Orbit Conceptual Vehicle Designs Using the SABRE Engine, AIAA Paper 2016-5320.
[61] Baoxi Wei, Wenhui Ling, Feiteng Luo, et al., Propulsion Performance Research and Status of TRRE Engine Experiment, 2017. AIAA Paper 2017-2531.
[62] Baoxi Wei, Wenhui Ling, Gang Qiang, et al., Analysis of key technologies and propulsion performance research of TRRE engine, J. Propuls. Technol. 38 (2) (2017) 298–305 (in Chinese).
[63] Andreadis Dean, Scramjet engine enabling the seamless integration of air & space operations, in: The Industrial Physicist, Aug./Sept., 2004.
[64] R. Lederer, W. Kruger, Nozzle Development as a Key Element for Hypersonic, 1993. AIAA Paper 93-5058.
[65] C.L.W. Edwards, W.J. Small, J.P. Weidner, et al., Studies of Scramjet/Airframe Integration Techniques for Hypersonic Aircraft, 1975. AIAA Paper 75-58.
[66] William H. Heiser, David T. Pratt, Hypersonic Airbreathing Propulsion, American Institute of Aeronautics and Astronautics, Washington, 1994, pp. 387–414.
[67] L.Z. Dumitrescu, Minimum length axisymmetric laval nozzles, AIAA J. 13 (4) (1975) 520–531.
[68] T. Zebbich, Z. Toubi, Supersonic Two-Dimensional Minimum Length Nozzle Design at High Temperature. Application for Air, 2006. AIAA Paper 2006-4599.
[69] Jianwei Mo, Jinglei Xu, Zhibin Quan, Kaikai Yu, Lv Zheng, Design and cold flow test of a scramjet nozzle with nonuniform inflow, Acta Astronaut. (2015) 92–105, 2018.
[70] M. Goeing, Nozzle Design Optimization by Method of Characteristics, 1990. AIAA Paper 90-2024.
[71] Shanguang Guo, Zhenguo Wang, Yuxin Zhao, Design of a continuously variable Mach-number nozzle, J. Cent. S. Univ. 22 (2015) 522–528.
[72] J.D. Hoffman, Design of compressed truncated perfect nozzles, J. Propul. Power 3 (2) (1987) 150–156.
[73] Zhibin Quan, Jinglei Xu, Jianwei Mo, Design of nonlinearly compressed SERN profile, J. Propuls. Technol. 33 (6) (2012) 951–955 (in Chinese).
[74] Jianwei Mo, Jinglei Xu, Zhibin Quan, Experimental and numerical study of the truncated SERN, J. Propuls. Technol. 33 (6) (2012) 940–945 (in Chinese).
[75] Qiang Zhao, Jinglei Xu, Zhipeng Fan, Experimental and numerical study of truncation method for thrust nozzle profile based on balance of thrust and friction, J. Propuls. Technol. 35 (2) (2014) 151–156 (in Chinese).
[76] S.K. Damira, A.G. Marathe, K. Sudhakar, et al., Parametric Optimization of Single Expansion Ramp Nozzle, 2006. AIAA Paper 2006-5188.
[77] Jianping Li, Wenyan Song, Ying Xing, et al., Influences of geometric parameters upon nozzle performance in scramjet, Chin. J. Aeronaut. 21 (2008) 506–511.
[78] Taiyuan Gao, Kai Cui, Xiuping Wang, et al., Aerodynamic optimization and evaluation for the three-dimensional afterbody/nozzle integrated configuration of hypersonic vehicles, Chin. Sci. Bull. 57 (8) (2012) 849–857.
[79] Xuzhao He, Yong Zhang, Guangyuan Wang, Hongli Ni, Jialing Le, Automated design optimization of single expansion ramp nozzle for hypersonic vehicle, J. Propuls. Technol. 28 (2) (2007) 148–151 (in Chinese).
[80] Qing Wang, Liangxian Gu, Chunlin Gong, Mutil-objective optimization design of geometry-variable nozzle for scrmajet, J. Propuls. Technol. 34 (3) (2013) 294–299 (in Chinese).
[81] Wendian Zhang, Yibai Wang, Yu Liu, Lizi Qin, Xiaoyuan Zhang, Miaosheng He, Regression study of scramjet nozzle performance on contour parameters, J. Aero. Power 28 (9) (2013) 2029–2036 (in Chinese).
[82] Lina Pang, Jinglei Xu, Zhipeng Fan, Optimization design and experimental study of cowl configuration of SERN, J. Propuls. Technol. 35 (10) (2014) 1297–1302 (in Chinese).
[83] Wei Huang, Zhenguo Wang, Derek B. Ingham, et al., Design exploration for a single expansion ramp nozzle (SERN) using data mining, Acta Astronaut. 83 (2013) 10–17.
[84] Hongchao Song, Xin Li, Lucheng Ji, Research on the optimization of unilateral expansion nozzle based on the discrete adjoint method, J. Eng. Thermophys. 38 (9) (2017) 1849–1854 (in Chinese).
[85] Rui Li, Jinglei Xu, Guangtao Song, Scramjet nozzle designed by chaotic particle swarm optimization and space-marching godunov scheme, J. Spacecraft Rockets 57 (4) (2020) 851–856.
[86] Lan Tang, Jinglei Xu, Jianwei Mo, Kaikai Yu, Weicheng Qi, Reverse design of asymmetric nozzle based on given wall pressure distribution, J. Aero. Power 32 (6) (2017) 1515–1522 (in Chinese).

[87] Lan Tang, Jinglei Xu, Jianwei Mo, Kaikai Yu, Weicheng Qi, Reverse design and experimental validation of single expansion ramp nozzle based on given wall pressure distribution, J. Propuls. Technol. 37 (12) (2016) 2226–2232 (in Chinese).

[88] Kaikai Yu, Jinglei Xu, Lv Zheng, et al., Inverse design methodology on a single expansion ramp nozzle for scramjets, Aero. Sci. Technol. 92 (2019) 9–19.

[89] Kaikai Yu, Yile Chen, Shuai Huang, et al., Optimization and analysis of inverse design method of maximum thrust scramjet nozzles, Aero. Sci. Technol. 105 (2020), 105948.

[90] Lv Zheng, Jinglei Xu, Yu Yang, Jianwei Mo, A new design method of single expansion ramp under geometric constraints for scramjets, Aero. Sci. Technol. 66 (2017) 129–139.

[91] G.V.R. Rao, Exhaust nozzle contour for optimum thrust, Jet Propuls. 28 (3) (1958) 377–382.

[92] M.J. Zucrow, J.D. Hoffman, Gas Dynamics, vol. 2, John Wiley and Sons, New York, 1976.

[93] Lv Zheng, Jinglei Xu, Kaikai Yu, Guangtao Song, Experimental and numerical investigations of a scramjet nozzle at various operations, Aero. Sci. Technol. 96 (2020), 105536.

[94] Philippe Novelli, Laurent Serre, Wolfgang Koschel, Francois Falempin. European Efforts towards Hypersonic Propulsion Mastering. AIAA Paper 2003-2604.

[95] Yanfeng Niu, Jinglei Xu, Jianwei Mo, Design and Performance analysis of the three-dimensional asymmetric nozzle with lateral expansion, J. Aero. Power 31 (11) (2016) 2717–2722.

[96] Yanfeng Niu, Jinglei Xu, Jianwei Mo, Investigation of effects of lateral expansion on performance of single expansion ramp nozzle, J. Propuls. Technol. 38 (3) (2017) 519–524 (in Chinese).

[97] Lv Zheng, Jinglei Xu, Jianwei Mo, Design of a three-dimensional scramjet nozzle considering lateral expansion and geometric constraints, Acta Astronaut. 141 (2017) 172–182.

[98] Adam Siebenhaar, Thomas J. Bogar. The Impact of Round Combustor on TBCC Propulsion and Hypersonic Cruise Vehicle. AIAA Paper 2006-7986.

[99] H.D. Thompson, S.N.B. Murthy, Design of optimized three-dimensional nozzles, J. Spacecraft Rockets 3 (9) (1966) 1384–1393.

[100] Lynn E. Snyder, H. Doyle Thompson, Three-dimensional thrust nozzle design for maximum axial thrust, AIAA J. 9 (10) (1971) 1891–1892.

[101] S.M. Dash, P.D. Del Guidice, Three Dimensional Nozzle-Exhaust Flow Field Analysis by a Reference Plane Technique, 1972. AIAA Paper 72-704.

[102] X.Q. Xing, M. Damodaran, Design of Three-Dimensional Nozzle Shapes Using Hybrid Optimization Techniques, 2004. AIAA Paper 2004-26.

[103] X.Q. Xing, M. Damodaran, Design of Three-Dimensional Nozzle Shape Using NURBS, CFD and Hybrid Optimization Strategies, 2004. AIAA Paper 2004-4368.

[104] Katsuypshi Tanimizu, David J. Mee, Raymond J. Stalker, et al., Nozzle design study for a quasi-axisymmetric scramjet-powered vehicle at Mach 7.9 flight conditions, Shock Waves 23 (2013) 453–460.

[105] Katsuypshi Tanimizu, David J. Mee, Raymond J. Stalker, et al., Thrust nozzle design study for a quasi-axisymmetric scramjet-powered vehicle [J], J. Propul. Power 27 (1) (2011) 40–48.

[106] Frederic S. Billig, Ajay P. Kothari, Streamline tracing: technique for design hypersonic vehicles, J. Propul. Power 16 (3) (2000) 465–471.

[107] T.M. Taylor, D. VanWie, Performance Analysis of Hypersonic Shape-Changing Inlets Derived from Morphing Streamline Traced Flowpath, 2008. AIAA Paper 2008-2623.

[108] Xin Lu, Lianjie Yue, Yabin Xiao, et al., Design of Scramjet Nozzle Employing Streamline Tracing Technique, 2009. AIAA Paper 2009-7248.

[109] Jianwei Mo, Jinglei Xu, Rui Gu, et al., Design of an asymmetric scramjet nozzle with circular to rectangular shape transition, J. Propul. Power 30 (3) (2014) 812–819.

[110] Jens Kunze, Michael K. Smart, Rowan Gollan, A design method for three-dimensional scramjet nozzles with shape transition, J. Propul. Power 38 (1) (2022) 3–17.

[111] Lv Zheng, Jinglei Xu, Jianwei Mo, Design and analysis on three-dimensional scramjet nozzles with shape transition, Aero. Sci. Technol. 71 (2017) 189–200.

[112] Sandra L. Snelling, Effect of Nonuniform Entrance Flow Profile on Hypersonic Nozzle Pitching Moment, AD-A244050, 1991.

[113] Xiaodong Wang, Jialing Le, Effect of temperature profile at entrance on flow-fields of nozzle, J. Propuls. Technol. 23 (4) (2002) 283–286 (in Chinese).

[114] Schindel Leon, Effect of nonuniform nozzle flow on scramjet scramjet performance, J. Propul. Power 15 (2) (1998) 363–364.

[115] Zhibin Quan, Jinglei Xu, Bin Li, Xin Li, Jianwei Mo, Cold flow experiment and numerical simulation on nonuniform entrance flow of scramjet nozzle, Acta Aeronautica Astronautica Sinica 34 (10) (2013) 2308–2315 (in Chinese).

[116] Pu Zhang, Jinglei Xu, Zhibin Quan, et al., Effects of nonuniform Mach-number entrance on scramjet nozzle flowfield and performance, Acta Astronaut. 129 (2016) 201–210.

[117] Dan Zhao, Ephraim Gutmark, Philip de Goey, A review of cavity-based trapped vortex, ultra-compact, high-g, inter-turbine combustors, Prog. Energy Combust. Sci. 66 (2018) 42–82.

[118] Wenxuan Li, Dan Zhao, Linqing Zhang, Xiong Chen, Proper orthogonal and dynamic mode decomposition analyses of nonlinear combustion instabilities in a solid-fuel ramjet combustor, Therm. Sci. Eng. Prog. 27 (2022), 101147.

[119] Wenxuan Li, Dan Zhao, Xiong Chen, et al., Numerical investigations on solid-fuel ramjet inlet thermodynamic properties effects on generating self-sustained combustion instability, Aero. Sci. Technol. 119 (2021), 107097.

[120] Oussama Rizkalla, Chinitz Wallace, I. John, Erdos, Calculated chemical and vivrational nonequilibrium effects in hypersonic nozzles, J. Propul. Power 6 (1) (1990) 50–57.

[121] J.J. Sangiovanni, T.J. Barber, Role of hydrogen/air chemistry in nozzle performance for a hypersonic propulsion system, J. Propul. Power 9 (1) (1993) 134–138.

[122] K.B. Wilson, J. H. Doty Analyizing, Hypersonic Nozzles Using an Efficient CFD Algorithm and Thermodynamic Model, 1994. AIAA Paper 94-0023.

[123] Yue Huang, Peiyong Wang, Yang Dou, Fei Xing, Chemical non-equailibrium flow analysis of H_2 fueled scramjet nozzle, Case Stud. Therm. Eng. 5 (2015) 89–97.

[124] T.A. Gronland, J.L. Cambier, Sensitivity to Physical Modelling for Nozzle/Afterbody Flowfields, 1996. AIAA Paper 96-4547.

[125] Xinyue Wang, Zhenpeng Yang, Yanqing Wang, Effects of non-equilibrium chemical reaction flow on the performance of scramjet nozzle, J. Aero. Power 24 (5) (2009) 1022–1027 (in Chinese).

[126] R.J. Stalker, N.K. Truong, R.G. Morgan, A. Paull, Effects of hydrogen-air non-equilibrium chemistry on the performance of a model scramjet thrust nozzle, Aeronaut. J. 11 (2004) 575–584.

[127] M.J. Candon, H. Ogawa, Numerical analysis and design optimization of supersonic after-burning with strut fuel injectors for scramjet engines, Acta Astronaut. 147 (2018) 281–296.

[128] Cheng Cheng, Lizi Qin, Yu Liu, Nozzle chemical non-equilibrium effects based on strut combustor, J. Beijing Univ. Aeronaut. Astronaut. 39 (1) (2013) 31–36 (in Chinese).

[129] Xiaoyuan Zhang, Lizi Qin, Yu Liu, Miao Sheng He, Effects of radical recombination on scramjet nozzle performance, J. Propuls. Technol. 34 (5) (2013) 589–594 (in Chinese).

[130] Xiaoyuan Zhang, Lizi Qin, Hong Chen, Xuzhao He, Yu Liu, Radical recombination in a hydrocarbon-fueled scramjet nozzle, Chin. J. Aeronaut. 27 (6) (2014) 1413–1420.

[131] Earl R. Keener, Frank W. Spaid, Hypersonic nozzle-afterbody experiment: flow visualization and boundary-layer measurement, J. Spacecraft Rockets 33 (3) (1996) 326–332.

[132] Shigeya Watanabe, Scramjet nozzle experiment with hypersonic external flow, J. Propul. Power 9 (4) (1993) 521–528.

[133] Frank W. Spaid, Earl R. Keener, Hypersonic nozzle-afterbody experiment: flowfield surveys, J. Spacecraft Rockets 33 (3) (1996) 333–338.

[134] V. Thiagarajan, S. Panneerelvam, E. Rathakrishnan, Numerical flow visualization of a single expansion ramp nozzle with hypersonic external flow, J. Visual 9 (1) (2006) 91–99.

[135] C. Hirschen, A. Gulhan, W.H. Beck, et al., Experimental study of a scramjet nozzle flow using the pressure-sensitive-paint method, J. Propul. Power 24 (4) (2008) 662–672.

[136] C. Hirschen, A. Gulhan, W.H. Beck, et al., Experimental Study of the Interaction between Internal and External Flows of a Scramjet Nozzle Using Various Diagnostic Techniques, 2007. AIAA Paper 2007-5088.

[137] C. Hirschen, A. Gulhan, Infrared thermograph and pitot pressure measurements of a scramjet nozzle flowfield, J. Propuls. Power 25 (5) (2009) 1108–1120.

[138] Xuzhao He, Si Qin, Feng Wei, Jialing Le, Test of non-uniform nozzle plume for air-breathing hypersonic vehicle, Acta Aeronautica Astronautica Sinica 38 (3) (2017), 120199 (in Chinese).

[139] P. Gruhn, A. Henckels, G. Sieberger, Improvement of the SERN nozzle performance by aerodynamic flap design, Aero. Sci. Technol. 6 (2002) 395–405.

[140] P. Gruhn, A. Henkels, S. Kirschstein, Flap contour optimization for highly integrated SERN nozzles, Aero. Sci. Technol. 4 (2000) 555–565.

[141] Eric J. Gamble, Haid Dan, Hydraulic and Kinematic System Model for TBCC Dynamic Simulation, 2010. AIAA Paper 2010-6641.

[142] M. Angel, Espinosa. Integration of Turbine Engines in Hypersonic Airbreathing Vehicles, 2003. AIAA Paper 2003-4408.

[143] Dustin Wood, Brian Landrum, Demaneuf Olivier, Analysis of the Rocket Plug Nozzle Combined Cycle Propulsion System, 2008. AIAA Paper 2008-5168.

[144] Dustin Wood, Preliminary Analysis of the Rocket Plug Nozzle Combined Cycle (RPNCC) Propulsion System, 2009. AIAA Paper 2009-202.

[145] Jianwei Mo, Jinglei Xu, Songsong Qiao, Numerical study of the over-under TBCC exhaust system, J. Propuls. Technol. 34 (4) (2013) 463–469.

[146] Yanfeng Niu, Jinglei Xu, Baocheng Xu, Kuangshi Chen, Numerical and experimental study of over-under TBCC exhaust system flow structure, J. Propuls. Technol. 38 (12) (2017) 2686–2691.

[147] Feng Wang, Design and Performance Study of a Ma0-7 Over-under TBCC Exhaust System, Nanjing University of Aeronautics and Astronautics, 2018 (in Chinese).

[148] Lv Zheng, Jinglei Xu, Guangtao Song, Design and performance analysis of an exhaust system for an over-under turbine based combined cycle operating at Mach 0-6, Aero. Sci. Technol. 94 (2019), 105386.

[149] Wenda Hua, Jinglei Xu, Design approach and experiment of three-dimensional over/under TBCC exhaust system, J. Aero. Power 33 (9) (2018) 2268–2277.

[150] Baocheng Xu, Jinglei Xu, Xiao Wang, et al., Flowfield and performance analysis of a three-dimensional TBCC exhaust nozzle, J. Eng. Gas Turbines Power 139 (2017), 112602-1-9.

[151] W. Engblom, Numerical Prediction of SERN Performance Using WIND Code, 2003. AIAA Paper 2003-4410.

[152] Y. Yu, J. Xu, J. Mo, M. Wang, Numerical investigation of separation pattern and separation pattern transition in overexpanded single expansion ramp nozzle, Aeronaut. J. 118 (1202) (2014) 399–424.

[153] Yu Yang, Jinglei Xu, Jianwei Mo, Mingtao Wang, Principal parameters in flow separation patterns of over-expanded single expansion ramp nozzle, Eng. Appl. Comput. Fluid Mech. 8 (2) (2014) 274–288.

[154] Y. Yu, J. Xu, K. Yu, J. Mo, Unsteady transitions of separation patterns in single expansion ramp nozzle, Shock Waves 25 (2015) 623–633.
[155] Chengjun He, Jianqiang Li, Zhaolin Fan, Yaohua Li, Jinmin Liang, Rongzhao Gao, Lei Miao, Flow separation unsteadiness in single expansion ramp nozzle, J. Aero. Power 34 (11) (2019) 2339–2346 (in Chinese).
[156] Chengjun He, Jianqiang Li, Zhaolin Fan, Yaohua Li, Rongzhao Gao, Jinmin Liang, Lei Miao, Experimental investigation of wall pressure unsteadiness in an over-expanded single expansion ramp nozzle, J. Propuls. Technol. 41 (3) (2020) 537–543 (in Chinese).
[157] Chengjun He, Jianqiang Li, Jiangtao Huang, Yaohua Li, Xian Chen, Unsteadiness of flow separation in an asymmetric supersonic nozzle, Acta Aeronautica Astronautica Sinica 43 (1) (2022), 124930 (in Chinese).
[158] Jianhui Ge, Jinglei Xu, Mingtao Wang, Jianwei Mo, Prediction of flow separation in asymmetric ramp nozzle, Acta Aeronautica Astronautica Sinica 33 (8) (2012) 1394–1399 (in Chinese).
[159] Mingtao Wang, Jinglei Xu, Yu Yang, Zhipeng Fan, Jianwei Mo, Numerical simulation and experimental validation of SERN separation with optimized turbulence model based on DOE method, J. Propuls. Technol. 35 (3) (2014) 289–295 (in Chinese).
[160] S. Yungster, C.J. Trefny, Computational Study of Single-Expansion-Ramp Nozzle with External Burning, 1994. AIAA Paper 94-0024.
[161] H.T. Lai, Computation of H2/Air Reacting Flowfields in Drag-Reduction External Combustion, 1992. AIAA Paper 92-3672.
[162] Scott C. Asbury, Christopher L. Gunther, Craig A. Hunter, A Passive Cavity Concept for Improving the Off-Design Performance of Fixed-Geometry Exhaust Nozzle, 1996. AIAA Paper 96-2541.
[163] Li Zhou, Hua Xiao, Zhanxue Wang, Zengwen Liu, Effects of passive cavity on high pressure ratio single expansion ramp nozzle, J. Aero. Power 30 (8) (2015) 1811–1817 (in Chinese).
[164] Eric Gamble, Haid Dan, Improving Off-Design Nozzle Performance Using Fluidic Injection, 2004. AIAA Paper 2004-1206.
[165] Eric J. Gamble, DeFrancesco Rich, Haid Dan, Fluidic Nozzle to Improve Transonic Pitch and Thrust Performance of Hypersonic Vehicle, 2005. AIAA Paper 2005-3501.
[166] Lv Zheng, Jinglei Xu, Jianwei Mo, Numerical investigation of improving the performance of a single expansion ramp nozzle at off-design conditions by secondary injection, Acta Astronaut. 133 (2017) 233–243.
[167] Zhanxue Wang, Shaobo Qi, Li Zhou, Jingwei Mo, Performance improvement study of over-under TBCC exhaust system by secondary flow injection, J. Propuls. Technol. 39 (4) (2018) 760–767 (in Chinese).
[168] James C. McDaniel, Harsha Chelliah, Christopher P. Goyne, et al., US National Center for Hypersonic Combined Cycle Propulsion: an Overview, 2009. AIAA Paper 2009-7280.
[169] Jianwei Mo, Jinglei Xu, Liuhuan Zhang, Design and experiment study of an over-under TBCC exhaust system, J. Eng. Gas Turbines Power 36 (2014), 014501-1-8.
[170] Lv Zheng, Jinglei Xu, Jianwei Mo, Study of the unsteady mode transition process for an over-under TBCC exhaust system, Acta Astronaut. 136 (2017) 259–272.
[171] Lv Zheng, Jinglei Xu, Kaikai Yu, Guangtao Song, Jianwei Mo, Experimental and numerical investigations on the mode transition process of an exhaust system for an over-under turbine based combined cycle, Aero. Sci. Technol. 107 (2020), 106265.
[172] Feng Wang, Jinglei Xu, Yangsheng Wang, Study of flowfield characteristics of an over-under TBCC exhaust system during mode transition process, J. Exp. Fluid Mech. 33 (3) (2019) 68–75 (in Chinese).